7SDX0 Bachelor End Project SED
Unit Structural Engineering and Design

Cycle bridge

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Introduction

Building lightweight and using more prefabricated elements is becoming more popular these days (Saint-Gobain Tape Solutions, 2022). These methods help with sustainable construction and decarbonization, which are crucial factors in decreasing climate change and increase the well-being inside buildings. There are numeral reasons for building lightweight constructions.

Generally lighter materials usually use less energy overall compared to heavier ones throughout their lifespan. Examples of construction materials that fit this description include timber, lightweight steel framing, structural insulated panels, and other pre-made products.

In addition to using less energy, lighter building materials also need fewer resources compared to building with concrete, steel, or traditional brick by brick methods. When concrete is still needed for a foundation for example, the amount of concrete could be decreased due to a lighter construction on top. This approach can contribute to creating very energy-efficient buildings with a small carbon footprint throughout their lifespan.

Using lightweight construction can make indoor spaces healthier and more comfortable, which, in turn, can enhance well-being and productivity. Specifically designed materials help control energy and temperature effectively, responding quickly to sudden temperature changes and cooling down rapidly overnight in warmer areas.

New and advanced building materials, along with lightweight technologies, provide a fresh outlook and show us how to create strong and flexible structures with unique designs. For example larger spans of structures could be realized because of the lower loads of lightweight materials used for the structure.

Being able to easily get lightweight construction materials is a big advantage. It makes production simpler and helps with transporting solutions, especially to faraway places. Using a lot of prefabrication ensures things are consistently high quality, makes processes efficient, and saves time by cutting down on initial on-site work. Using lightweight construction can also make a positive economic impact. It saves money by speeding up processes, using more pre-made parts, being more efficient, and cutting down on labor and transport costs.

A challenge in which lightweight building is needed, is building on a bridge. A bridge is on first instance built to allow passage over an obstacle, typically something that would be challenging or impossible to cross otherwise (Wikipedia, 2022). A bridge could for example be used for traffic to cross a river. In this case the loads applied on the bridge are loads coming from traffic. This means that a building on such a bridge should not exceed the loads of the traffic. Therefore lightweight building is needed.

Furthermore, constructing on a bridge poses challenges. Accessing the bridge with cranes and other vehicles is difficult. To address this issue, building with prefabricated elements in a factory is more efficient, ensuring a quicker construction process.

In this report a building will be designed which is placed on top of a bridge over the Dommel which is located at the TU/e campus. By the use of lightweight building and prefabrication of elements a design will be made. This design will be worked out in terms of structural design. Design decisions will be made during the process and explained to make the design more clear and understandable.

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1. Plot and environment analysis

To get a closer view on the environment in which the bridge is placed, a proper analysis of the plot is mandatory. In this chapter, the environment will be analysed with respect to different factors, such as view, space, roads, water, structure of the surface, greenery, etc.

In Figure 1 below, a wide range view of the TU/e campus and some surroundings. The bridge is circled in red. To create and impression of the surroundings, the environment could be divided into three parts; the north side, east side and the south/west side.

The north side of the bridge is a nature reserve with a river called the ‘Dommel’ going through it. It separates the TU/e campus from the busy road called the John F Kennedylaan. The Dommel is a shallow long river which partly flows through Eindhoven.

The east side of the bridge is the TU/e campus itself. It’s a campus with large buildings meant for education. There are mostly cyclist and pedestrians and almost no cars. Sometimes there is an event or construction at the campus, but overall there’s a calm atmosphere.

The south/west side of the bridge is the centre of the city of Eindhoven, with the train station most close to the bridge. The busy road together with the train station gives a noisy atmosphere.

The overall surface of the plot is flat which means the view is not disturbed by hills. However, a lot of high buildings are to be seen when standing on the bridge. The direct environments around the bridge is a flat surface with walking and cycling paths and some greenery.

In terms of transport, the John F Kennedylaan is the closest possible road for large vehicles to access the plot. This street is marked in blue in Figure 1 The bridge could also be entered by coming from the TU/e campus. However, there are height limitations for transport when driving over the campus. The maximum clearance height is 4.20 meter. Next to that driving over the campus with large vehicles and high masses is not efficient.

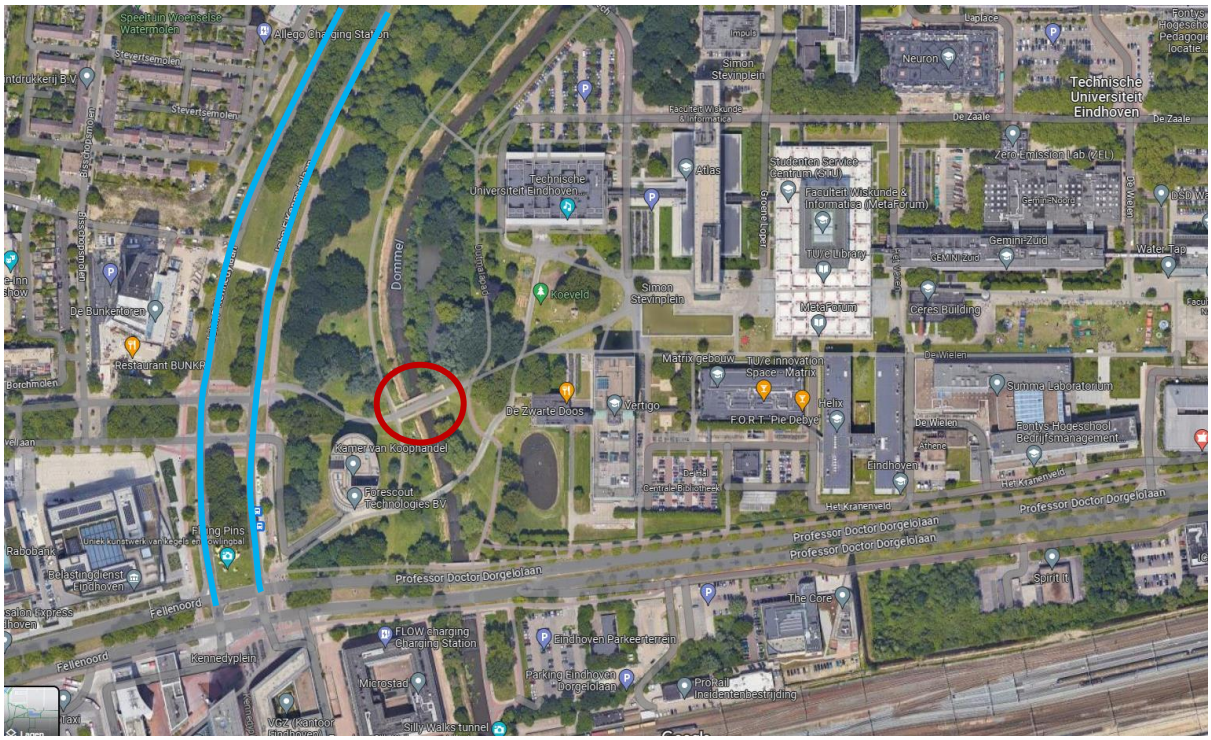


Figure 1: The environment around the bridge (Google Earth, 2023)

2. Analysis of the current bridge

To design a structure/building on the bridge, the current bridge over the Dommel should be analysed. Because of the lack of information on this bridge, a lot of assumptions should be made. In this chapter, assumptions will be made according to an excursion to the site.

The current bridge over the Dommel is shown in Figure 2. The bridge exists of a concrete slab placed on a few beams. These beams are placed on a foundation. On top of the concrete slab, two railings are placed which provide safety. Figure 3 shows what the bridge looked like in October 2010. The difference between the bridge in 2010 and the bridge now is the amount of grass on the bridge. A new path is made in the middle of the bridge used by cyclists. The sides of the bridge are used as walking paths.

The measurements are made according to Google Earth. In Figure 5 the measurements are estimated and shown in a floorplan. Figure 5 also gives a three dimensional view of the bridge.

One of the assumptions made is that the bridge is used to drive trucks over. The width of the path is way larger than the width that cyclists need to pass the bridge. Assumed is that the bridge was needed for trucks to access the TU/e Campus. This assumption could be used to determine the maximum weight on the bridge. A calculation is made for when one loaded truck is standing still on the bridge.

The width of the bridge is 14 meter. 5.5 meter is used for traffic. There are four models for the loads on the bridge;

1. concentrated and evenly distributed loads, covering the majority of the effects of freight and passenger car traffic.
2. a single axle load with specific contact surfaces of the wheels
3. special vehicles
4. crowds of people

The first model is separated into two subsystems;

1. concentrated double axel loads, where the load per axle is given by the formula;

$$a_Q \cdot Q_k$$

2. Evenly distributed loads, with the following loads per square meter of theoretical lane;

$$a_q \cdot q_k$$

The first subsystem gives a value of 264 kN. This would give a maximum point load of 250 kN for the bridge.

According to the second subsystem 7.92 kN/m^2 is the maximum load applicable on the bridge. Therefore 8.0 kN/m^2 is chosen as the maximum load on the bridge.

The other models are assumed to have a higher maximum load which means that those are not relevant to the maximum load used for designing a building on top of the bridge.

When building right above the foundation, the loads could be higher. There is no information about the strength and depth of the foundation.







Figure 2: Different views of the bridge over the Dommel



Figure 3: Top view of the bridge over the Dommel (Google Earth, 2023)



Figure 4: Perspective view of the bridge over the Dommel (Google Earth, 2010)

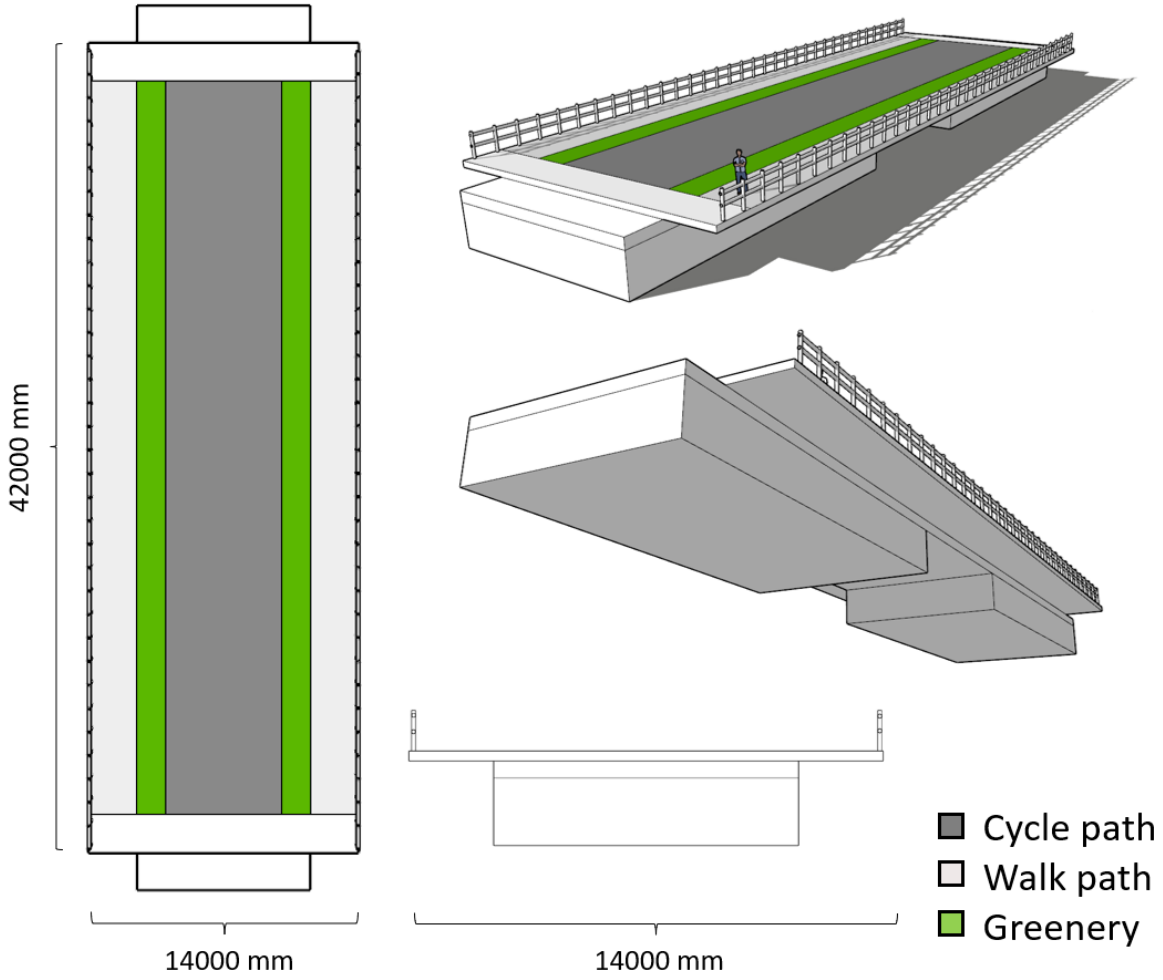


Figure 5: Top view and perspective views of the bridge with dimensions

3. Requirements

The goal of this project is to build a structure on top of the bridge over the Dommel. A certain function should be added to the building on top of the bridge. There are set requirements which must be met according to the assignment and requirements which will be added for the project. The following requirements must be met:

- Only use the existing bridge as support
- As a basis for a building that creates a link between the city of Eindhoven and TU/e
- This can be: a link with black box, a straight link with Atlas (research) or a link with nature reserve
- Maintain cycle route
- Building contains an interesting (visible) construction
- All modifications to the bridge are permitted. However, no additional foundation.

Next to these requirements, additional requirements came up after analyzing the environment and bridge itself.

If a certain function will be applied to the building, people will probably be there for a longer period of time. Therefore a toilet should be included, which means electricity and water should be accessible in the building. Staying longer in the building also means a parking spot is desired. While the bridge is meant for cyclists and pedestrians a bicycle storage would be enough.

The main function of the bridge is still a function of transport. This means that traveling over the bridge by pedestrians and cyclists is still prioritized. To not change the transformation too much the cycling path should be wide enough and should not have a slope to make passing the bridge not more difficult then it is.

The maximum load applied on the bridge is calculated in the analysis of the bridge. This turned out to be 8 kN/m^2.

While entering the building it is preferable to have a roof in order to not stand in the rain for example. Therefore the entrance should be roofed. The room inside will probably be quite large which means a small space at the entrance would prevent the air inside to change drastically. A revolving door would take the least space and is therefore an efficient option.

The additional requirements are divided into hard- and soft requirements. The hard requirements are the requirements that are expressed in present or absent. The soft requirements are stated as principles for the building. Below the additional hard- and soft requirements are stated in short;

Hard requirements:

- Toilet should be included (water and electricity)
- Bicycle storage
- Indoor space
- No slope for cyclists
- Maximum load applied on the bridge = 8 kN/m^2
- Roofed cycle/walking path
- Function: Exhibition/take away
- Multifunctional space

Soft requirements:

- Lightweight structure and materials
- Transparent building
- Building with length of bridge
- Ability to look at exhibition from outside
- Fast building method
- Interesting building structure

4. Function: Exhibition/café

To find an appropriate function for a building on top of the bridge, the environment could be a determining factor. As the requirements states that the building should create a link between the city of Eindhoven and the TU/e. ‘’This can be: a link with black box, a straight link with Atlas (research) or a link with nature reserve’’. The chosen function is an exhibition/café.

The bridge now only functions as a way for people to visit the TU/e campus. These people are mostly students or teachers. Adding a building on top of the bridge gives the opportunity to expand this and try to get more people coming over the bridge to enjoy the TU/e campus and its interesting functionality.

As a link between the TU/e and the city of Eindhoven, an exhibition space could show models and projects made at the TU/e. This way people who are normally not visiting the campus could see where the TU/e is meant for. The bridge connects the people from Eindhoven to the TU/e.

A transparent building would help with attracting people. When people are seeing the exhibition partly from outside, they could get interested and visit the building. There they could witness the exhibition together with coffee and a snack. Therefore a small café can be included. As used for the toilet, the electricity and water should then also be used for a bar with some electronic devices and water for cleaning etc. The café also helps with attracting people and help to make them stay longer. People could sit down and relax as well as admiring the inventions and projects that are exhibited.

The façade of the building could work as an art showcase. A visual of this idea is shown in Figure 6 However, it should be kept in mind that cyclists and pedestrians should be able to get over the bridge safe and easily. Especially at the peak hours it should be possible for the students and teachers to not have delays due to the building which takes over the bridge. Therefore wide paths are preferred



Figure 6: Art showcase (almere Buiten kunstetalage Graag Regenboog de Tjaske, 2016)

5. Functional design

Integrating the requirements into a functional design is possible in multiple ways. In this chapter three different design concepts will be discussed, after focusing on some references. One of the variants will be chosen and worked out furthermore in the report. This chosen variant is changing during the process. The design choices made in the progress will be explained as well.

5.1 References

Inspiration for the variants will come by the use of references. Interesting building structures are explored to be translated into ideas for the building on the bridge, with the exhibition/café function.

The first reference is the bus station in Woensel (Figure 7). This bus station exists of three large portals with relatively long cantilevers. Because of the large spans between the elements, lots of space is created under the structure. This is meant for people to not stand in the rain while waiting for the bus. However, due to these large spans, also large profiles are needed for the beams and columns, which results in high material use.

Large open spaces are useful to make a building open and transparent. Non loadbearing windows could be placed at the sides of the structure to create an inside structure with a lot of light and transparency. The roof in this case does not have to carry a lot of load, because it is not meant for people to get on top. The important thing for this structure is to make sure that rainwater could be transferred to the ground without having the problem of people walking through a waterfall before getting under the structure to stay dry.



Figure 7: Bus station in Woensel (BUSSTATION WOENSEL, Rindert van den Toren)

The second reference is the KeenSystems headquarters in Gemert (Figure 8). This building has a huge overhang together with one small beam supported by two columns at the end to compensate this overhang. To make this overhang possible, a steel floor-high truss is used. On the left top of Figure 8 this principle is shown. Around this truss aluminum window frames are placed in which the windows are placed.

The structure gives the impression of a floating box. The view of this building from the inside shows a large green area as seen in the bottom right of Figure 8 a calm atmosphere is created in the building even though it is surrounded by busy roads. Also the truss itself is not blocking the windows that much as seen in the right top and left bottom of Figure 8. It does take some of the view, but it still gives a friendly feeling and is not disturbing.

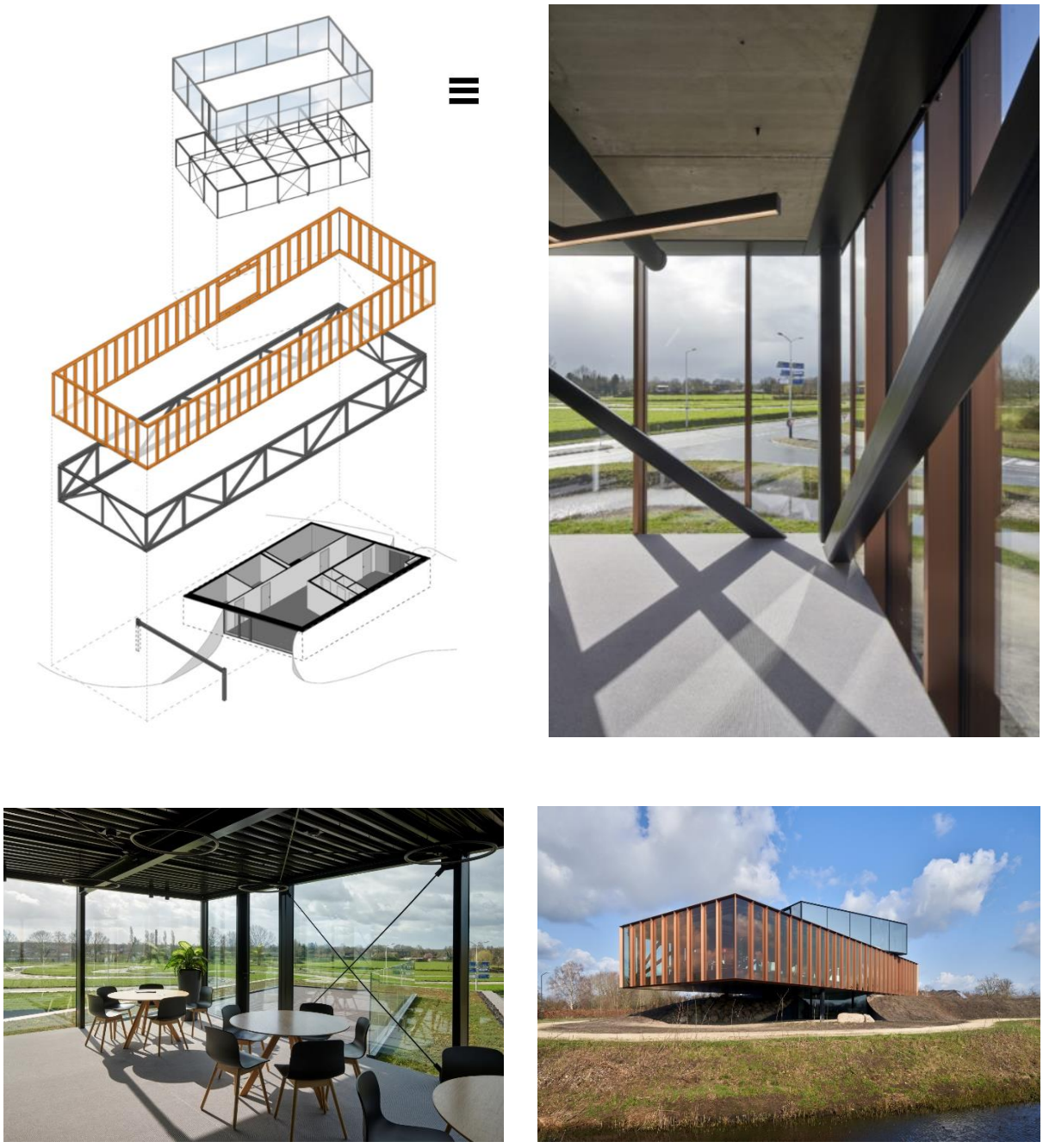


Figure 8: KeenSystems headquarters in Gemert (KeenSystems, 2023)

The last reference is the former office of Philips in Eindhoven (Figure 9). This building is made out of concrete cores with a large structure hanging at the sides. The cores have large V-shaped elements at the top on which steel tension columns/cables are hanging. These columns/cables are supporting the floor partly. The building has more space at the ground floor due to this hanging structure and the raised floors.

The cores create large point loads on the ground for which a strong foundation is needed with large piles. For the bridge this is quite challenging. The amount of steel and concrete has a lot of weight as well which is not ideal for the bridge.

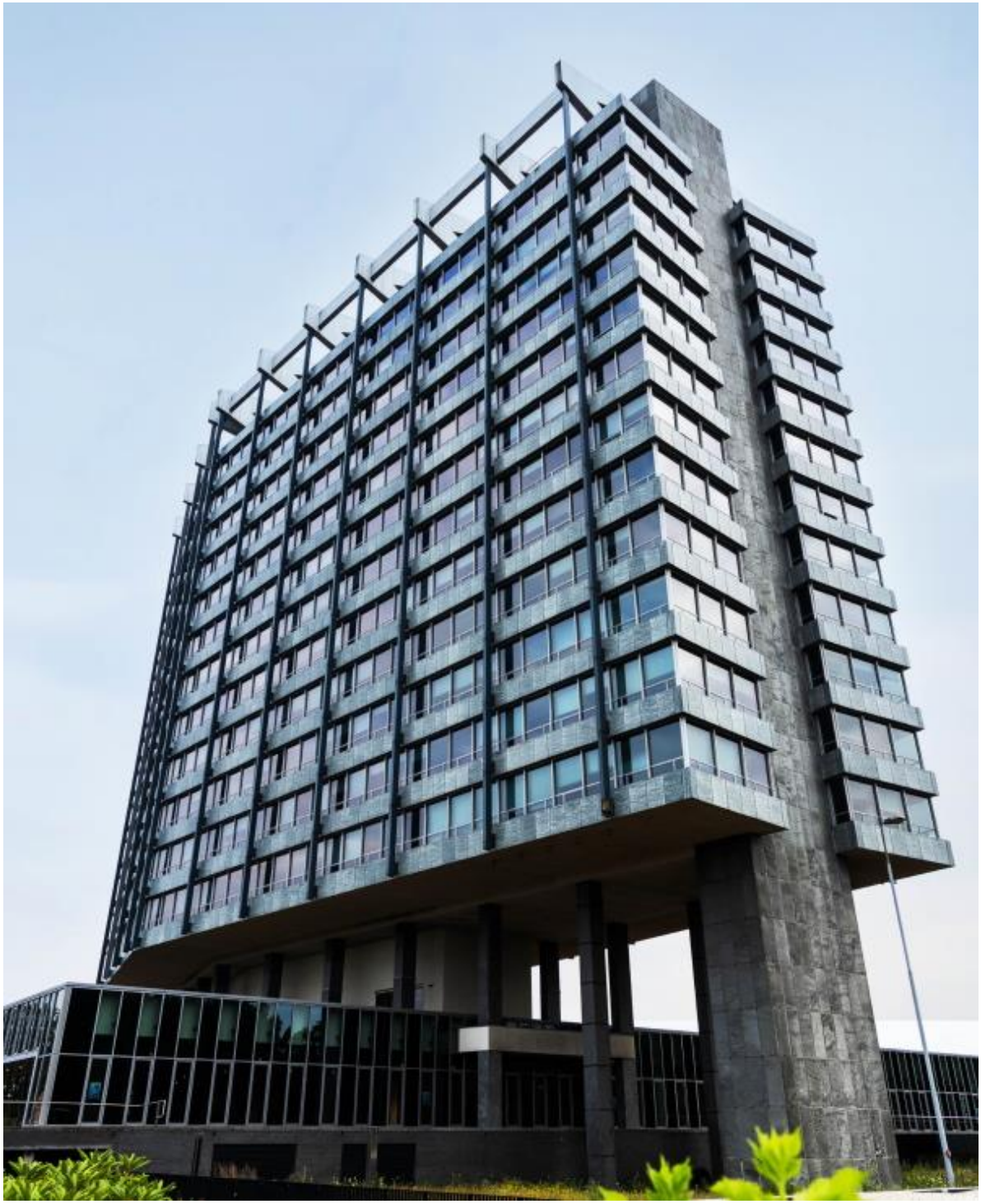


Figure 9: Philips office (Ten Brinke, 2023)

5.2 Three design concepts

With the help of the references, some design variants could be made. All variants will be considered with their advantages and disadvantages. Out of these three variants one or a combination of more will be chosen.

The first variant is shown in Figure 10. This variant is based on the bus station in Woensel, with the use of large timber portals. The building is made over the length of the bridge. This way a lot of space is used efficiently. The cantilevers on both sides are chosen to give the people walking and biking over the bridge a cover for the rain. The entrance will be in the middle of the building as seen in Figure 11. However, due to these overhangs, daylight is blocked. Therefore, windows are made on top of the building. This way daylight could still enter the building as much as possible.

The idea of the exhibition space is shown in Figure 10 The exhibition space is right next to the windows on both sides of the building. This way pedestrians and cyclists could see the models and projects from outside while looking through the window. This may take interest of the people to get inside and watch the exhibition from the inside with a more calm atmosphere. In the middle there is a bar with a small take-away to get something to eat or drink. The height of the building is meant for large objects to be exhibited. At the 'head' of the building the entrance is made for storage and to put the exhibition materials in.

The routing is to be seen in Figure 13. The main reason for the routes being at the sides of the bridge and not in the middle is because of the views at the sides. When the routing is through the middle of the building, it could feel enclosed by the building around it and the views (especially the one to the nature reserve) is blocked. The bikers have to cycle at the outer side of the bridge to keep the transport fluent. When they want to visit the building they could step of their bike and walk over the footpath and enter the building in the middle. For pedestrians the path is close to the building, which makes it easier to look through the glass and see the exhibition.

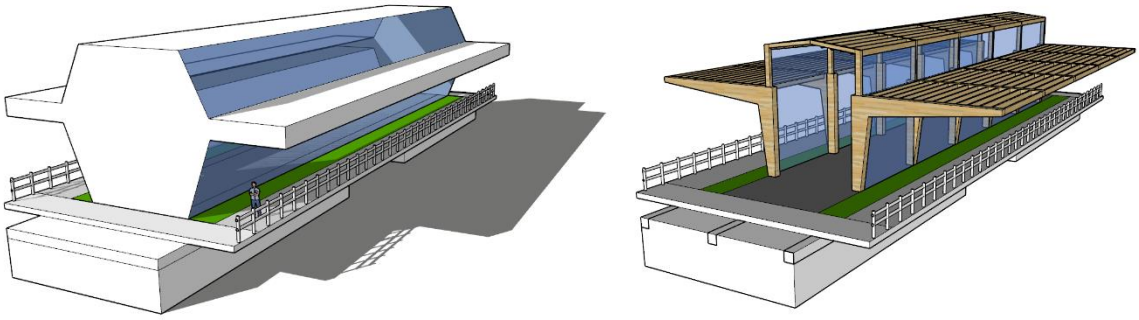


Figure 11: Variant 1 perspective view of the building and the building structure

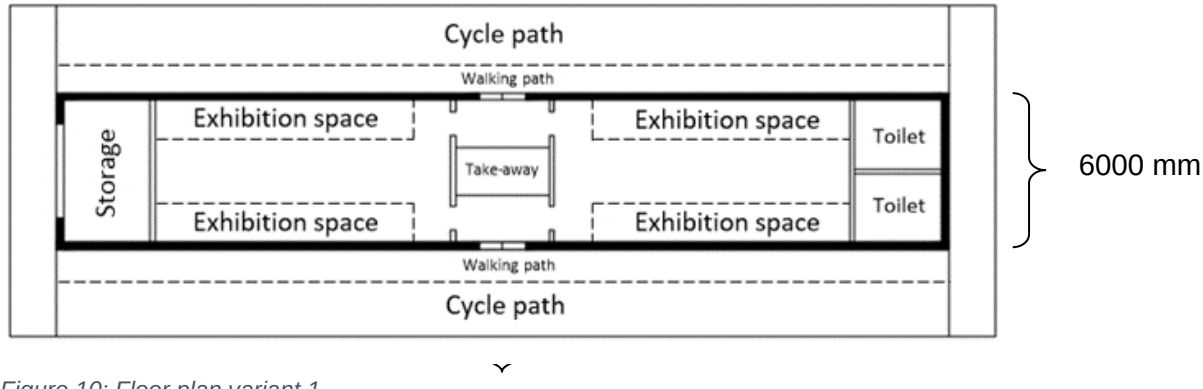


Figure 10: Floor plan variant 1

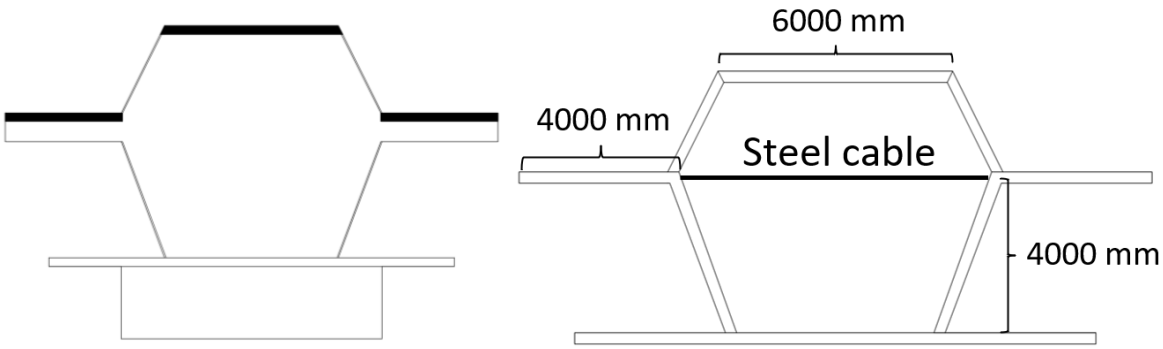


Figure 12: Sections of variant 1

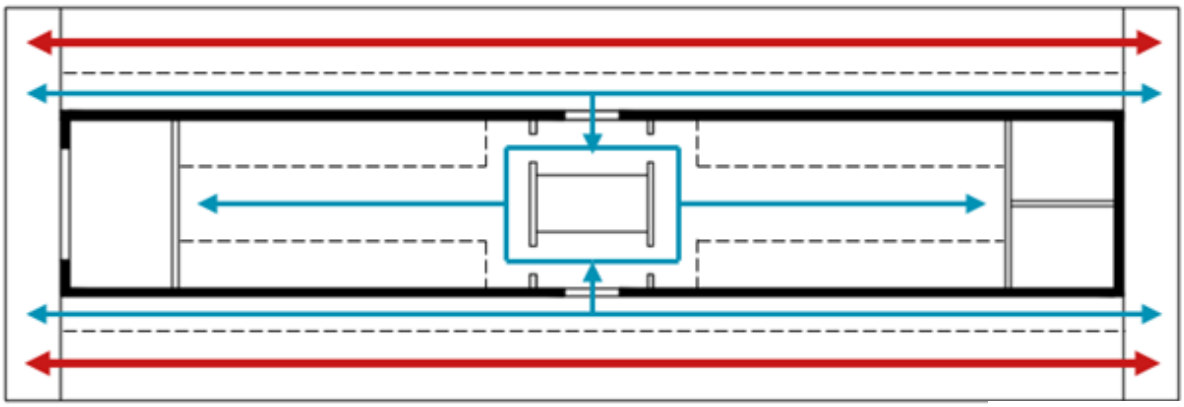


Figure 13: Routing of variant 1

The second variant is based on the KeenSystems headquarters. This variant is shown in Figure 14. The design is made out of five boxes stacked upon each other. Two boxes at the outer sides with both one overhanging box on top, and one ‘core’ box in the middle. The sides of the bridge are still functioning as the paths for the pedestrians and the cyclists as seen in Figure 15. The overhanging boxes are partially faced to the nature reserve. The two boxes on top all have glass facades to get a wide view. The middle box function as a core and has the elevators and stairs inside.

The ground floors next to the core are meant for the exhibition where again glass facades are used to make the exhibition visible from the outside. On the first floor tables are placed for people to relax and eat and drink something, while enjoying the view.

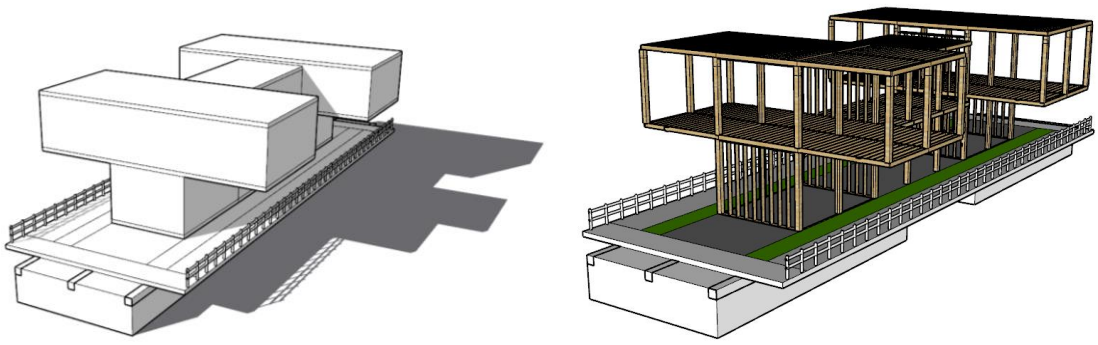


Figure 14: Perspective view of the building and the building structure of variant 2

The structure exists of timber frame construction. Timber frame walls with timber floor beams, are used as the load bearing structure. For the top boxes beams and columns are used to make the room as transparent as possible by using glass for the facades.

The first floor in the core of the building is lower than the first floor with the glass facades and the table as to be seen in the sections in Figure 16. In the floorplan the ramps and stairs to come from the core to the side boxes is shown as well. The reason for the height difference between the core and the side boxes is made to make these boxes stand out more and give more depth into the building.

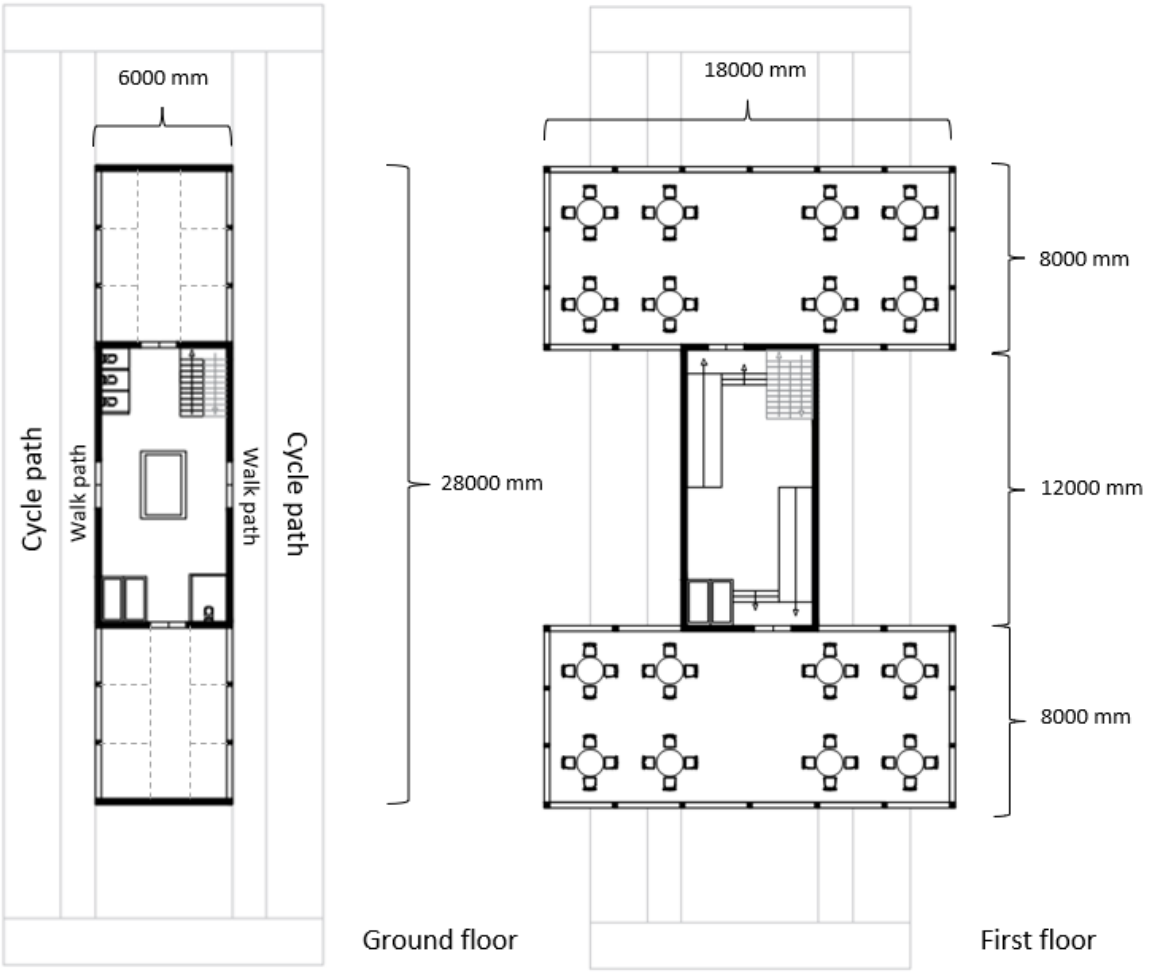


Figure 15: Floorplans variant 2

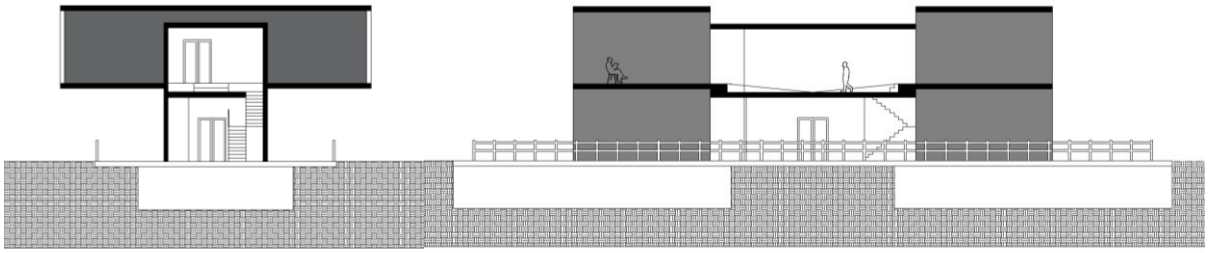


Figure 16: Sections of variant 2

The last concept is based on a raised building standing on a certain amount of columns. A lot of space is open below the building which makes it easy for cyclists and pedestrians to cross the bridge. All façades are transparent, which makes it possible to look at the environment at all sides. Also an outdoor space is placed on both sides facing to the Dommel.

The structure is a column and beam structure placed on a table structure. The table structure exists of large columns with a platform on top. In Figure 17 the structure is to be seen.

In the floorplan in Figure 18 it is visible that the exhibition is all placed in the middle of the room. The reason for this is that the exhibition now is not visible from outside below the building. Next to that the view would be blocked by the exhibition when standing inside. Therefore tables are placed to sit and relax.

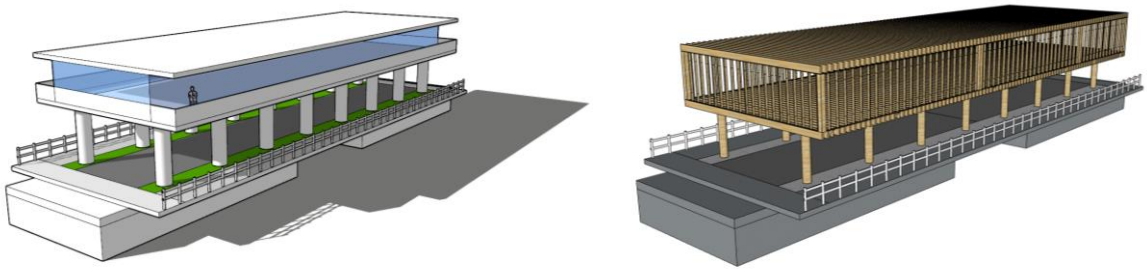


Figure 17: Perspective view of the building and the building structure of variant 3

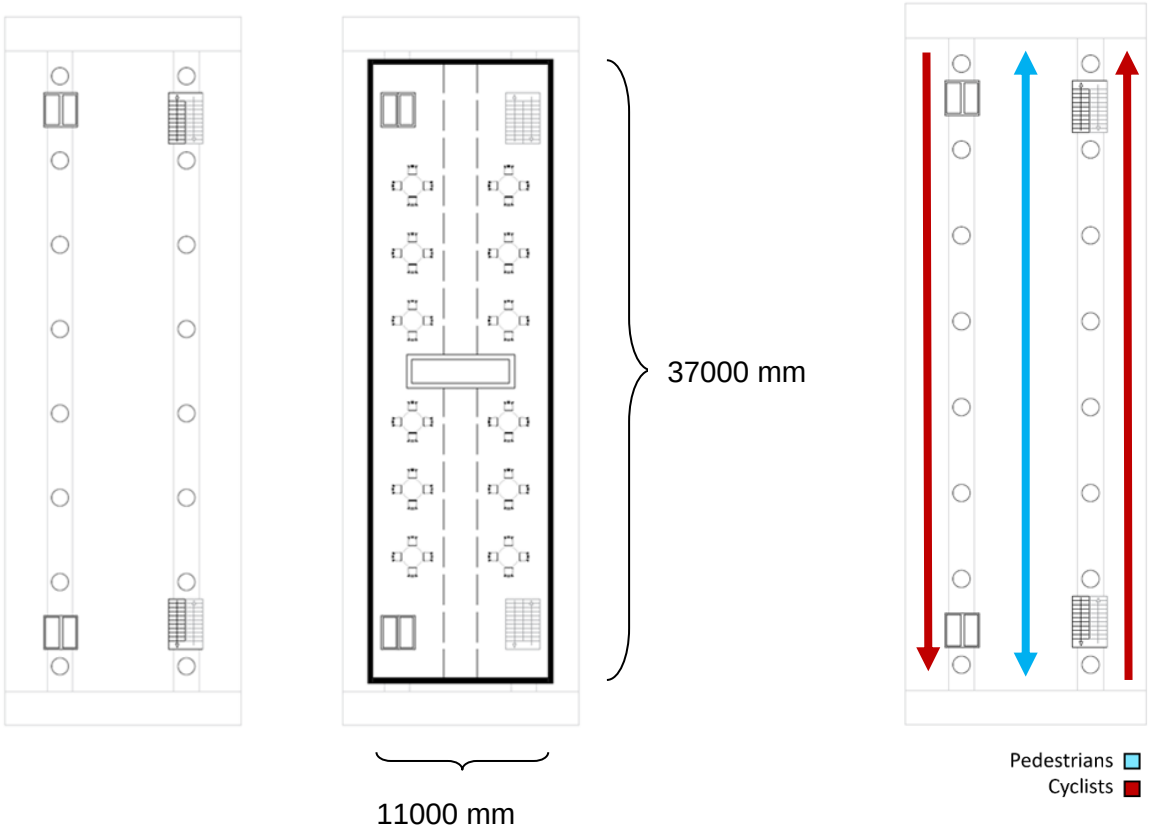


Figure 18: Floorplan and routing of variant

5.3 Chosen concept

All concepts have their own advantages and disadvantages. With the use of the requirements the variants could be assessed. The cycle and pedestrian paths at the sides are best to keep the view for the people crossing the bridge. This concept is to be seen in both the first and second variant. Furthermore the transparent facades are beneficial for the views, exhibition and daylight. All variants have a lot of windows which make the variants transparent. However, for the third variant, it's very hard to look to the exhibition inside, because of the raised floor.

One of the disadvantages of the first and third variant are the high point loads which are placed on the spanning part of the bridge and therefore not directly placed on top of the foundation.

An important part of the design is the challenging structure. When looking at the variants, the second variant would have the most interesting structure because of the cantilevers and different boxes in the building. The first variant has these overhangs as well, but the structure is repetitive and therefore not challenging enough. The last variant has the least interesting structure.

The challenging structure is decisive for the chosen variant. Therefore the second design is chosen to continue with. However, the idea of a hanging building as seen in the Philips building is something that could make the building even more interesting. If the boxes at the ground floor which are placed next to the core are removed. The box on top of that has an even more outstanding effect. To make this possible, the roof could have a strong cantilevering structure on which the glass boxes could hang. In Figure 19 this is visualized.

By the use of large timber trusses the floor could hang and have a thin floor where the people could stand on. Because of the steel cables hanging under the trusses, the façade could be made out of non-loadbearing glass which makes the box transparent.

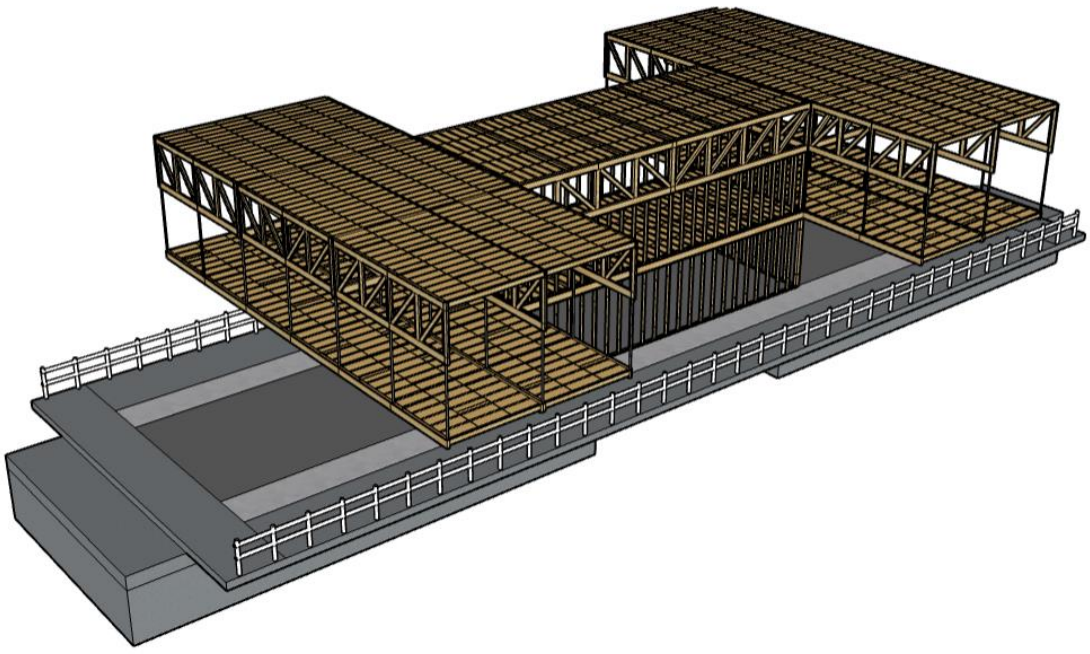


Figure 19: Structure of the chosen variant

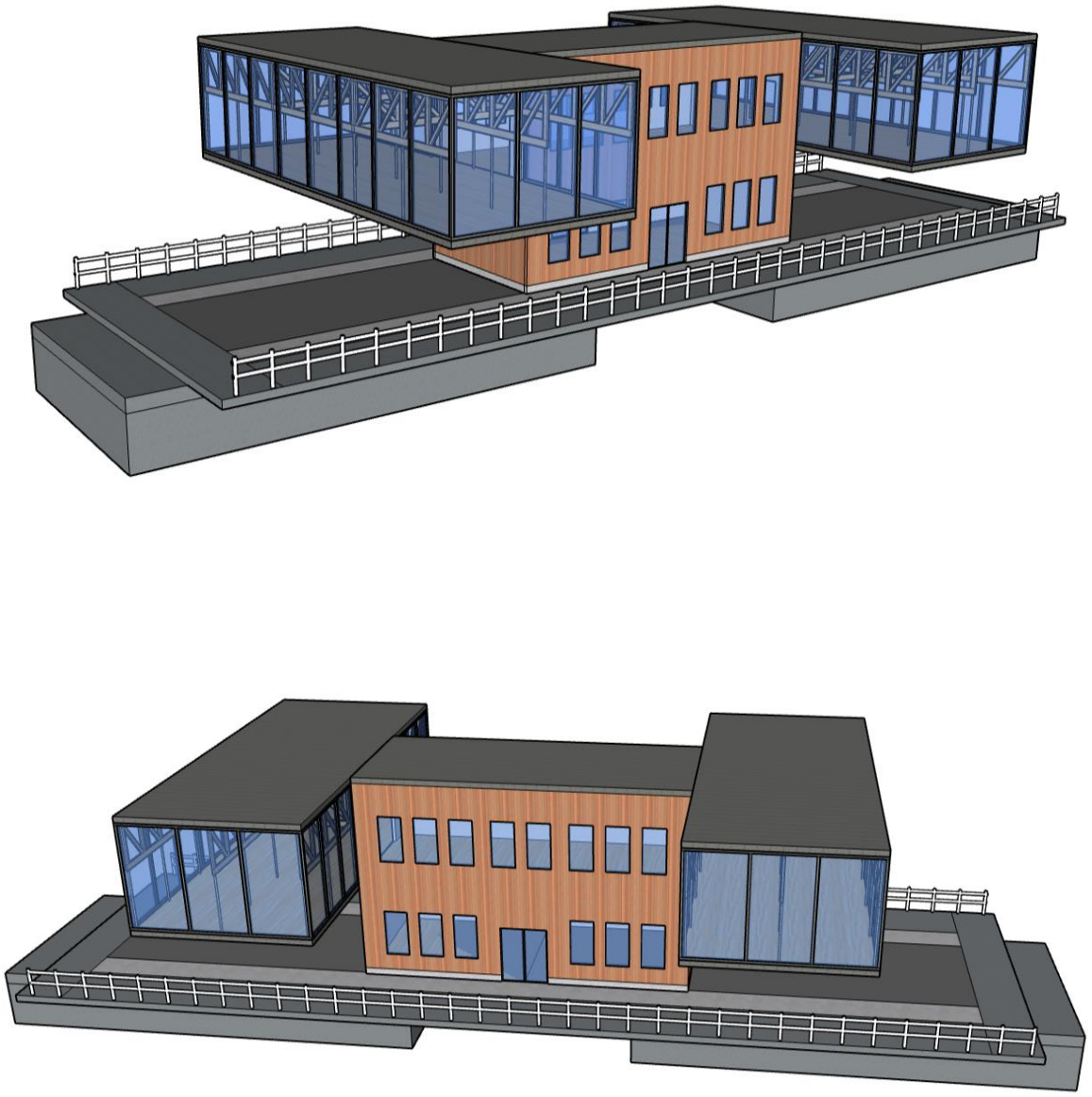


Figure 20: Perspective views of the chosen variant

The building needs infrastructure as mentioned before. Water and electricity should be able to reach the building. However, the problem is that pipes cannot come over the bridge. The reason for this is that these pipes will be an obstacle for the cyclists and pedestrians. The solution for this problem is to let the pipes and electricity cables go underneath the bridge. To make this work, holes should be made in the concrete bridge underneath the building. Under the building a crawling space is located where the pipes and electricity cables could come in. In Figure … below the floor plans, sections and elevations are shown. In the Appendix some renders are given for the building.

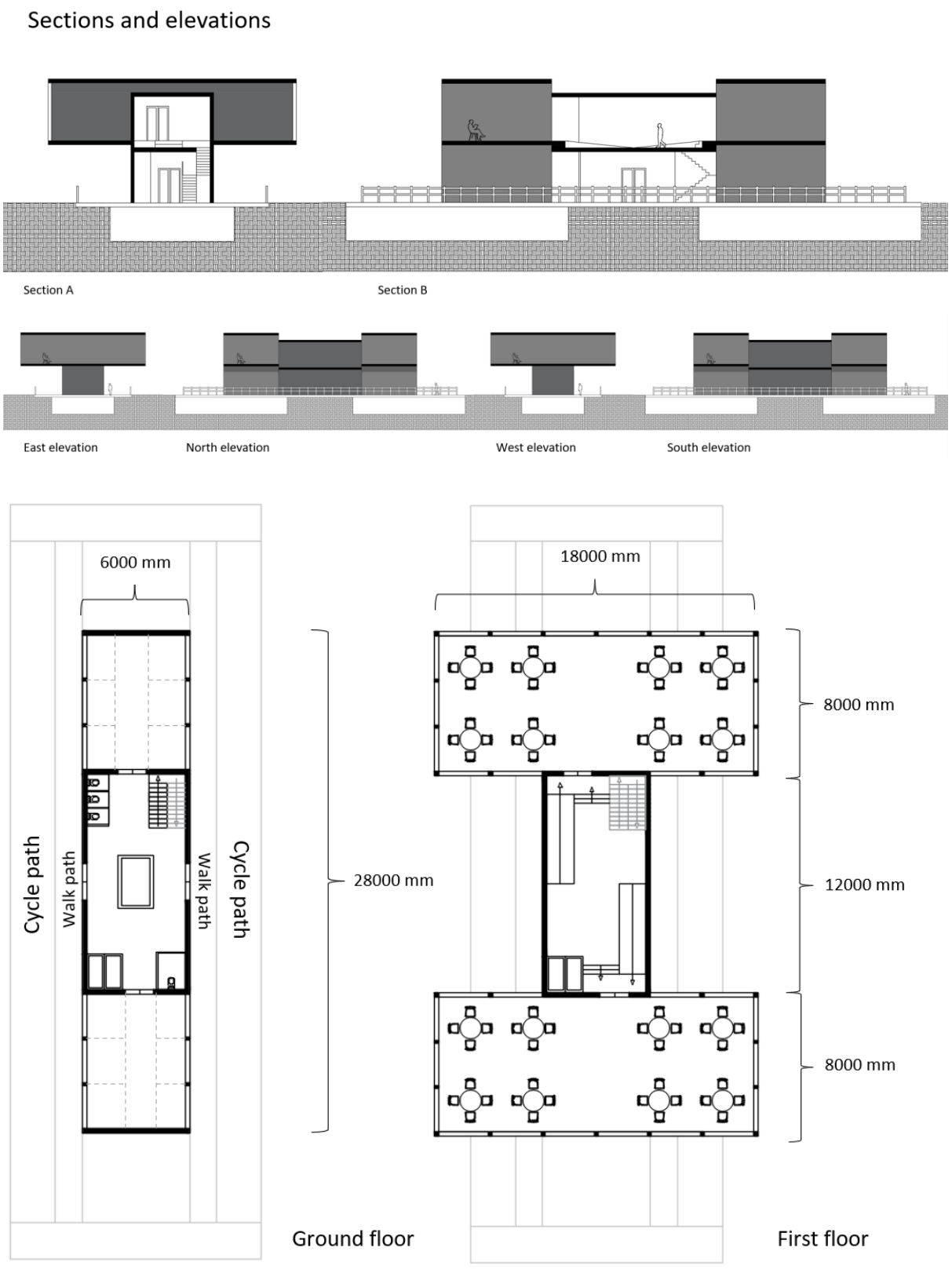


Figure 21: Floor plans, sections and elevations of the chosen variant

6. Structural design

For the chosen design, the structural design should be elaborated. In this chapter the structure will be explained more specifically. Therefore load transfer, materialization, dimensioning and stability are relevant. Also the weight of the building will be calculated to see if the foundation is capable of resisting the forces.

6.1 Materialization

The material choice for the design is based on the bridge. The bridge has a maximum load capacity of 8 kN/m^2. This means that the building should be lightweight. The design of the building has two floors which means lightweight materials and building methods should be used to make sure the bridge does not collapse.

For the structure of the chosen design a few materials are most common to use. These materials are, among other materials; steel, concrete, and timber. These materials will be investigated shortly to see which material is most efficient for the building.

Steel has several positive attributes that makes it a widely used material in construction. Steel is recognized for its exceptional strength, with in particular its high tensile strength. Next to that, steel structures has a prolonged lifespan and resilience against environmental conditions, thereby reducing maintenance costs over time. Also The steel can be shaped and fabricated into various forms, providing architects and engineers with greater flexibility in design. Recyclability is another positive point of steel. Steel contributes to sustainable construction practices and reduces the environmental impact of the industry. Lastly the durability and low maintenance requirements of steel often lead to cost savings in the long run.

However, there are certain drawbacks associated with the use of steel. Without proper protection, steel is vulnerable to corrosion. The production of steel involves significant energy consumption and carbon emissions, contributing to environmental concerns. Furthermore, steel is denser than other construction materials and steel has a high thermal conductivity, which is associated with additional insulation for energy efficiency.

Concrete has some positive points as well; it is mainly known for its high compressive strength, concrete provides robust support for structures, ensuring long-term durability. Also concrete is fire-resistant, making it a preferred material for structures where fire safety is crucial. As well as for steel architects and builders benefit from concrete's ability to be molded into various shapes and sizes, offering flexibility in design. Concrete's thermal mass helps regulate indoor temperatures, contributing to energy efficiency in buildings. Lastly concrete structures generally require minimal maintenance, reducing long-term maintenance costs.

There are some drawbacks for using concrete as well; while strong in compression, concrete can be brittle and may not perform as well under tension, leading to potential cracking. The production of cement, a key component of concrete, involves high energy consumption and significant carbon emissions, contributing to environmental concerns. Also concrete is heavy, which can pose challenges in projects with strict weight limitations. Furthermore Concrete requires time to cure and gets its full strength over some time. At last the aesthetic options with concrete may be perceived as limited compared to some alternative materials.

The material timber comes with various advantages. First of all, timber is a renewable resource. Growing trees absorb carbon dioxide, and timber continues to store carbon throughout its life, making it a climate-friendly material. Also timber requires less energy to produce compared to materials like steel or concrete, resulting in lower carbon emissions

during manufacturing. As for both other materials, timber offers flexibility in design, allowing for creative architectural solutions. Different from the other two materials, timber is lightweight and can be easier to handle during construction, contributing to faster building processes. The last benefit of timber is the protection of itself in the event of fire in a natural way: the outer layer chars and thus protects the inside. Because wood releases energy slowly, it takes a long time to burn completely.

There are some drawbacks to using timber as well. Without proper treatment, timber is sensitive to pests and decay, impacting its longevity. Also timber has lower load-bearing capacity compared to steel or concrete, affecting its use in certain structural applications. Lastly, timber structures may require periodic maintenance, such as sealing or painting, to ensure durability.

After this short summary of the three building materials, one of them is best for this design. The best material for most of the structure is timber. Especially in accordance with the maximum weight on the bridge. Timber has the lowest density which is best for a structure on this bridge. This lightweight property is useful for the easier handling of the construction on the bridge. Prefabricating the elements would help even more. Also the renewability, low carbon emission, low embodied energy and fire safety are useful benefits for the design.

However, when using timber, the disadvantages should be taken into account as well. Maintenance should be taken into account for example. But also the structure should be made in accordance with the load bearing capacity of timber, which means larger dimensions should be used for the cross section, or lower spans should be applied in the building.

For the tension cables, steel will be used. Steel has a higher tensile force and is easier to connect to timber when used in tension. Due to this property, the steel cable could have small sectional areas which gives more space in the building. In terms of fire resistance, timber ‘protection’ could be placed around the steel cable. This would also be better for architectural reasons, while the timber has a more aesthetic view. For the connections between the timber, steel will be used as well. Also mainly for the tensile strength and the stronger connections.

For the windows, glass will be used which is used for most buildings. These glass windows will be placed in aluminum window frames, instead of timber, which would also be possible. Aluminum is highly durable and resistant to corrosion, rot, and pests, making it a long-lasting material. Timber, on the other hand, is as mentioned before sensitive to decay and insect damage over time. Also aluminum window frames require minimal maintenance. They do not need painting, sealing, or regular upkeep, unlike timber frames, which needs periodic painting or staining to maintain their appearance and structural function. Even though timber is a lightweight material, aluminum is even more lightweight, which gives less load and easier mountability. This at the expense of still having structural stability and strength. Window openings could be larger and profiles could be smaller compared to timber. At last, aluminum is resistant to weather elements, including rain and sunlight. It does not warp, crack, or swell, which can be concerns with timber frames under certain weather conditions.

A negative point of the aluminum window frames is especially the higher embodied energy and higher carbon emissions. However aluminum is highly recyclable, contributing to sustainable building practices. Recycled aluminum retains its properties, making it an environmentally friendly choice.

6.2 Load transfer

The vertical load transfer is important to see how the structure works. This will be used for the dimensioning of the elements in the structure. With the use of hand calculations and Technosoft calculations, all elements of the structure will be calculated. Therefore a particular order is used. In Figure 22 the vertical load transfer is shown schematically. The order in short exists of;

1. Floor beam 1
2. Floor beam 2
3. Steel cable
4. Roof beam
5. Truss 1
6. Truss 2
7. Column

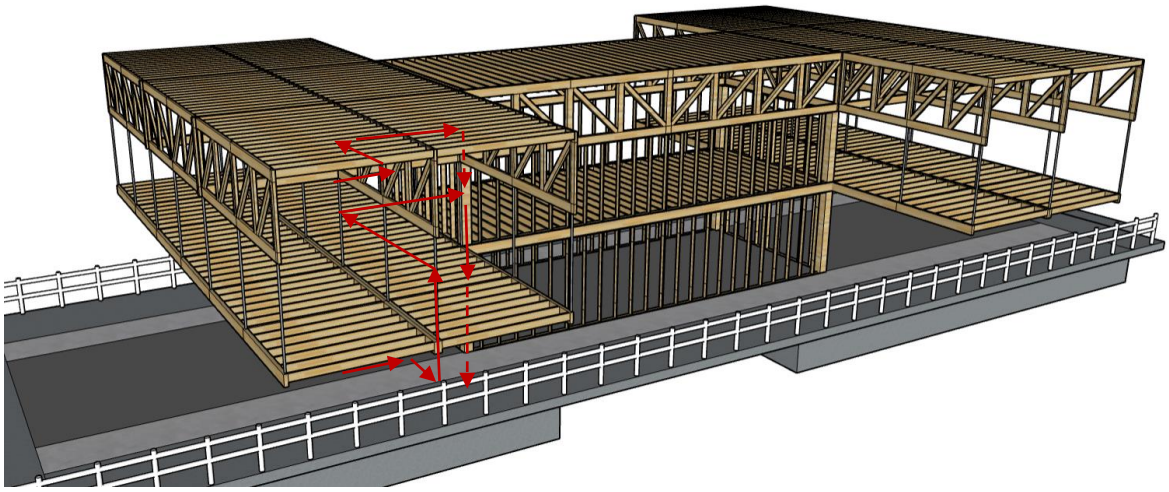


Figure 22: Vertical load distribution

Floor beam 1

The first floor beam is the one that is directly supporting the people and other loads on top. In Figure 23 a zoom in is given to show this beam with the force distribution included.

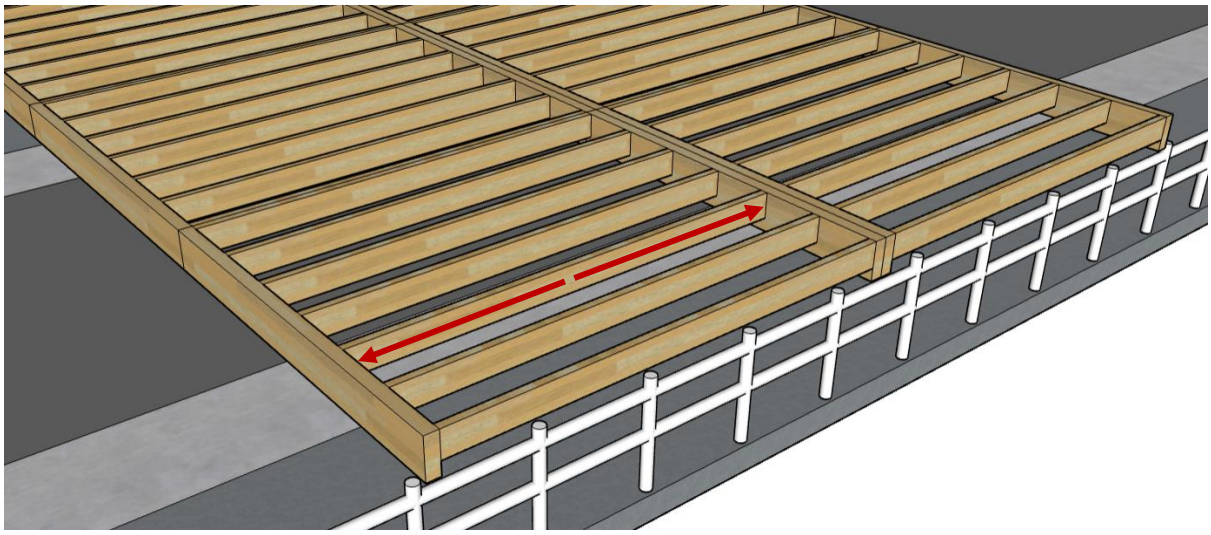


Figure 23: Floor beam 1

The beam is spanning between two other beams which are attached to the steel cable. The schematization of this beam is given in Figure 24. The beam is 3.6 meter long.

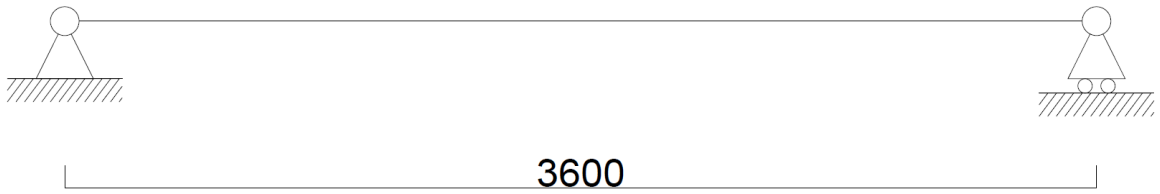


Figure 24: Mechanical scheme of floor beam 1

Floor beam 2

This floor beam is directly attached to the steel cables and supports seven times floor beam 1. In Figure 25 a zoom in is given again. In Figure 26 a schematization of floor beam 2 is given. This beam is 3 meter long.

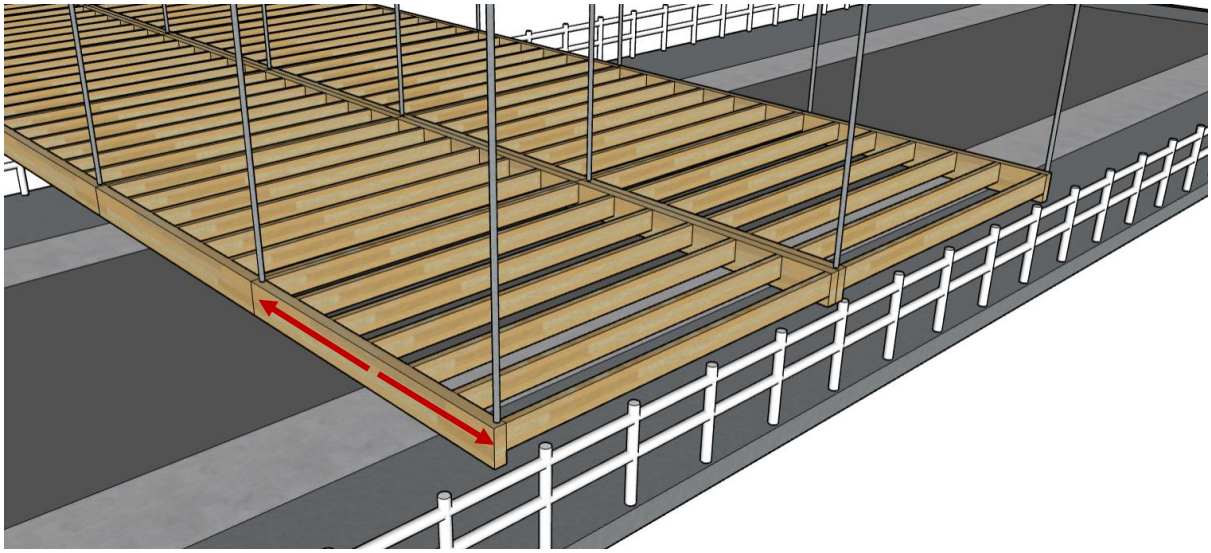


Figure 25: Floor beam 2

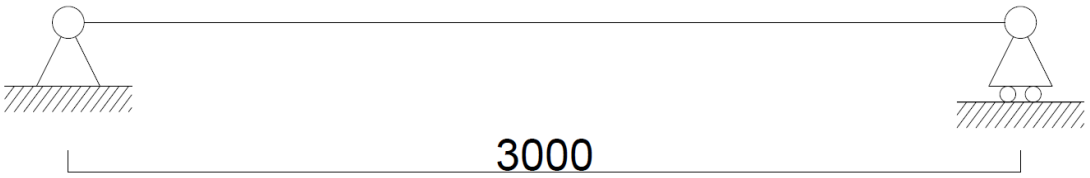


Figure 26: Mechanical scheme of floor beam 2

Steel cable

The steel cable is attached to floor beam 2 at the bottom and truss 1 at the top. The steel cable is under tension and supports the elements which exists of floor beam 1 and floor beam 2. In Figure 27 a zoom in of the steel cable is given.

In Figure 27 a schematization of the steel cable is given. The cable is 4 meter long and has a timber finishing (not shown in Figure 27). The force going through the cable is parallel to the cable itself.

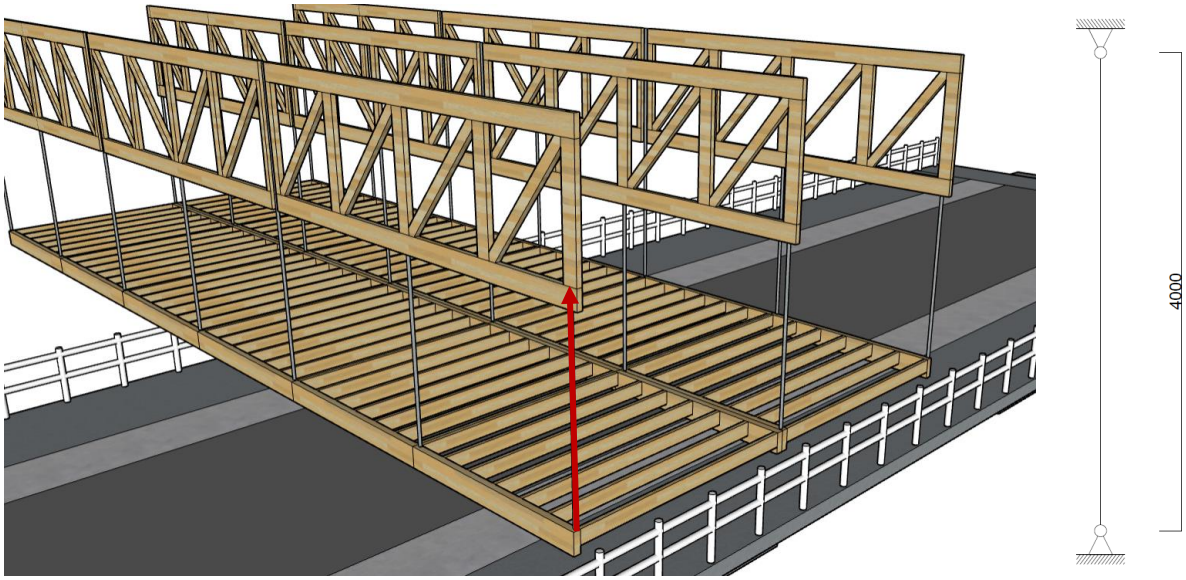


Figure 27: Steel cable and the mechanical scheme

Roof beam

In Figure 28 the roof beam is visualized. This beam is spanning between the trusses. These trusses are the ones that are called truss 1. The length of these beams is the same as for floor beam 1. Also the force distribution is the same as for floor beam 1. The only difference is the variable load, because of the roof function. In Figure 29 the schematization is given. This is the same as for floor beam 1.

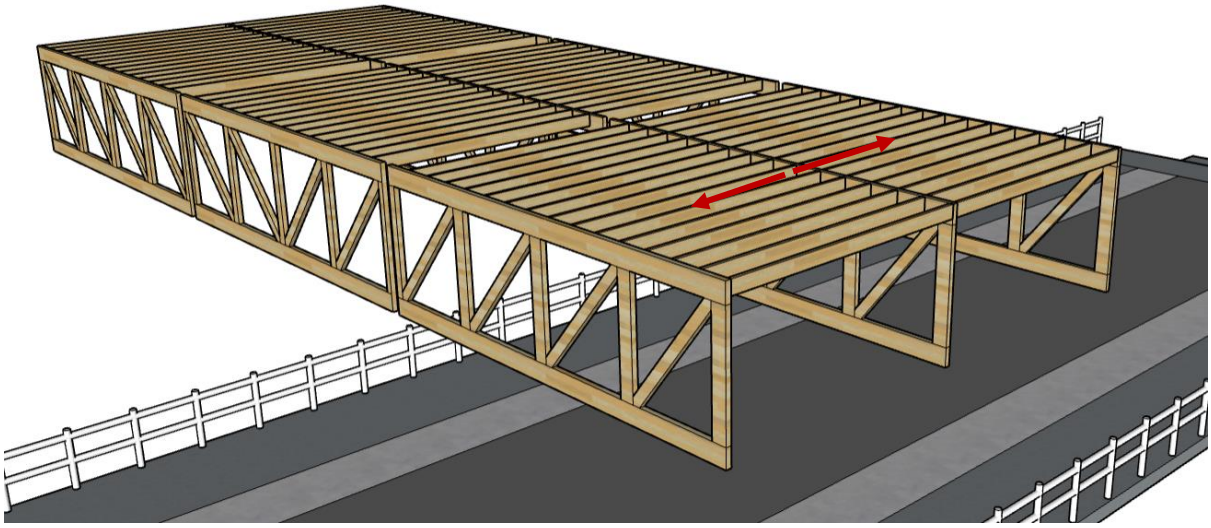


Figure 28: Roof beam

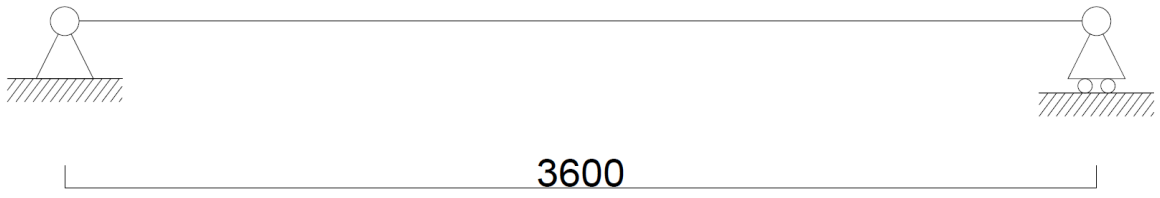


Figure 29: Mechanical scheme of the roof beam

Truss 1

Truss 1 is the truss that is supporting the steel cables and the roof beams. In Figure 30 a zoom in is given of the truss. This truss is supported by truss 2 which will be seen in the next step. The truss is 18 meter long in total. However, cuts are made which divides the truss into three parts of 6 meter. This is needed in terms of connection to truss 2. In Figure 31 the schematization of the truss is given. The truss is 1.5 meter high.

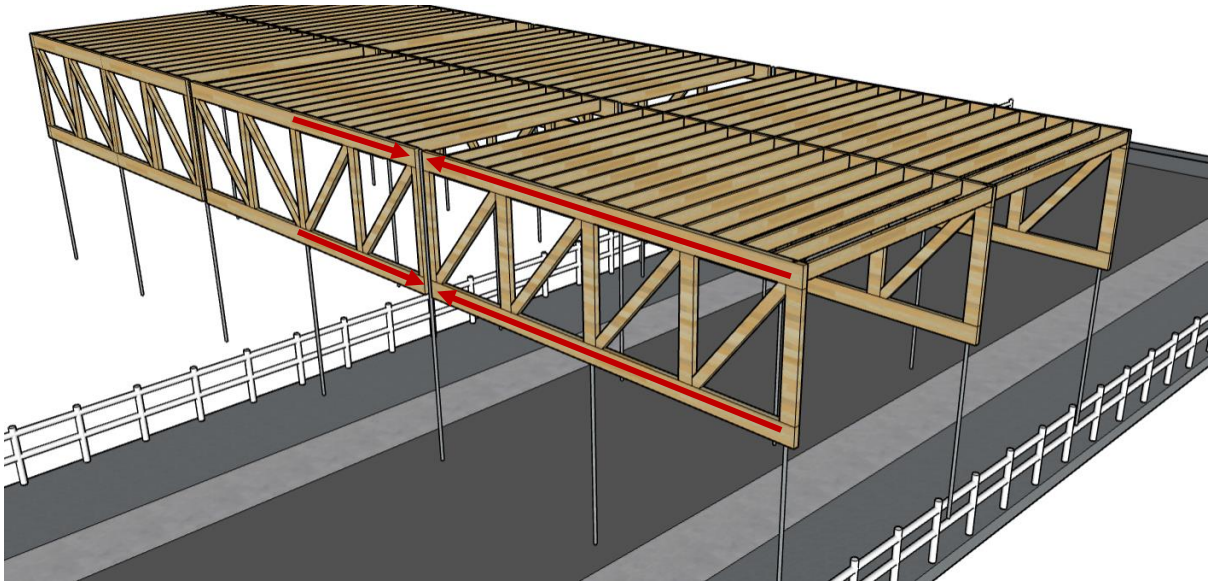


Figure 30: Truss 1

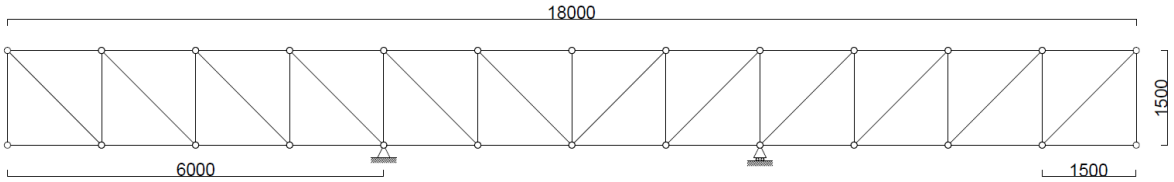


Figure 31: Mechanical scheme of truss 1

Truss 2

The second truss is placed perpendicular to truss 1. This truss is also supporting truss 1 at the ends. In Figure 32 a zoom in of truss 2 is shown.

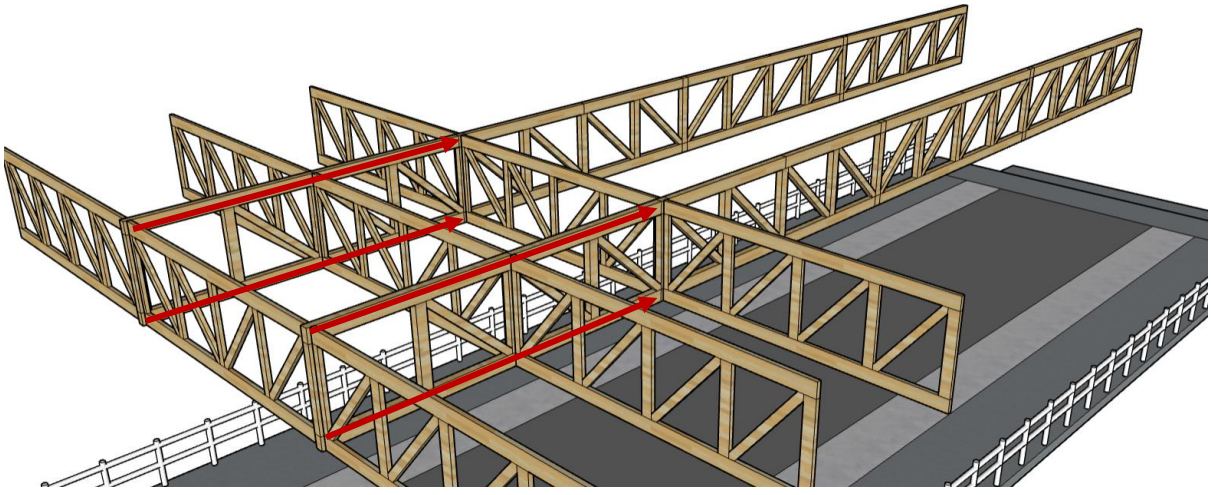


Figure 32: Truss 2

Truss 2 is supported by columns which will be explained in the next step. The truss is 28.8 meter long and 1.8 meter high. In Figure 33 the schematization of the truss is given. The truss is supported on two supports which are 14.4 meter. There is a cantilever on both sides of the truss, which is spanning 7.2 meter. This truss is not cut through like truss 1.

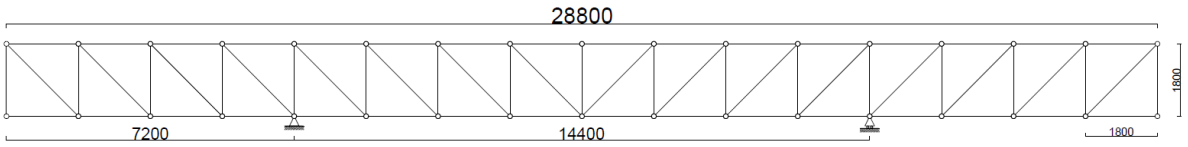


Figure 33: Mechanical scheme truss 1

Column

The last element is the column. The column is supporting truss 2. It is bringing the forces of the whole structure to the foundation. There are large forces placed on this column which is why it is placed directly on the foundation and not on the spanning part of the bridge. In Figure 34 the column is given schematically. The column is 5 meter and is loaded in compression.

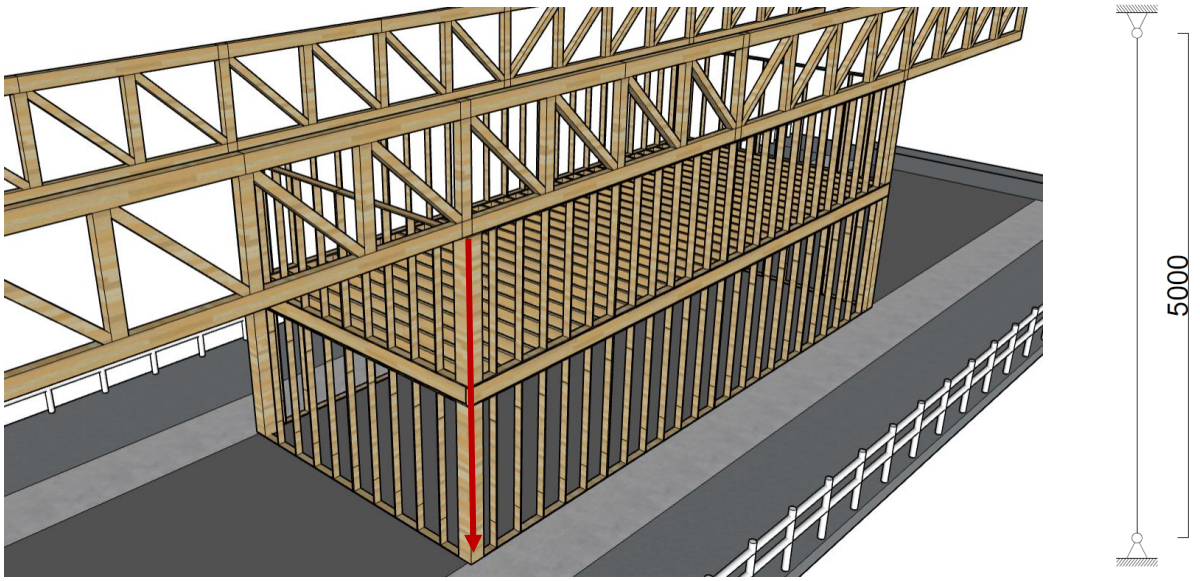


Figure 34: Column and the mechanical scheme

The box in the middle which is seen in Figure 35 is not distributing forces coming from the boxes at the sides. This is all done by the columns placed in the corners. Therefore the box in the middle is timber frame structure which is quite general. The forces are not extraordinary high which means using the rules of thumb would be enough, and in depth calculations are not relevant. The box is lighweight and distributed on the spanning part of the bridge.



Figure 35: Core box

6.3 Dimensioning

With the use of the load transfer each element could be calculated properly. First the acting loads should be calculated. This will be divided into the variable and permanent loads. With the use of these loads, the floor beam 1 could be calculated. With floor beam 1, floor beam 2 could be calculated and so on. First the hand calculations will be done for each element. Then this will be checked with Technosoft. The dimensions will be based on the Ultimate Limit State (ULS) and Serviceability Limit State (SLS). The ULS is based on the maximum capacity of the timber. The SLS is based on the deflection. There are no restrictions for the deflection, but in this case L/250 (for cantilever the maximum deflection is L/125) is taken as the maximum deflection, according to the old version of 'NEN-EN 1995: Eurocode 5, Ontwerp en berekening van houtconstructies'.

Loads

The loads are given in Table 1 below. The variable loads are retrieved from NEN-EN 1991: Eurocode 1, Belastingen op constructies. The permanent loads are determined by different sources which give the densities of different materials. These are transformed to the amount of weight in kN over the area m^2. For all elements the snow and wind are not taken into account for the calculation of the dimensions. Assumed is that the people on the roof, which is stated as 1.0 kN/m^2, is the maximum load

Table 1: Variable and permanent loads

Variable loads/live loads	**kN/m²**	**Permanent loads/dead loads floor**	**kg/m²**	**kN/m²**
Roof	1,00	Floor finishing	15,00	0,14715
Snow	0,70	Estrich Fermacell floor	26,00	0,25506
Wind	0,46	OSB	10,00	0,09810
Water	-	Rockwool	-	-
Floor (exhibition function)	2,80	Battens	-	-
		Aluminum finishing	16,38	0,160688
		Timber beams	15,00	0,14715
		Total	**82,38**	**0,808148**

Floor beam 1

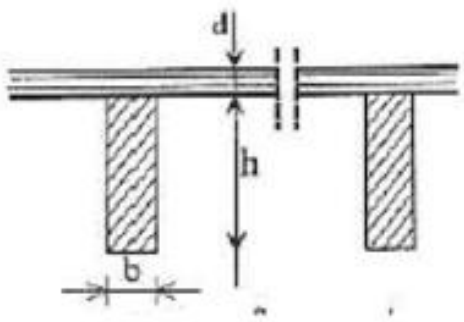
The length of the timber beam for the floor is chosen in accordance with the transportability and standard dimensions of timber beams. 3600 mm is transportable with a truck when using it in a prefabricated system of floor parts of 3600x6000 mm.

The assumption is that the height of the beam will be between 180 mm and 240 mm. The corresponding width will be chosen according to the standard dimensions of timber beams (Table 2).

$$h_{floorbeam} = \frac{1}{15}L - \frac{1}{20}L \qquad \frac{1}{15} \cdot 3600 = 240\ mm \qquad \frac{1}{20} \cdot 3600 = 180\ mm$$

To make a proper estimation of the exact dimensions of the timber beam, the acting loads should be used which are given above. Minimum dimensions are preferred due to economic and environmental reasons.

Sawn timber floor beams



$$\frac{L}{h} = 15 - 20$$

Figure 36: Rules of thumb for the height of a beam

The center to center (c.t.c.) distance is assumed to be 488 mm. This dimension is chosen considering the placing of the OSB boards on top of the floor beams. The OSB boards with standard dimensions (2440x1220 mm) are best connected to the floor beams when the ends of the OSB boards are placed in the middle of one of the floor beams as seen in Figure 37.

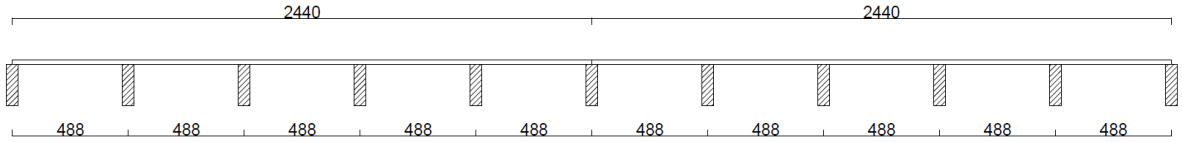


Figure 37: C.t.c. distance of the beams

The height and width of the beam are determined in Figure 82 seen in the Appendix. To get to a standard dimension of timber beams Table 2 is used. For the Excel sheet in the Appendix the information on the next four pages are used. The formulas are taken from 'NEN-EN 1995: Eurocode 5, Ontwerp en berekening van houtconstructies'. However there is a clear overview given in the course 'Dimensioning of structures' which is to be seen in all figures and tables on the next four pages. All relevant factors for the calculated elements in the figures and tables are circled by red circles. The first dimension which fulfills the Unity Check is a c18 beam of 200x63 mm.

Table 2: Standard dimensions of timber beams

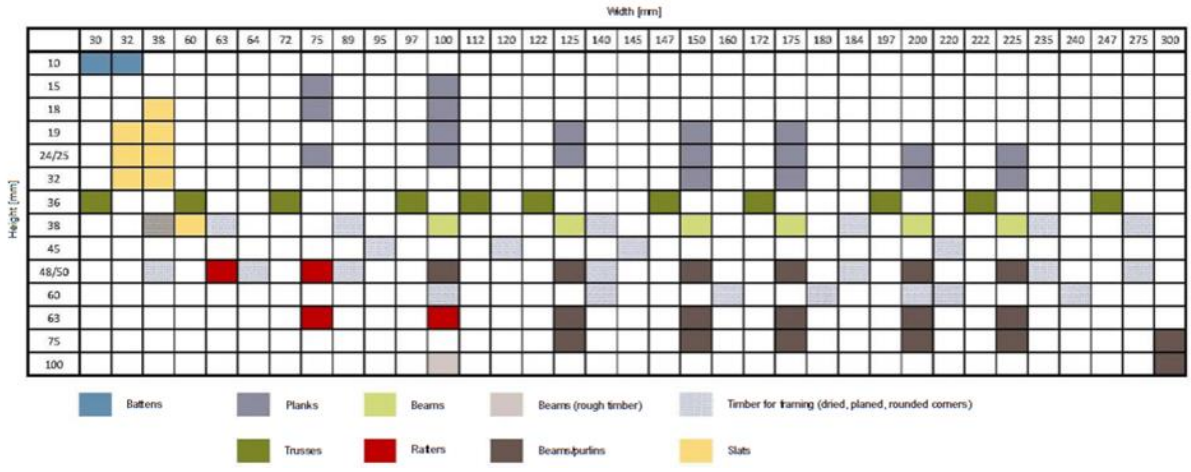


Table 3: Climate classes

Climate class	$\omega_{average}$ [%]	Description
1	12	Standard indoor conditions
2	20	Outdoor, covered structures
3	>20	- poorly ventilated spaces (indoor) - fully exposed to outdoor conditions (not covered) - structures in and underneath water

Table 4: ψ -factors for variable loads.

Load type	Description	ψ_0	ψ_1	ψ_2
Category A	dwellings	0.4	0.5	0.3
Category B	offices	0.5	0.5	0.3
Category C	congresses, meeting places, theatres, conferences	0.6 / 0.4[a]	0.7	0.6
Category D	shopping	0.4	0.7	0.6
Category E	storage	1.0	0.9	0.8
Category H	roofs	0.0	0.0	0.0
	snow	0.0	0.2	0.0
	wind	0.0	0.2	0.0

Table 5: Values of mod_k

Material	Standard	Climate-class (table 5.2)	Load duration class				
			permanent	long	medium-long	short	very short
Sawn timber	EN 14081-1	1	0.60	0.70	0.80	0.90	1.10
		2	0.60	0.70	0.80	0.90	1.10
		3	0.50	0.55	0.65	0.70	0.90
Glued laminated wood	EN 14080	1	0.60	0.70	0.80	0.90	1.10
		2	0.60	0.70	0.80	0.90	1.10
		3	0.50	0.55	0.65	0.70	0.90
LVL	EN 14374 , EN 14279	1	0.60	0.70	0.80	0.90	1.10
		2	0.60	0.70	0.80	0.90	1.10
		3	0.60	0.55	0.65	0.70	0.90
Plywood	EN 636						
	Parts 1, 2 and 3	1	0.60	0.70	0.80	0.90	1.10
	Parts 2 and 3	2	0.60	0.70	0.80	0.90	1.10
	Part 3	3	0.50	0.55	0.65	0.70	0.90
OSB	EN 300						
	OSB/2	1	0.30	0.45	0.65	0.85	1.10
	OSB/3, OSB/4	1	0.40	0.50	0.70	0.90	1.10
	OSB/3, OSB/4	2	0.30	0.40	0.55	0.70	0.90

Table 6: Material factors y_m

material	Material factor γ_m
sawn timber	1.30
glued laminated timber	1.25
LVL, plywood, OSB	1.20
connections	1.30
metal plate connectors	1.25

Table 7: Height factors h_k and length actor l_k

Material	tension parallel to the grain direction and bending
Sawn timber with $\rho_k \leq 700$ kg/m^3	$1.0 \leq k_h = \left(\frac{150}{h}\right)^{0.2} \leq 1.3$
Glued laminated timber	$1.0 \leq k_h = \left(\frac{600}{h}\right)^{0.1} \leq 1.1$
LVL (laminated veneer lumber)	Depending on variations to be determined according to EN 14374
Wood-based panels	1.0

Note: the reference length for the length factor k_l is L = 3000 mm (element length).

Table 8: Load duration classes

Load duration class	Cumulative duration of the characteristic load	Examples
Permanent	Longer than 10 years	Dead load
Long	6 months - 10 years	Storage
Medium-Long	1 week - 6 months	Life loads on floors
Short	Less than 1 week	Snow, wind (The Netherlands)
Instantaneous		Accidental load, wind (Belgium)

Table 9: k_{def} factors for wood and wood-based materials

		Climate class		
		1	2	3
Sawn timber	EN 14081-1	0.6	0.8	2.0
Glued laminated timber	EN 14080	0.6	0.8	2.0
LVL	EN 14374, EN 14279	0.6	0.8	2.0
Plywood	EN 636			
	Part 1	0.8	-	-
	Part 2	0.8	1.0	-
	Part 3	0.8	1.0	2.5
OSB	EN			
	OSB/2	2.25	-	-
	OSB/3, OSB/4	1.50	2.25	-

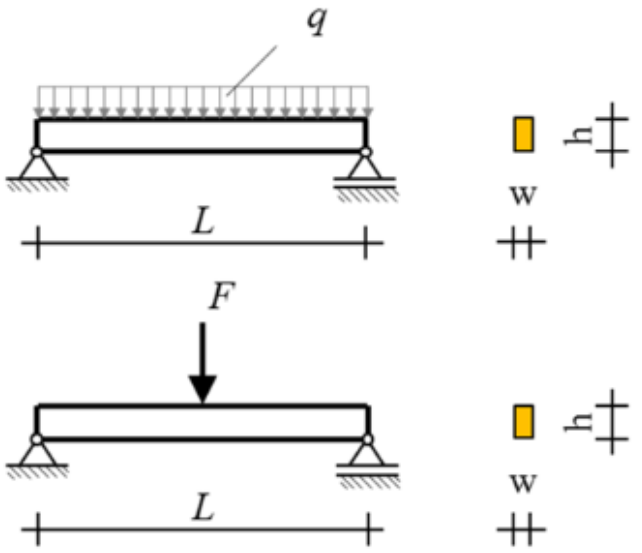
Table 10: Strength classes for wood

Strength class			C18	C24	D30	D40	D50	D70	GL24h
Ultimate Limit States (ULS)	$f_{m,k}$	N/mm²	18	24	30	40	50	70	24
	$f_{t,0,k}$	N/mm²	10	14.5	18	24	30	42	19.2
	$f_{t,90,k}$	N/mm²	0.4	0.4	0.6	0.6	0.6	0.6	0.5
	$f_{c,0,k}$	N/mm²	18	21	24	27	30	36	24
	$f_{c,90,k}$	N/mm²	2.2	2.5	5.3	5.5	6.2	12.0	2.5
	$f_{v,k}$	N/mm²	3.4	4.0	3.9	4.2	4.5	5.0	3.5
	ρ_k	kg/m³	320	350	530	550	620	800	385
	$E_{m,0,k}$	N/mm²	6,000	7,400	9,200	10,900	11,800	16,800	9,600
Seviceability Limit States (SLS)	$E_{m,0,mean}$	N/mm²	9,000	11,000	11,000	13,000	14,000	20,000	11,500
	$E_{m,90,mean}$	N/mm²	300	370	730	870	930	1,330	300
	G_{mean}	N/mm²	560	690	690	810	880	1,250	650

Table 11: Klasse van belaste oppervlakte

Klasse van belaste oppervlakte	q_k [kN/m²]	Q_k [kN] 0.1x0.1 [m²]
Klasse A (wonen en huishoudelijk gebruik)		
A-vloeren	1,75	3
A-trappen	2,0	3
A-balkons	2,5	3
A-ontsluitingsweg Q_k op 0.5 x 0.5 [m]	2,0	3
Klasse B (kantoorruimten)		
B-kantoorruimten	2,5	3
B-ontsluitingsweg	3,0	3
Klasse C (bijeenkomstruimten, sport, personenvervoer)		
C1-tafels	4,0	7
C2-vaste zitplaatsen	5,0	7
C3-zonder obstakels voor rondlopende mensen	5,0	7
C4-fysieke activiteiten	5,0	7
C5-grote mensenmassa's	5,0	7
C-ontsluitingsweg	5,0	7
Klasse D (winkelruimten)		
D1-kleinhandel	4,0	7
D2-warenhuizen	4,0	7
D-ontsluitingsweg	4,0	4
Klasse E		
E1-winkels	5,0	7
E1-bibliotheken	2,5	3
E1-ontsluitingsweg	4,0	4
E2-industrieel	3,0	7
E2-ontsluitingsweg	4,0	4
Klasse F/G		
F - Voertuigen < 25 kN	2	10
G - Voertuigen 25 kN – 120 kN	5	40
G - Voertuigen > 120 kN		
H-Daken		
H (niet toegankelijk) $0 \geq \alpha < 15°$	1	1,5/2,0
$15 \geq \alpha < 20°$	4-0,2α	

Mechanics



$$V = \frac{q \cdot L}{2}$$

$$M = \frac{q \cdot L^2}{8}$$

$$w = \frac{5}{384} \cdot \frac{q \cdot L^4}{EI}$$

$$V = \frac{F}{2}$$

$$M = \frac{F \cdot L}{4}$$

$$w = \frac{F \cdot L^3}{48 \cdot EI}$$

Safety factors

Consequence Class	CC1	$\gamma_G = 1.08$	$\gamma_Q = 1.35$
Consequence Class	CC2	$\gamma_G = 1.20$	$\gamma_Q = 1.50$

Design load

$$q_d = \gamma_g \cdot G_k + \gamma_q \cdot Q_k$$

Final deflection

$$w_{fin} = w_G \cdot \left(1 + k_{def}\right) + w_Q \cdot \left(1 + \psi_2 \cdot k_{def}\right)$$

Load including creep

$$q_{creep} = G_k \cdot \left(1 + k_{def}\right) + Q_k \cdot \left(1 + \psi_2 \cdot k_{def}\right)$$

Unity Check stresses

$$UC = \frac{\sigma_{m,d}}{f_{m,d}} \qquad UC = \frac{\sigma_{v,d}}{f_{v,d}}$$

Unity Check deflection

$$UC = \frac{w_{fin}}{w_{max}} \qquad w_{max} = 0.004 \cdot L$$

Moment of resistance

$$W = \frac{1}{6} \cdot b \cdot h^2$$

Moment of inertia

$$I = \frac{1}{12} \cdot b \cdot h^3$$

Design value / characteristic value

$$f_{m,d} = \frac{f_{m,k}}{\gamma_M} \cdot k_{mod} \qquad f_{v,d} = \frac{f_{v,k}}{\gamma_M} \cdot k_{mod}$$

Bending stress

$$\sigma_{m,d} = \frac{M_d}{W}$$

Shear stress

$$\sigma_{v,d} = \frac{3}{2} \cdot \frac{V_d}{b \cdot h}$$

Figure 38: Formulas for timber calculations

In Figure in the Appendix the 200x63 mm beam is checked in Technosoft with the acting loads. The maximum moment corresponds with the maximum moment determined in the hand calculation which is around 4.0 kNm. The deformation in Technosoft (approximately 10.1mm) is slightly lower than the deformation which came out of the hand calculations (approximately 12.5 mm). The Unity Check for the moment is 0.93 and the Unity Check for the deflection is 0.87. This is considered to be sufficient. In Figure 39 the mechanical scheme of floor beam 1 is shown with the permanent load. In Figure 40 the mechanical scheme of floor beam 1 is shown with the variable load.

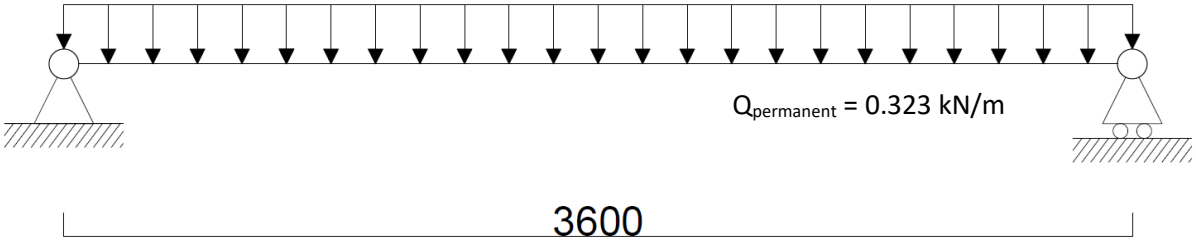


Figure 39: Permanent load floor beam 1

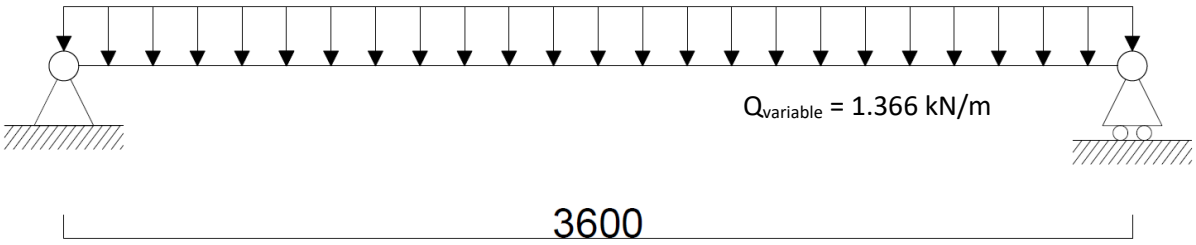


Figure 40: Variable load floor beam 1

For floor beam 1 a detailed explanation for the explanation for all factors is given with figures tables and references. For the rest of the calculations the results will be given without these details. Assumed can be that all factors are the unless stated otherwise.

Floor beam 2

The timber c18 200x63 beams calculated above are connected perpendicular to a timber beam which is attached to steel cables. This timber beam should be capable of holding 7 members of the timber c18 200x63 beam. To determine the dimensions of this timber beam, the reactions forces of the beams will be placed on top of the beam as point loads, which are the reaction forces of the beam calculated before.

The rules of thumb gives the following results for the height of the beam:

$$h_{floorbeam} = \frac{1}{15}L - \frac{1}{20}L \qquad \frac{1}{15}\cdot 3000 = 200\,mm \qquad \frac{1}{20}\cdot 3000 = 150\,mm$$

However, known is that the loads on top of the beam are way higher than in the beam calculated before. Therefore the assumption is that the beam should be higher than 200 mm. In Figure 87 in the Appendix the height and width of the beam are calculated. The mechanical scheme is given in Figure 41 with the permanent force. The variable force on the same beam is shown in Figure 42. For the calculation, a beam of standard dimensions c24 300 x 100 is used.

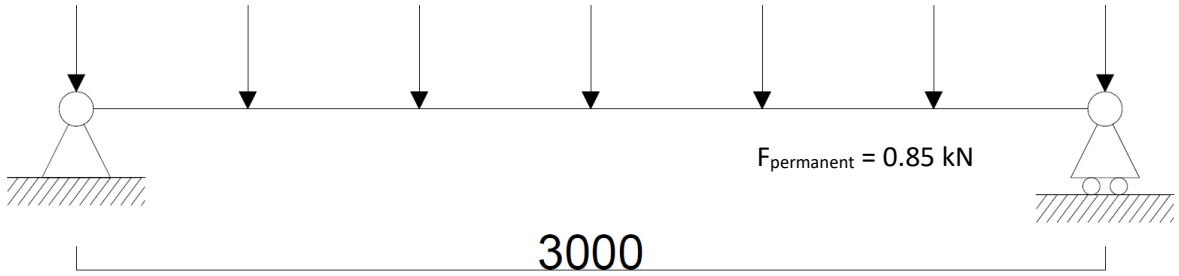


Figure 41: Permanent load floor beam 2

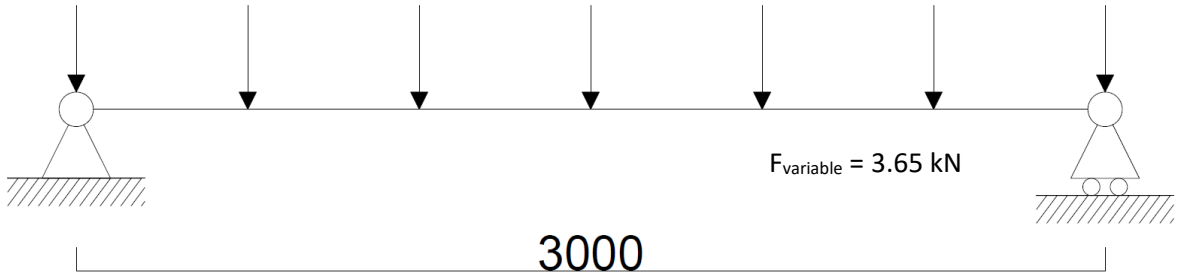


Figure 42: Variable load floor beam 2

In Figure 88 till Figure 91 in the Appendix the 300 x 100 mm beam is checked in Technosoft with the acting loads. The maximum moment corresponds with the maximum moment determined in the hand calculation which is around 14.8 kNm. The deformation in Technosoft (approximately 3.8 mm) is slightly lower than the deformation which came out of the hand calculations (approximately 5.0 mm). The Unity Check for the moment is 0.83 and the Unity Check for the deflection is 0.42. This is considered to be sufficient.

Steel cable

The steel cable which is loaded the most is one of the cables in the middle of the middle truss. This cable has a load of one floor panel of 3.0 x 3.6 meter. For the calculation of the diameter of the cable, both the variable force and the permanent force are applied without safety factors. It turns out that the steel cable needs a diameter of at least 15 mm. Because of not taking the safety factors into account, a diameter of 20 mm is chosen. In Figure 92 in the Appendix the calculation for the cable is shown. In Figure 43 the mechanical scheme of the steel cable is given. The Figure is turned 90 degrees to make the Figure more clear. The cable is not calculated by Technosoft.

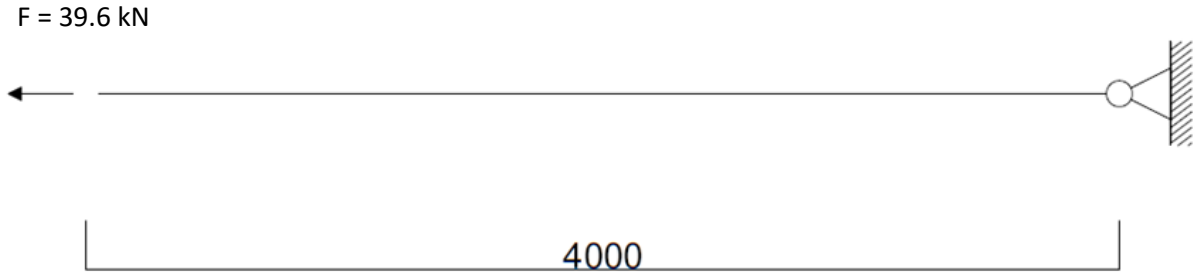


Figure 43: Steel cable with applied force

Roof beam

The roof beam is spanning over the trusses. This beam is 3.6 meter. The assumption is that the height of the beam will be between 180 mm and 240 mm. The corresponding width will be chosen according to the standard dimensions of timber beams (Table 2).

$$h_{floorbeam} = \frac{1}{15}L - \frac{1}{20}L \qquad \frac{1}{15} \cdot 3600 = 240\,mm \qquad \frac{1}{20} \cdot 3600 = 180\,mm$$

The loads on top in this case are lower then for the floor beams because of the function. This means that the rules of thumb are probably quite close to the actual dimensions of the beam.

In Figure 93 in the Appendix the height and width of the beam are calculated. The mechanical scheme is given in Figure 39 with the permanent force. The variable force on the same beam is shown in Figure 40. For the calculation, a beam of standard dimensions c18 175 x 50 is used.

In Figure 94 till Figure 97 in the Appendix the 175 x 50 mm beam is checked in Technosoft with the acting loads. The maximum moment corresponds with the maximum moment determined in the hand calculation which is around 1.9 kNm. The deformation in Technosoft (approximately 9.2 mm) is slightly lower than the deformation which came out of the hand calculations (approximately 13.4 mm). The Unity Check for the moment is 0.70 and the Unity Check for the deflection is 0.93. This is considered to be sufficient.

Truss 1

The first truss is calculated with respect to the maximum moment. The maximum moment is calculated. For the top chord it is assumed that the beam is in tension and for the bottom chord it is assumed that the beam is in compression. For easy connection all elements of the truss have the same dimensions. Assumed is that the calculated dimensions for the top chord which is loaded in tension is the highest. This is because the tensile force of timber is lower than the compressive force. For the top chord the maximum moment is divided by the height of the truss. When dividing the axial force by the tensile strength of timber, the area of the cross section found. This turned out to be 21145,89 mm^2 (See Figure 98) in the Appendix). However, no factors corresponding to timber are taken into account. This means that the dimensions should probably be way higher. For the Technosoft calculation the double of this number is assumed (42291.78 mm^2). A beam of c24 300 x 150 mm would be assumed to be enough. The mechanical scheme of the truss with the permanent loads and forces is given in Figure 44 The scheme with the variable loads and forces is shown in Figure 45. The steel cables on the sides are holding half of the weight from the other cables which is why the point loads at the sides are divided by 2.

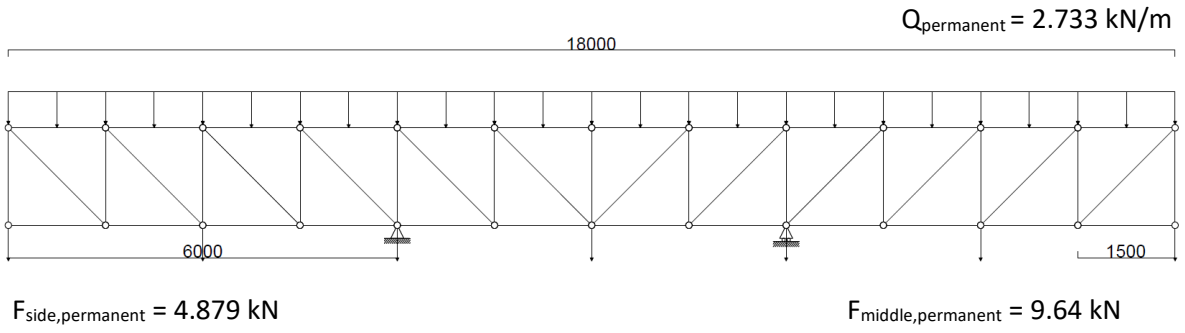


Figure 44: Permanent load truss 1

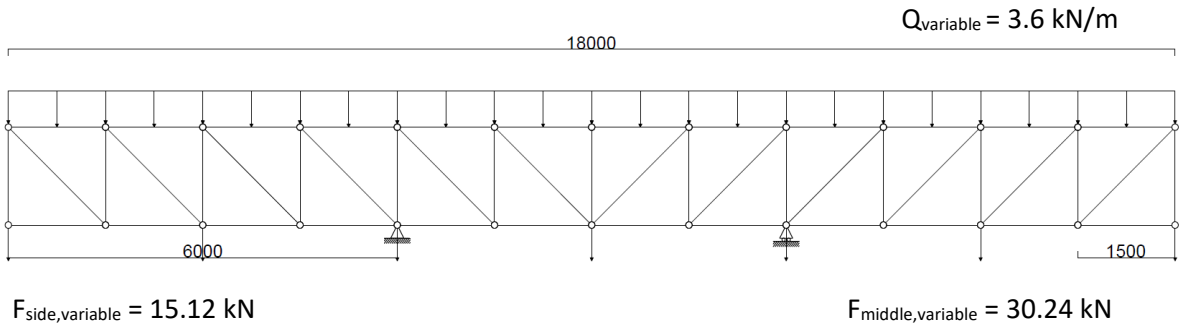


Figure 45: Variable load truss 1

With the use of Technosoft, the truss will be checked. The results of the Technosoft calculation are shown in Figure 99 till Figure 102 in the Appendix. If the truss would be a perfect truss, all elements would be hinged and all moments should be equal to 0. However assumed is that both chord are not hinged, but instead the bottom and top chord are one long beam. Still the normal forces are most important for the truss while dimensioning the elements. This is because, due to the hinges, all members are in tension or compression. As seen in Figure 101 in the Appendix, The normal forces are quite high. The maximum normal force is 333 kN. In Figure 98 in the Appendix, this normal force is filled in with the dimensions of 300 x 150 mm. It turns out the unity check is 0.95 for the normal force and 0.4 for the deflection which is considered to be sufficient. As seen in Figure 46 the trusses at the sides could be half of the truss in the middle, because they support half the load. These trusses have elements with dimensions 300 x 75 mm of c24.

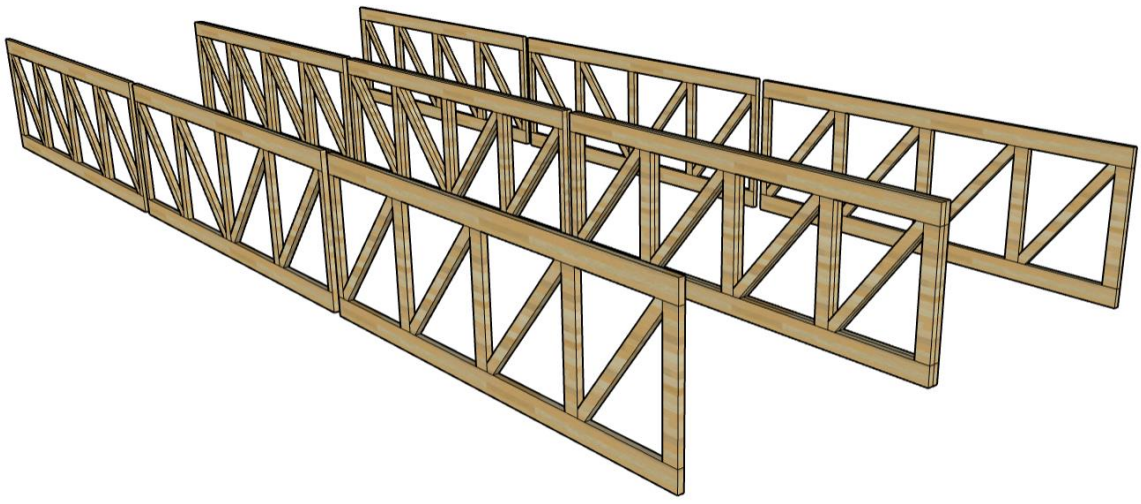


Figure 46: Difference in thickness truss 1

Truss 2

Truss 2 is calculated the same way as truss 1. However, for truss 1 it was doable to combine the loads and point loads into one average point load to calculate the maximum moment. This is more difficult for truss 2, which is why only a Technosoft calculation will be performed. In Figure 47 the mechanical scheme for truss 2 is shown with the permanent loads and in Figure 48 with the variable loads. In Figure … in the appendix the calculation for the dimensions of the truss elements is shown. The results from Technosoft are to be seen in Figure 104 till Figure 107. The normal forces are used to determine the dimensions.

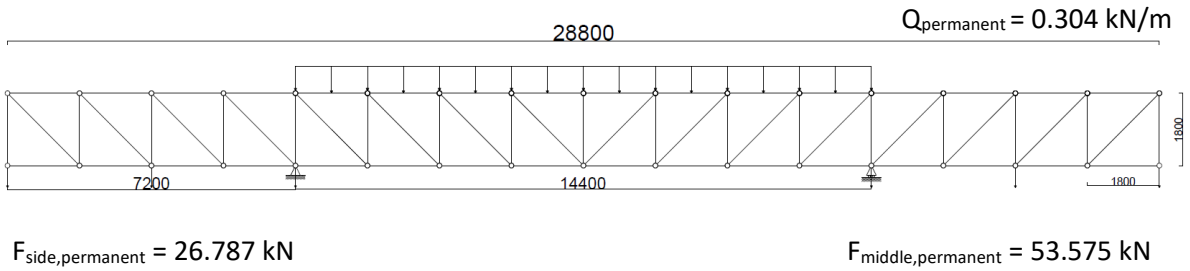


Figure 47: Permanent load truss 2

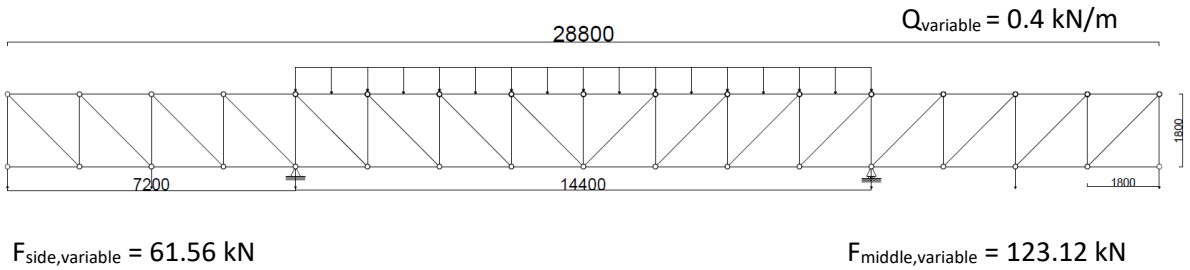


Figure 48: Variable load truss 2

As seen in Figure 106 in the Appendix, The normal forces are quite high again. The maximum normal force is 1012 kN. In Figure 103 in the Appendix, this normal force is filled in with the dimensions and timber type of c24 500 x 300 mm. It turns out the unity check is 0.96 for the normal force and 0.99 for the deflection which is considered to be sufficient. The dimensions of truss 2 are not standard dimensions in c24.

Column

The column is based on the reaction force coming from truss 2. This reaction force is equal to 659 kN. The column is loaded in compression and should have large dimensions. The area of the timber column should be 31380.95 mm^2. It turns out that the timber column needs to be 200 x 200 mm. In Figure 108 in the Appendix the calculation for the cable is shown. In Figure 49 the mechanical scheme of the timber column is given. The Figure is turned 90 degrees to make the Figure more clear. The column is not calculated by Technosoft.

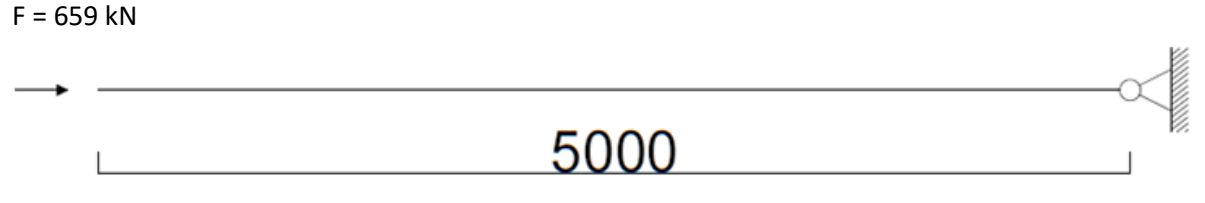


Figure 49: Applied force on the timber column

6.4 Stability

To distribute horizontal forces like wind, the structure should be stable. These forces have to be distributed to the foundation without the building collapsing. Different stability principles could be used for this.

For this building, two different principles could be used. The first one being the portals in both directions. Because of the timber frame structure, there are a lot of portals in the building built up out of columns and beams. This is used in both directions of the structure. In Figure 51 a section is shown in which the forces coming from wind are distributed. In Figure 50 a top view of the building is given to show the distribution of the wind forces. To make the portal more clear, a schematic view of the structure is given in Figure 52.

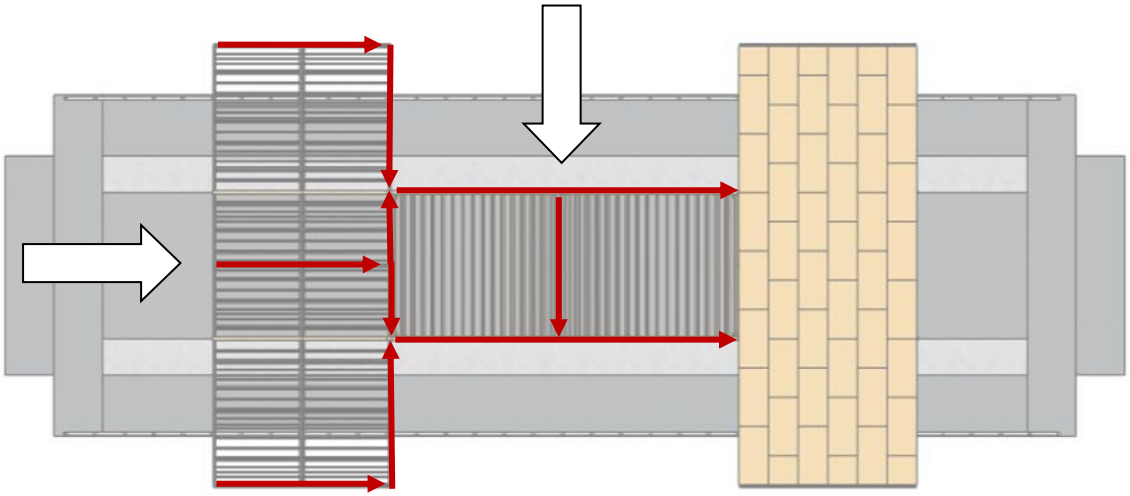


Figure 50: Floorplan of stability

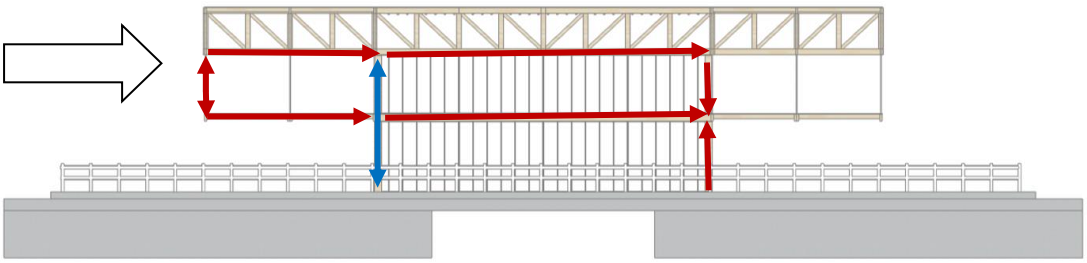


Figure 51: Section of stability

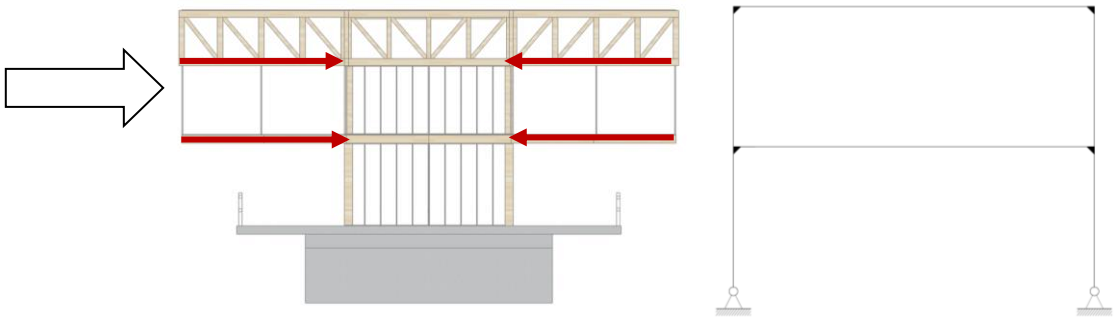


Figure 52: Section of stability and schematic view of the structure

The second principle is based on stiff timber frame walls. For the middle box in the building timber frame walls are used. These walls have OSB board attached on both sides (Figure 53). These OSB boards make the walls stiff. In Figure 54 it is shown what happens when the OSB is not attached to the timber frame panels and what happens when the OSB is attached.

When wind is exposed to the glass façade in the hanging boxes, the forces go to the timber floor and timber roof elements which have same principle as the walls. OSB is attached to both sides of the timber beams, which makes the floor stiff. The forces go from the floors to the walls of the middle box. In Figure 55 the wind force is shown on both sides of the glass hanging boxes.

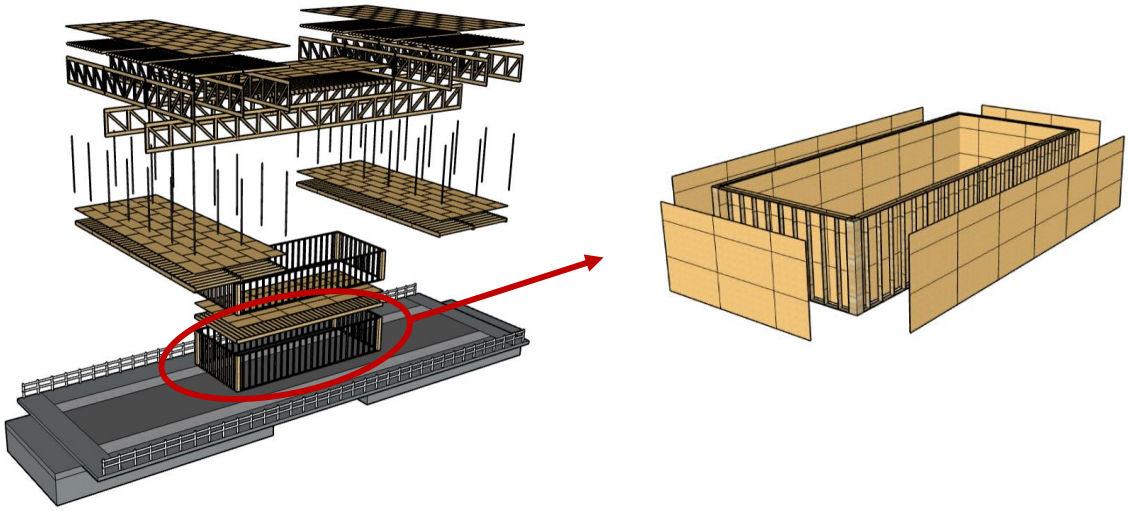


Figure 53: Stability of the wall components

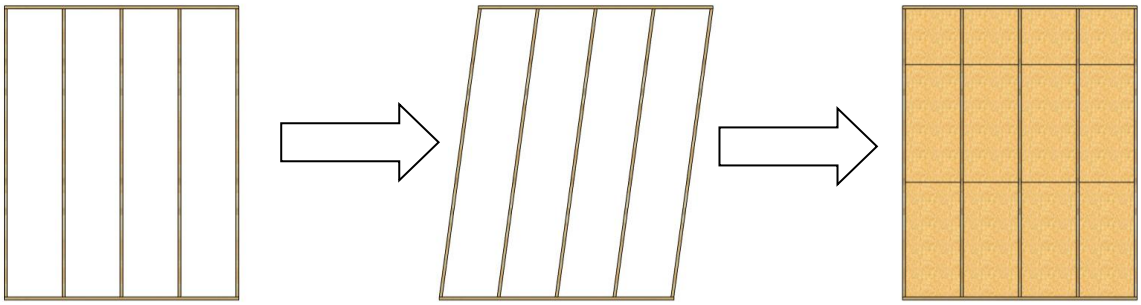


Figure 54: Stability of a wall component

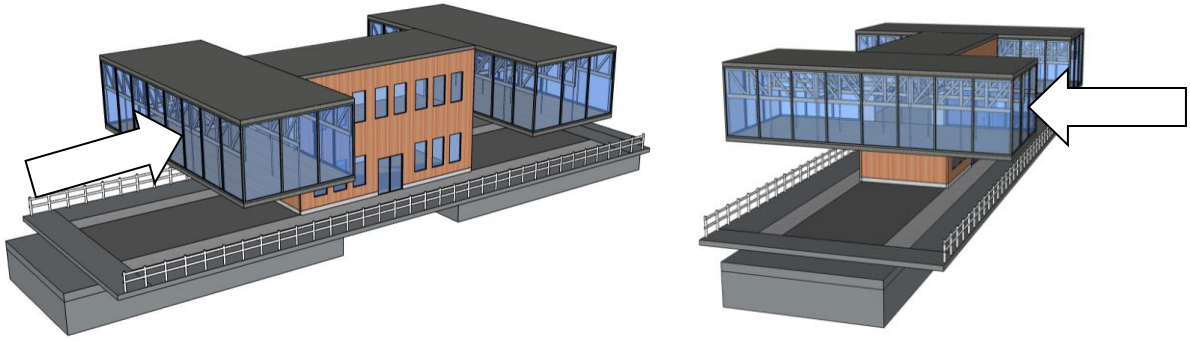


Figure 55: Wind forces on the glass facade

7. Construction process

Design decisions during the process are made mostly based on the way the building will be constructed on the plot. Most of the building will be made out of prefabricated elements. However, these elements should be able to fit in the transport vehicles which are probably trucks. For the construction on the plot, the fact that bridge is difficult to reach due to the Dommel, should be considered. In Figure 56 the exploded view of the building is given.

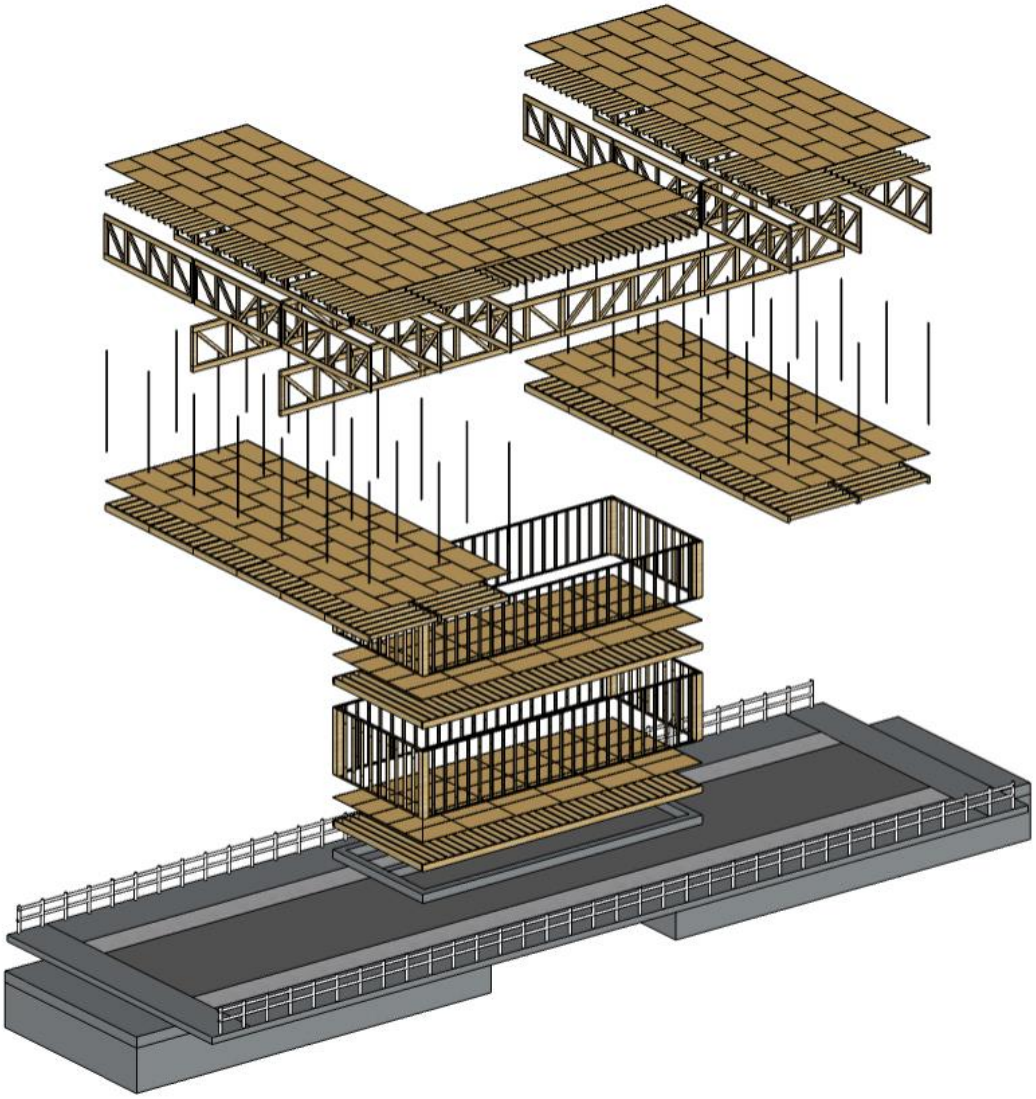


Figure 56: Exploded view of the construction

In the construction process the first thing that will be made is the foundation on top of the bridge. This is made out of reinforced concrete and is spanning over the bridge to both foundations. The reason for this is that the forces coming from the middle box could be distributed over the concrete beam into the foundation instead of having the loads of the middle box directly on top of the bridge. The other reason for placing the timber on top of a concrete strip, is to keep the timber away from the ground and keep it dry. In Figure 57 the concrete strips are is shown, which have a height of 40 centimeters. This height is chosen for the crawling space underneath.

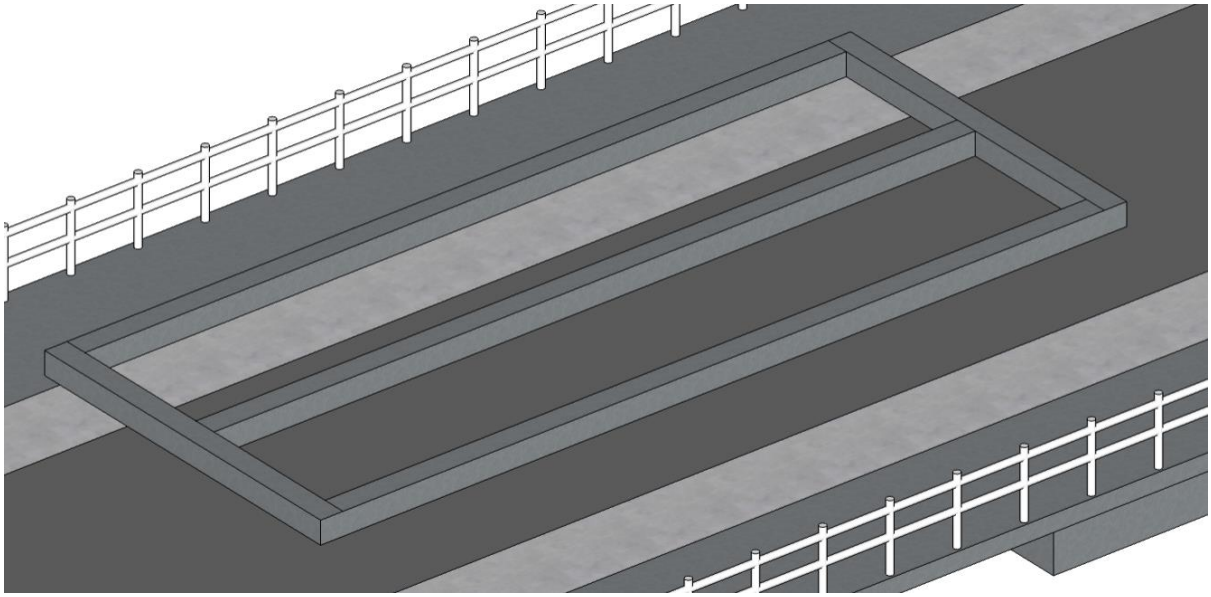


Figure 57: Concrete strips

On top of the concrete strips, the box in the middle will be placed. This box exists of wall elements of 3.6 x 3.0 meter. The floor elements are 3.0 x 6.0 meters and shown in Figure 59. This is except for the beams placed on the foundation strips. These beam elements are 3.0 x 3.6 meters and shown in Figure 59. The wall elements are 3.0 x 3.6 meters as well (Figure 60). The four columns in the corners are placed next to the wall elements. In Figure 58 the middle box is shown.

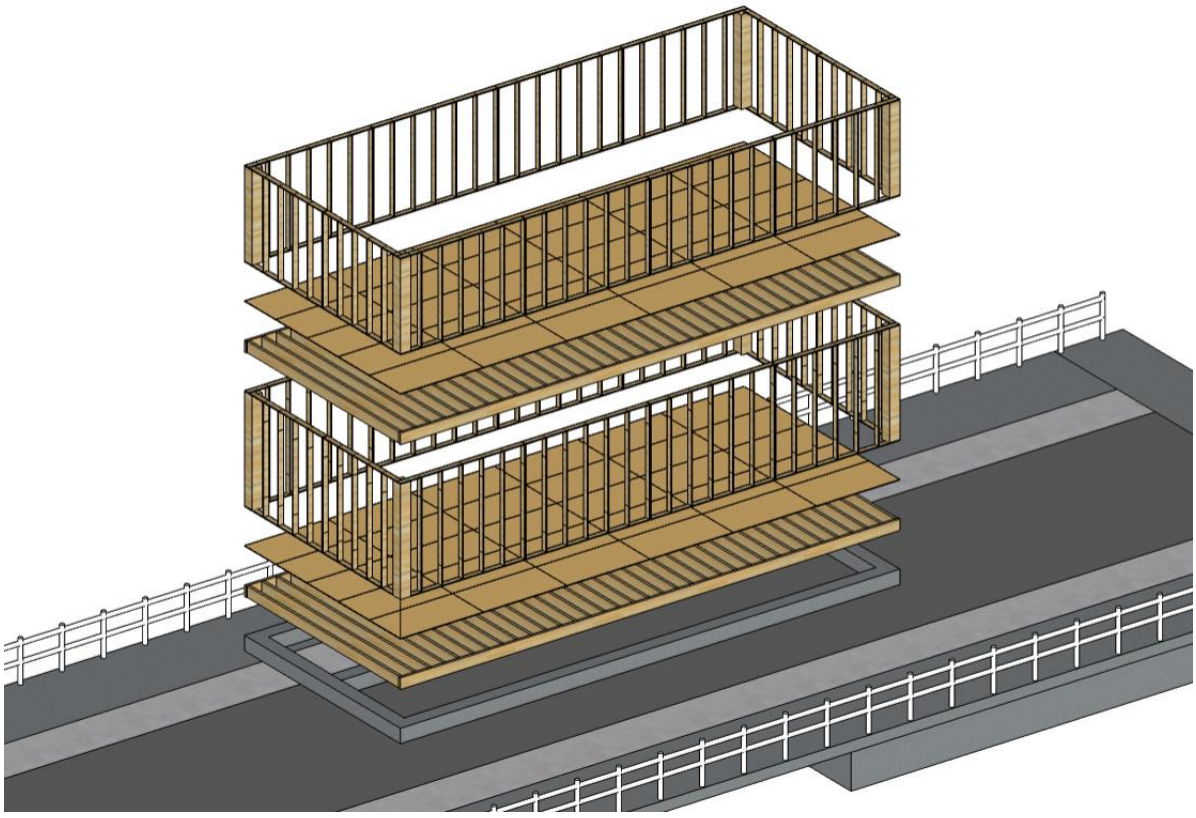


Figure 58: Exploded view middle box

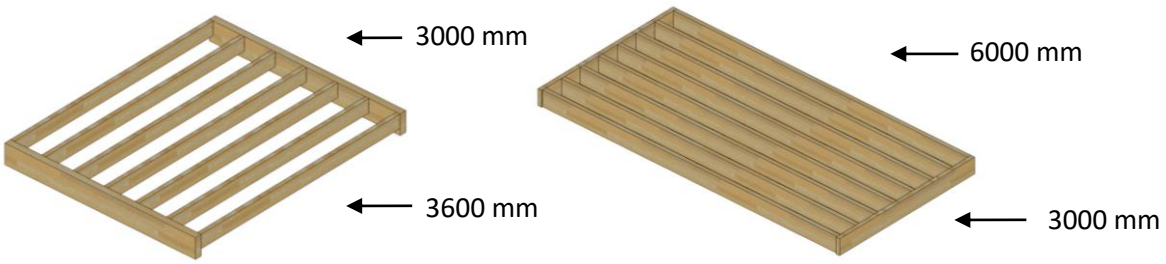


Figure 59: Floor elements

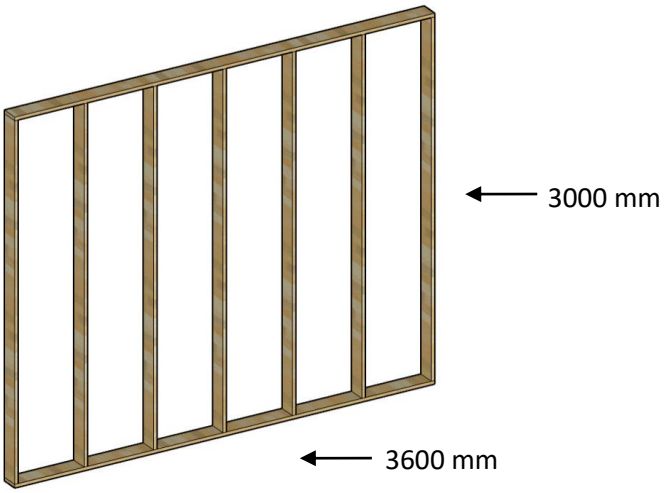


Figure 60: Wall element

After the box is placed, the large trusses could be placed on top of the four columns. The truss is 28.8 meter long and therefore to long for a truck to transport (Figure 61). This means the truss should at least be divided in two parts. Preferably the connection is made at the place where the moment is minimal. In Figure 62. The moment diagram of truss 2 is given.

As to be seen the moments are maximal at the supports and minimum at the ends of the truss. Making a connection at the end does not work while the parts would be to small and the connection with truss 1 is made at the end as well. Therefore the truss could be separated best in the exact middle of the truss, or preferably right next to the middle (not in a nod) while the moment is less there. This is even though the moment is quite high on average still the best place to connect the truss parts. In Figure 63 the truss is shown in the two parts.

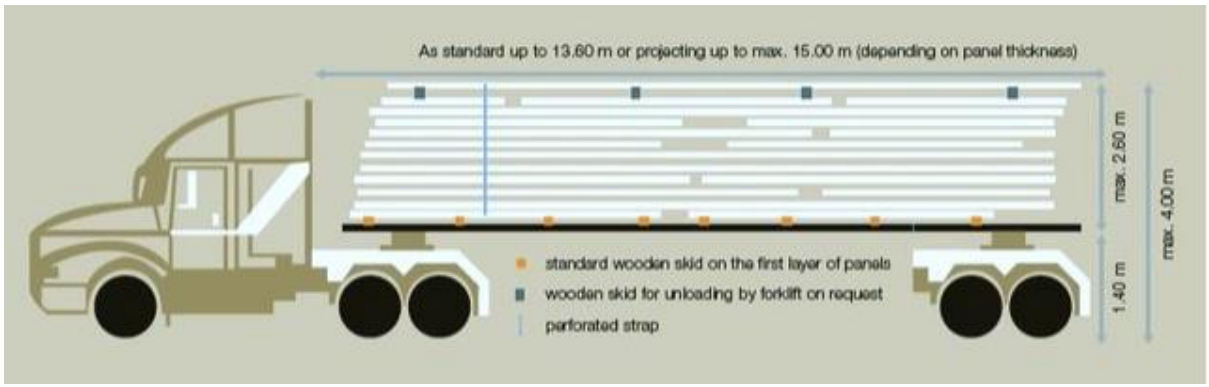


Figure 61: Dimensions of a truck

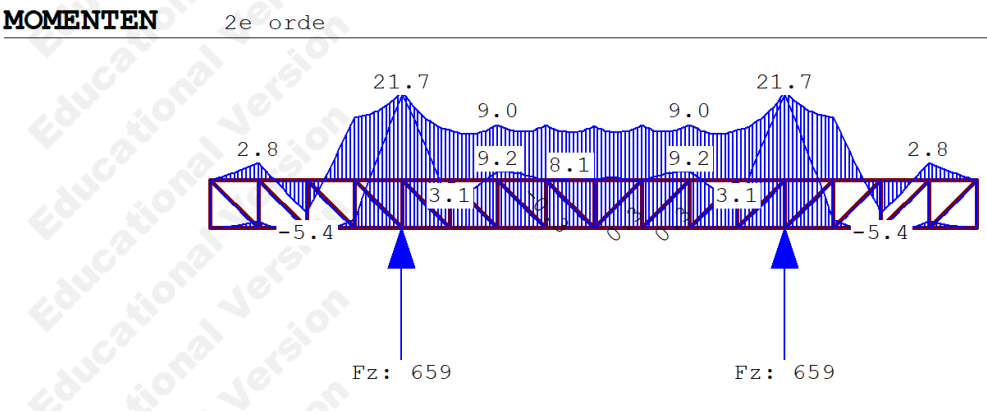


Figure 62: Moment diagram truss 2

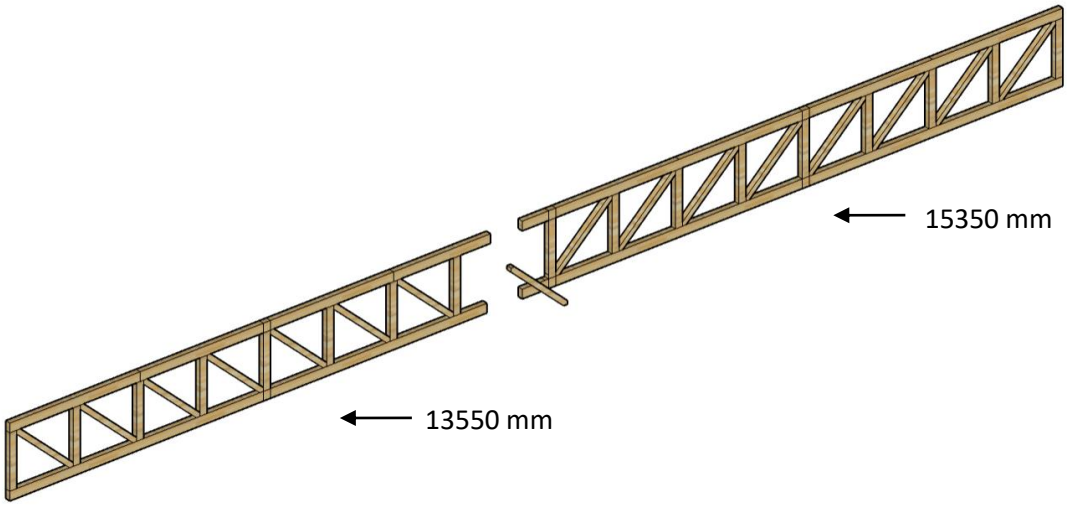


Figure 63: Cut through truss 1

The next part of the structure are the perpendicular trusses. These trusses are divided as well. Because of aesthetic reasons, this truss will not be placed on top of the other truss, but instead placed between truss 2. This means truss 1 should be divided on the spots where they are connected to truss 2. In Figure 65 this is shown. The problem that arises is that the moments are maximum at the connection points, which means these connections should be fixed and therefore capable to resist high moments. In Figure 64 the moment diagram of truss 1 is shown.

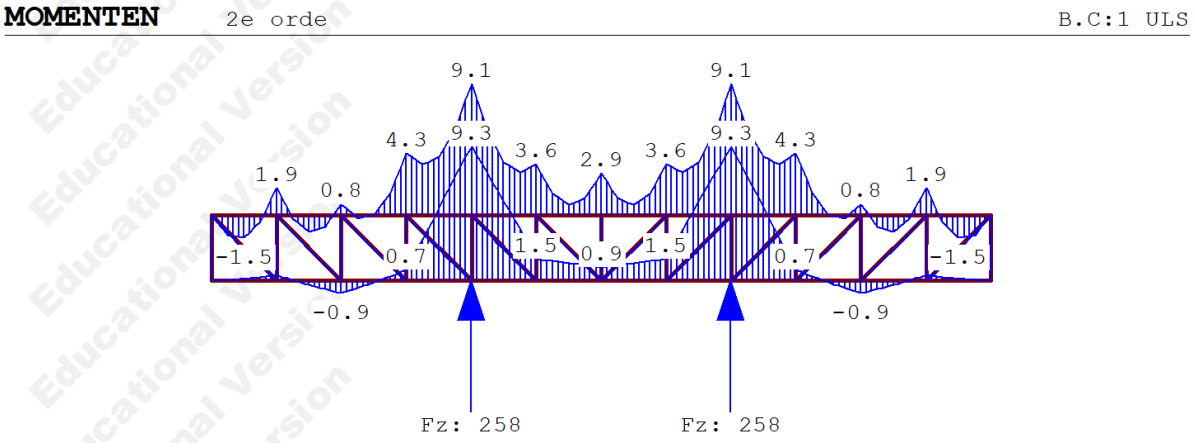


Figure 64: Moment diagram truss 1

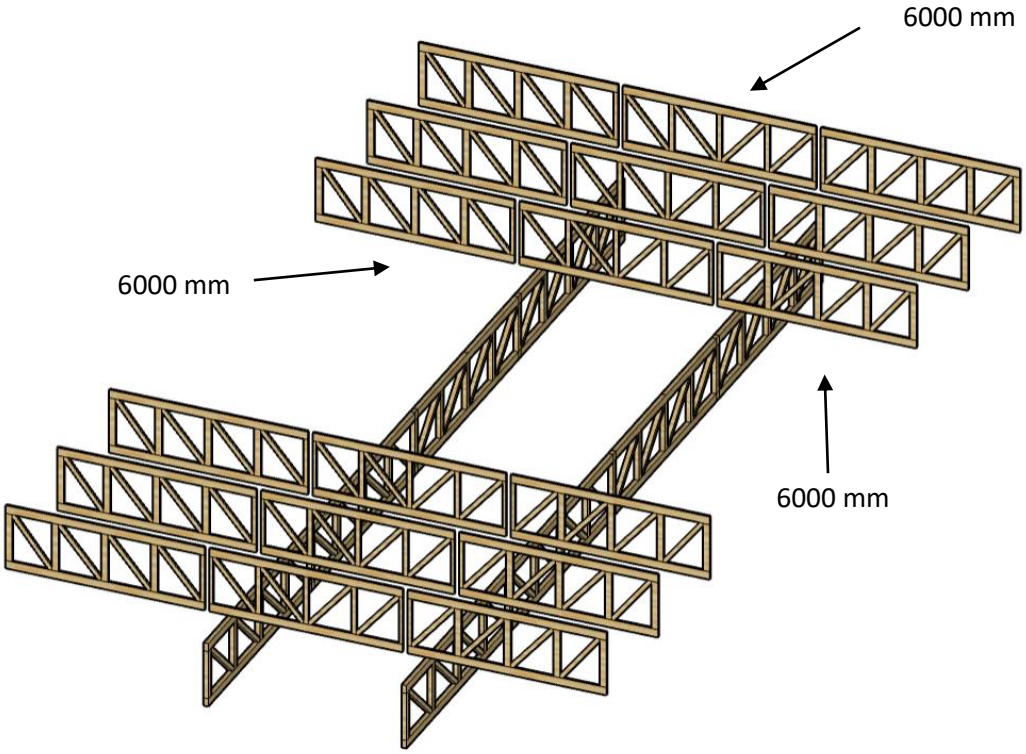


Figure 65: Exploded view truss 1 and 2

Now that the trusses are placed, the cables could be hanged underneath them. After this, the floor elements of the side boxes should be held up and be attached to the cables. The side boxes have the same floor elements as shown in Figure 59. In Figure 66 the built up of the side boxes is shown.

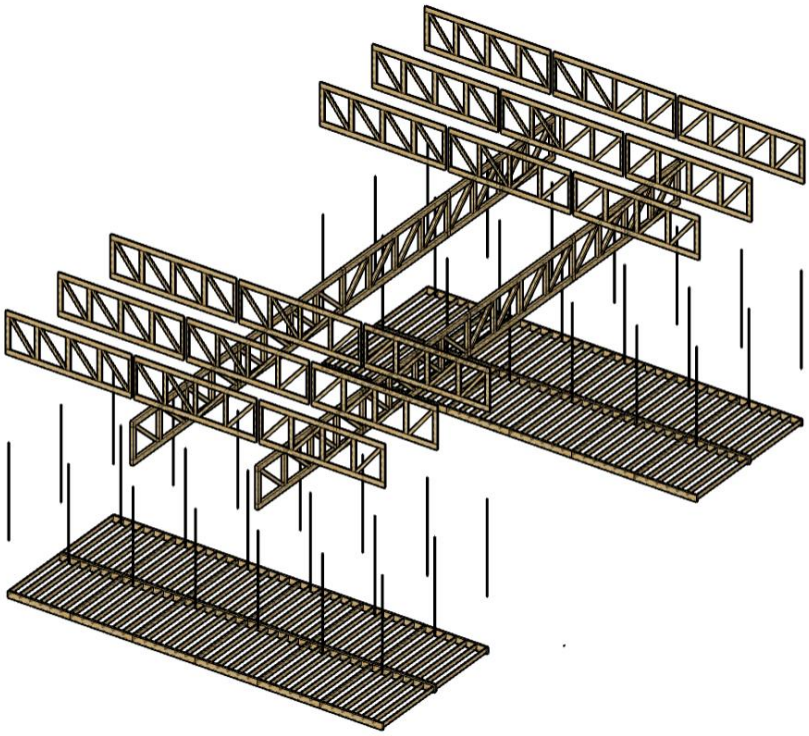


Figure 66: Trusses with hanging floor structure

On top of all the trusses the roof beams will be placed. This is the last step for the structure. The roof beams are spanned between the two trusses of type truss 2 and are 6 meter long. The other roof beams are spanned between the three trusses of type truss 1 and are 3.6 meter long.

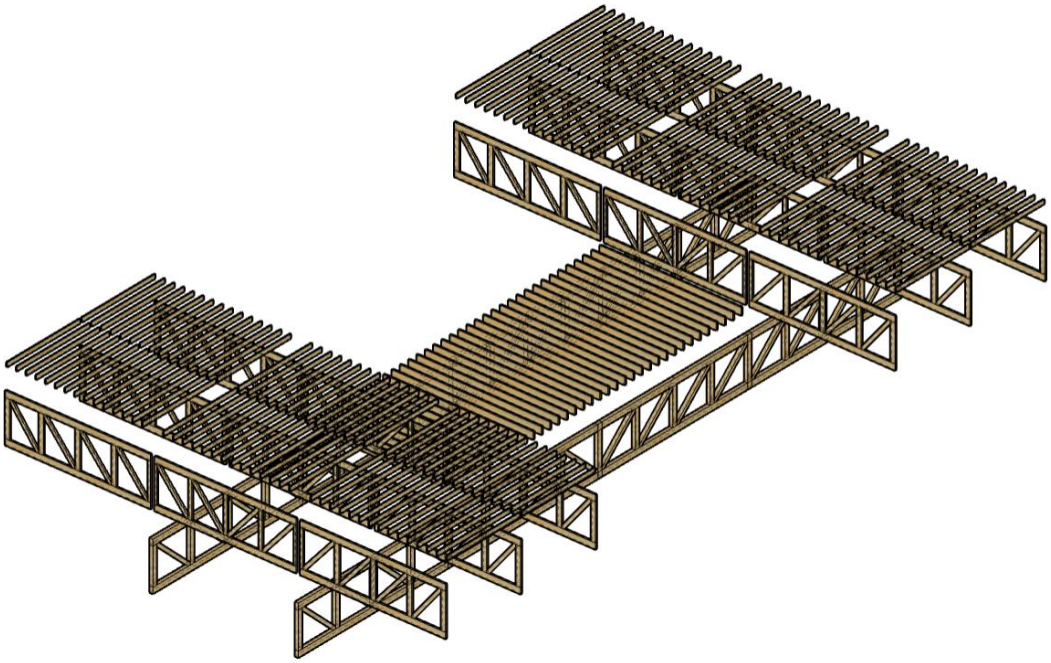


Figure 67: Roof beams on top of truss structure

When the floor elements are attached the structure of the building is built up. Now the windows and window frames could be placed and the façade of the middle box as well. In Figure 68 the finished building is shown.

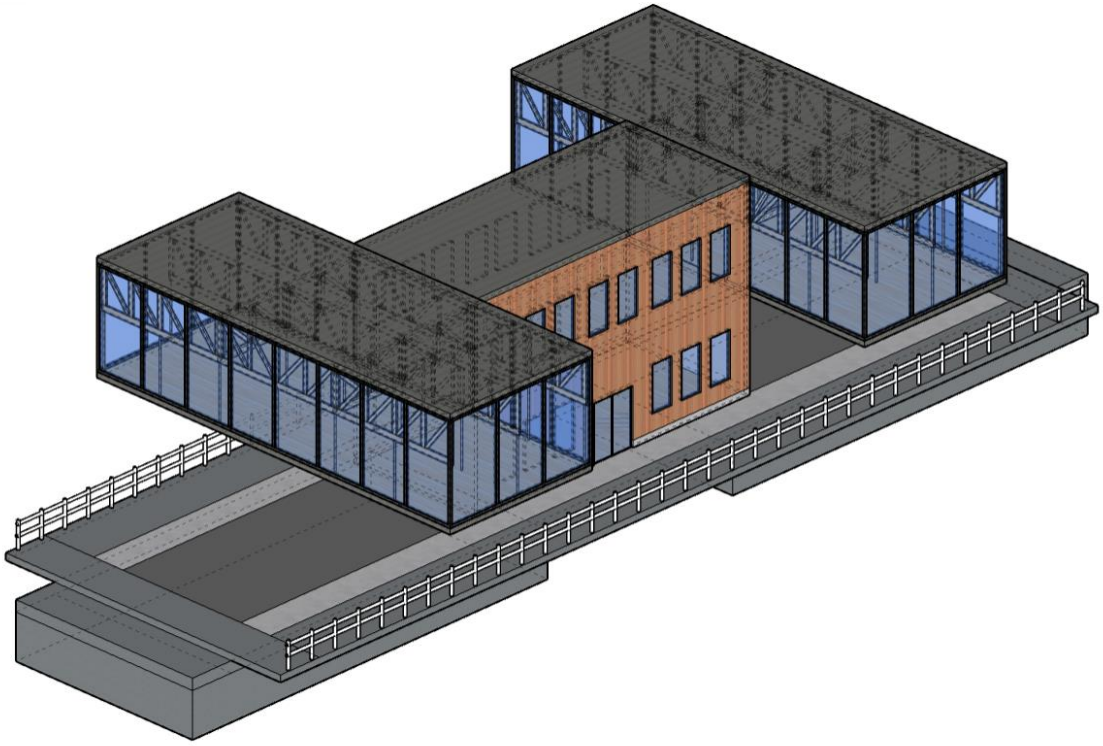


Figure 68: Complete view of the building

8. Detail

After focusing on the structure, the details of the structure are important. Connections between elements are based calculations as well. In this chapter one detail will be calculated and a few other details will be explained without calculations.

In Figure 69 the truss with the cables is shown. The middle truss exists of two trusses connected to each other, while the outer trusses are only one truss. The reason for this is that the middle truss has twice the load of the outer ones. One detail of this middle truss is calculated in this chapter, which is marked with the red circle in Figure. In Figure 70 a reference of the detail is shown. The plate which is now placed on top, will be placed between the timber members later. At the places of the holes, bolts will be placed.

The detail which will be worked out is shown in Figure 70. The detail shows the connection between the horizontal beam and the vertical and diagonal elements of the truss. The bolts will be calculated with respect to the failure mechanisms of the timber and the steel. Then the steel plates between the timber parts are calculated. For this calculation, Eurocode 5 is used.

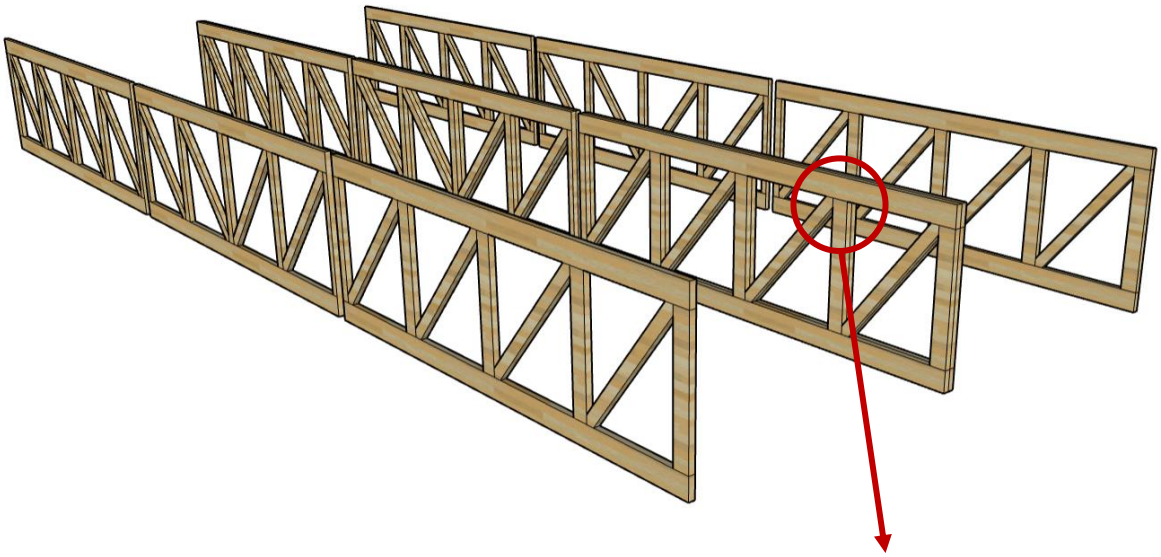


Figure 69: Truss 1 with connection



Figure 70: Reference connection and detail of connection

For the calculation of the connections shown in Figure 71 three failure mechanisms are taken into account (f, g and h). f is the failure of the timber, g is the failure of the bolt together with the timber and h is the failure of the bolt. The minimum value of the force which is leading to failure will be used to detemine the amount of bolts. For now the bolts are assumed to be M24 of class 8.8. This is because of the large normal force in the elements.

$$F_{v,Rk} = \min\begin{cases} f_{h,1,k} t_1 d & \text{(f)} \\ f_{h1,k} t_1 d \left[\sqrt{2 + \frac{4M_{y,Rk}}{f_{h,1,k} d t_1^2}} - 1 \right] + \frac{F_{ax,Rk}}{4} & \text{(g)} \\ 2{,}3\sqrt{M_{y,Rk} f_{h,1,k} d} + \frac{F_{ax,Rk}}{4} & \text{(h)} \end{cases}$$

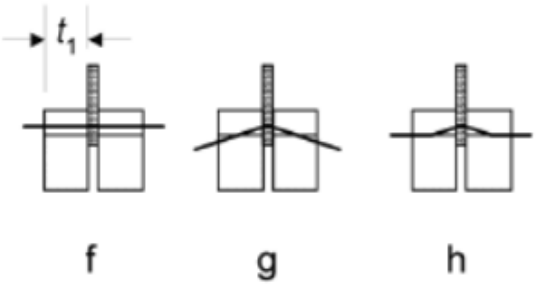


Figure 71: Failure mechanisms

$F_{v,Rk}$ is de karakteristieke sterkte per afschuifvlak per verbindingsmiddel;

$f_{h,k}$ is de karakteristieke stuiksterkte in het houten element;

t_1 is de kleinste waarde van de houtdikte van het element aan de zijkant of de hechtlengte;

t_2 is de kleinste waarde van de houtdikte van het element in het midden of de hechtlengte;

d is de diameter van het verbindingsmiddel;

$M_{y,Rk}$ is het karakteristiek vloeimoment van het verbindingsmiddel;

$F_{ax,Rk}$ is de karakteristieke uittreksterkte van het verbindingsmiddel.

First the failure of the timber will be calculated. For d the diameter is 24 mm (M24 bolts). The density of c24 is 350 kg/mm^2 (NF EN 338). t_1 is 75 mm. It turns out the maximum force in the bolt could be 39.26 kN. The calculation is shown below.

$$F_{v,Rk} = f_{h,1,k} t_1 d$$

$$d = 24\ mm$$

$$p_k = 350\ kg/m^3$$

$$t_1 = 75\ mm$$

$$f_{h,1,k} = 0.082 \cdot (1 - 0.01d) p_k$$

$$f_{h,1,k} = 0.082 \cdot (1 - 0.01 \cdot 24) \cdot 350 = 21.812\ N/mm^2$$

$$f_{h,1,k} t_1 d = 21.812 \cdot 75 \cdot 24 = 39261.6\ N$$

The second failure mechanism is the failure of the timber and the bolts. The properties for the first part of the calculation have the same values as the calculation for the first failure mechanism. For $M_{y,Rk}$, the maximum tension of the bolts is needed which is f_u = 800 N/mm^2. The Below the calculation for this failure mechanism is shown. It turns out the maximum force in the bolt is 38.23 kN.

$$f_{h,1,k} t_1 d \left[\sqrt{2 + \frac{4 M_{y,Rk}}{f_{h,1,k} d t_1^2}} - 1 \right] + \frac{F_{ax,Rk}}{4}$$

$$f_{h,1,k} = 26.404\ N/mm^2$$

$$t_1 = 75\ mm$$

$$d = 24\ mm$$

$$M_{y,Rk} = 0.45 f_u d^{2.6}$$

$$f_u = 800\ N/mm^2$$

$$d = 24\ mm$$

$$M_{y,Rk} = 0.45 \cdot 800 \cdot 24^{2.6} = 1395892\ Nmm$$

$$21.812 \cdot 75 \cdot 24 \left[\sqrt{2 + \frac{4 \cdot 1395892}{21.812 \cdot 24 \cdot 75^2}} - 1 \right] + \frac{0}{4} = 38227.9\ N$$

The last mechanism is based on the failure of the bolts. All values of the needed properties are given before. Therefore the formula below could be filled in. This formula gives the highest maximum force of all three formulas which is 62.17 kN

$$2.3 \sqrt{M_{y,Rk} f_{h,1,k} d} + \frac{F_{ax,Rk}}{4}$$

$$f_{h,1,k} = 26.404\ N/mm^2$$

$$d = 24\ mm$$

$$M_{y,Rk} = 0.45 f_u d^{2.6}$$

$$f_u = 800\ N/mm^2$$

$$M_{y,Rk} = 0.45 \cdot 800 \cdot 24^{2.6} = 1395892\ Nmm$$

$$2.3 \sqrt{1395892 \cdot 21.812 \cdot 24} + \frac{0}{4} = 62173.8\ N$$

It turns out that the force calculated in the second formula (g) is the lowest force (38.23 kN). When looking at the maximum normal force in the verticals in Figure 72 the highest value is 115 kN in tension. Dividing this by the maximum force gives the amount of bolts needed.

$$\frac{115}{38.23} = 3 \text{ bolts}$$

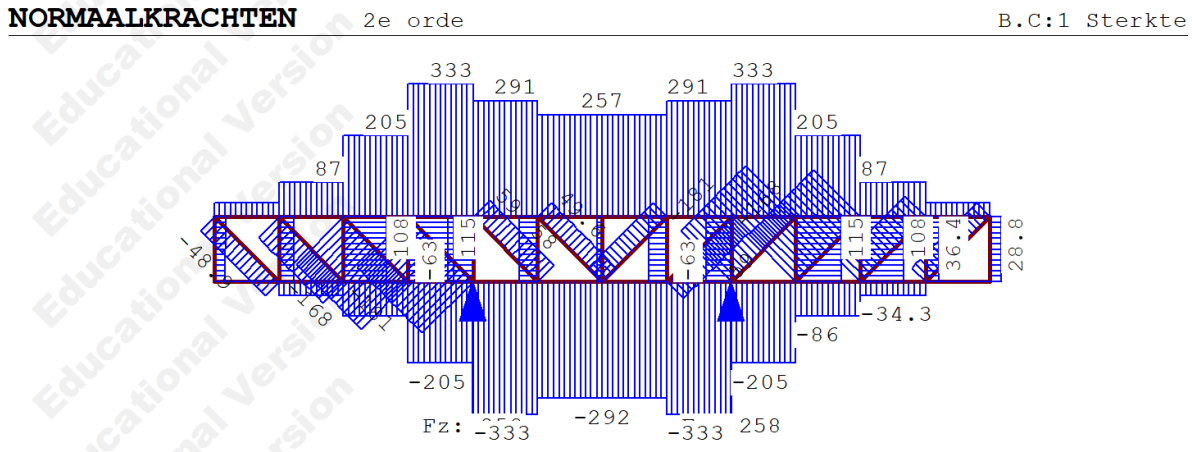


Figure 72: Normal force diagram of truss 1

This means the connection between the horizontal and vertical members should consist of three bolts in both members as seen in Figure 74. The diagonal in the detail is in compression which means the timber should be able to take all compression forces and the connection is only needed to prevent shear. This force is really low as to be seen in Figure 73. This means one M24 bolt is chosen to make sure the diagonal does not come out of the truss sideways.

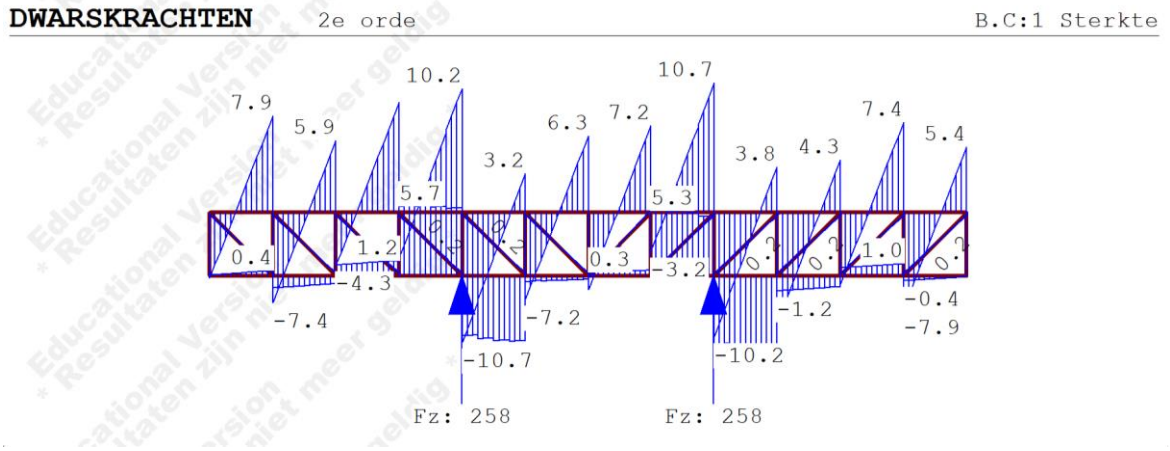


Figure 73: Shear force diagram of truss 1

Now that the bolts are calculated. The steel plate will be calculated. Therefore a twodimensional view is given in Figure 74. In Figure 74 at the right this detail is simplified and focused on the connection between the horizontal and vertical members of the truss. As mentioned before a large tension force is distributed here. The diagonal member is loaded in compression and will be ignored in this calculation.

The maximum tensile force will be checked on bearing of the bolts in the gusset plate. The gusset plate is assumed to be 10 mm in thickness. Therefore the following formula will be used:

$$F_{b,Rd} = \frac{k_1 a_b f_u dt}{y_{M2}}$$

$$k_1 = \frac{2.8e_2}{d_0} - 1.7; \frac{1.4p_2}{d_0} - 1.7; 2.5$$

$$a_b = \frac{e_1}{3d_0}; \frac{f_{ub}}{f_u}; 1.0$$

$$\frac{p_1}{3d_0} - \frac{1}{4}; \frac{f_{ub}}{f_u}; 1.0$$

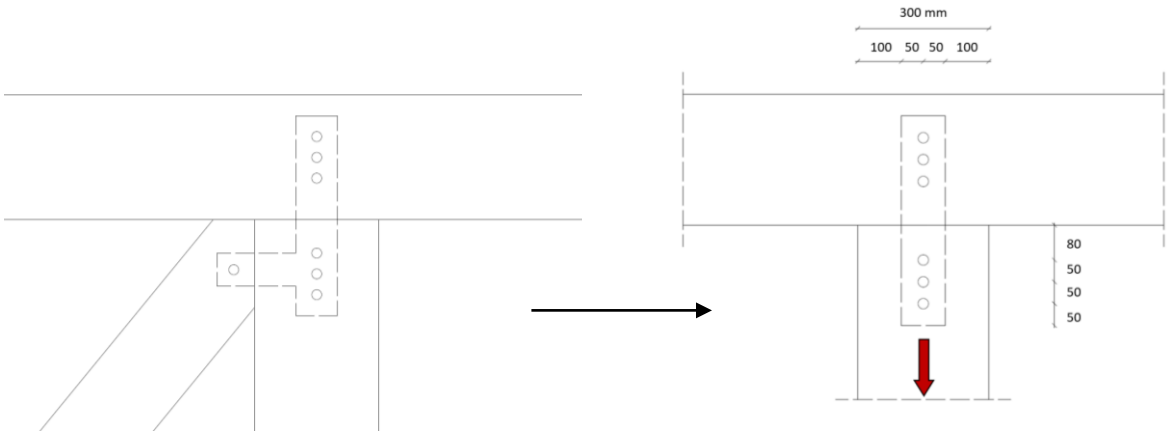


Figure 74: Detail of the connection and simplified detail of the connection

First factor k_1 is calculated. This should be the lowest value of the following three values:

$$k_1 = \frac{2.8e_2}{d_0} - 1.7; \frac{1.4p_2}{d_0} - 1.7; 2.5$$

$$k_1 = \frac{2.8e_{2,gusset\ plate}}{d_0} - 1.7$$

$$d_0 = 26\ mm$$

$$e_{2,gusset\ plate} = 75\ mm$$

$$k_1 = \frac{2.8 \cdot 75}{26} - 1.7 = 6.377$$

p_2 is not relevant because perpendicular to the load there are only edge bolts.

$$p_2 = 0$$

This means the minimum value of k_1 is:

$$k_1 = 2.5$$

Now factor a_b is calculated for the end bolts. The lowest value is taken for the following three values:

$$a_b = \frac{e_1}{3d_0}; \frac{f_{ub}}{f_u}; 1.0$$

$$a_b = \frac{e_1}{3d_0}$$

$$e_1 = 50$$

$$d_0 = 26\,mm$$

$$a_d = \frac{50}{3 \cdot 26} = 0.64$$

$$a_b = \frac{f_{ub}}{f_u} = \frac{800}{360} = 2.22$$

$$a_b = 1.00$$

The lowest value is 0.64 for the end bolts. Which means the maximum force for the end bolt could be calculated. This turns out to be 92.16 kN.

$$F_{b,Rd} = \frac{k_1 a_b f_u dt}{y_{M2}}$$

$$F_{b,Rd} = \frac{2.5 \cdot 0.64 \cdot 360 \cdot 20 \cdot 10 \cdot 10^{-3}}{1.25} = 92.16\,kN$$

Now factor a_b is calculated for the inner bolts. The lowest value is taken for the following three values:

$$a_b = \frac{p_1}{3d_0} - \frac{1}{4}$$

$$p_1 = 50$$

$$a_b = \frac{50}{3 \cdot 26} - \frac{1}{4} = 0.39$$

$$a_b = \frac{f_{ub}}{f_u} = \frac{800}{360} = 2.22$$

$$a_b = 1.00$$

The lowest value is 0.39 for the end bolts. Which means the maximum force for the inner bolts could be calculated. This turns out to be 56.16 kN.

$$F_{b,Rd} = \frac{2.5 \cdot 0.39 \cdot 360 \cdot 20 \cdot 10 \cdot 10^{-3}}{1.25} = 56.16\,kN$$

The maximum force is 56.16 kN. This means that for 3 bolts the maximum force is:

$$56.16 \cdot 3 = 168.48\,kN$$

This force is higher than the normal force which is 115 kN, which means the thickness of the gusset plate of 10mm is considered efficient.

9. Research

Truss 2 which is spanning over the buildings length and which is supporting the other trusses which are placed perpendicular has huge dimensions for normal c24 timber. Glulam would be better in this case. However, to keep costs low and the embodied energy as well, normal c24 timber would be preferred. Also the structure itself could be changed which means spans would get smaller or other materials would be used. A lot of possible solutions are there to face the problem. However the idea of the building would be desired to not change drastically in terms of architecture. In the work environment, architects have to work together with structural advisors. Therefore compromising is important. To zoom in on this problem a research question is determined. The research question goes as follows;

''How can the large-sized truss be modified best without making significant changes to the overall architecture?''

To answer this question will be found in by trying different options. First some general ways to make a truss more efficient will be given.

Optimizing the geometry of a truss would be a logical option for the design of a truss. Adjusting the span, height, and configuration of the truss to find an optimal shape. Therefore software tools for structural analysis and optimization could be used to find the most efficient geometry.

Also the selection of materials is important. Materials with high strength-to-weight ratios are chosen to reduce the weight of the truss without compromising strength. Innovative materials or composites could be considered that offer improved performance. The joint design could be optimized as well. By doing this efficient load transfer and minimal stress concentrations could be realized. Using advanced connection methods that reduce the need for additional elements. To optimize the load path, loads should be efficiently transferred through the truss to minimize bending and shear forces. Consider the arrangement of supports and loads to achieve a more direct load path.

Another way to improve a truss is by incorporating redundancy in the truss design to enhance safety. Redundant members can help distribute loads more evenly, preventing localized failures. Employing advanced analysis methods would also help in improving the efficiency. For example the finite element analysis (FEA) could be used to simulate and optimize the truss's behavior under different conditions. Using modulization and prefabricating elements would ensure the precision of the assembly of the truss.

This last point is already used in the design. The trusses and other elements will be assembled beforehand as much as possible. Optimizing the geometry is a logical solution. For example increasing the height of the truss, would already help in the reduction of the dimensions for each individual member. While the truss is placed in the roof and there is no floor on top of that, the height of the truss does not matter a lot. The only thing is that it creates more space in the building. However, a higher truss would lower the dimensions which gives the truss a lower weight. This is beneficial for the columns which now support less load and could be reduced in dimensions as well.

Changing the materials would not be preferred, because of the weight. When using steel for example, the dimensions of the elements would probably be lower. However the weight of steel is still high which would mean the building is getting too heavy.

Adding an extra member in the design, such as a brace, would help in the truss as it is giving the truss an extra support. This would also help for the redundancy.

Changing the connections of the truss from hinges into fixed connections would not help in the reduction of the normal forces. First of all, the highest normal forces are already in the top and bottom chords which are already connected by fixed connections. Only the diagonals would probably get lower normal forces, because they would not be needed if all connections are fixed.

Reducing the permanent load of the hanging floor structures would also not help enough for the dimensions to change. The permanent loads of the floor are already significantly lower than the variable loads on the floor. When for example taking half of the permanent load of the floor, the first timber floor beam would already not change in dimensions.

It turns out that two of the options are best applicable for the structure to reduce dimensions without changing the architecture drastically. For this research, two different options will be worked out:

1. Add bracing
2. Increase the truss height

1. Add Bracing

Bracing in the building would give the truss an extra support. Placing the braces at the bottom of the building would help, but also change the architecture drastically (see Figure 75). The idea of the side boxes ‘floating’ is now gone. However, placing the braces one elevation higher, would keep the same effect. These braces are now in the ‘floating box’ (Figure 76). The braces are not really user friendly now, because if someone does not watch out, it is possible to bump into one of them. Therefore, a solution could be, to make a small space around the two braces as some sort of threshold between the middle box and the side box.

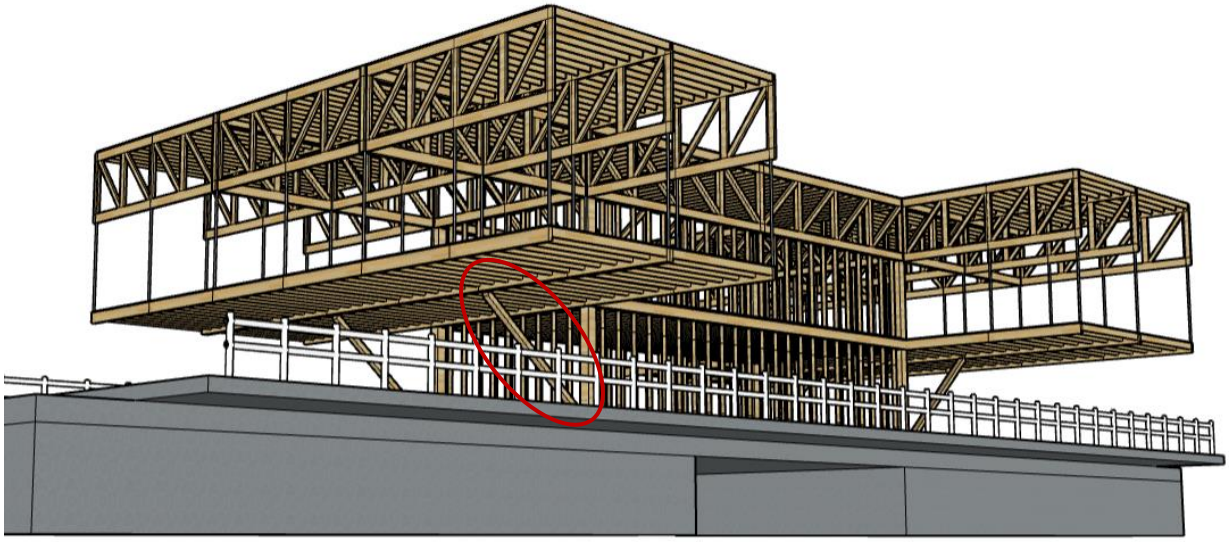


Figure 75: Structure with bracing on bottom

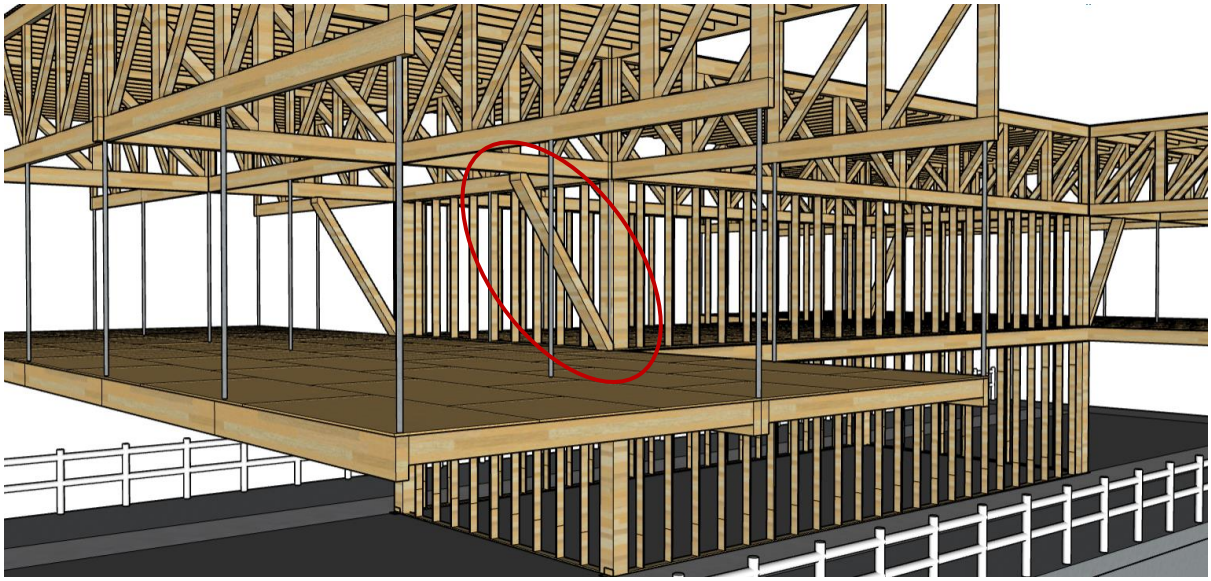


Figure 76: Structure with bracing in middle

Now with the use of Technosoft, the effect of the brace will be checked. In Figure 77 the normal force diagram of the truss is shown. In Figures 110 till 113 in the Appendix, the other diagrams are shown. For the calculation, c24 500 x 300 are used again. It turns out that when this section is used, the brace would fail und buckling. To prevent this from happening, a steel brace would help. Steel has a higher modulus of elasticity, which would allow the section from buckling when a large force is applied. In the new Technosoft calculation, HEA100 is used for the brace. As seen in Figure 78 the normal force is slightly higher in comparison with the timber brace.

The new maximum value of the normal force is used to calculate the new possible section for the timber elements in the truss. In Figure 79 the calculation is shown. As to be seen, a beam with section 300 x 300 from the class c24, would give a Unity Check of 0.86 for the normal forces.

The maximum normal force in the truss is 1.7 times lower when a steel brace is added. The cross section of the timber elements could therefore be 1.7 times lower to keep a decent Unity Check.

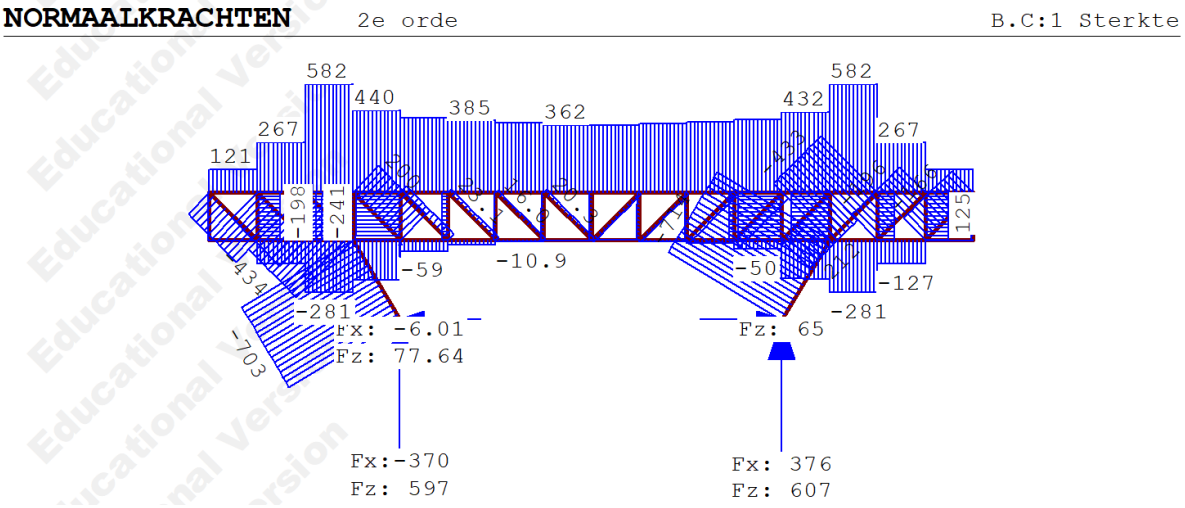


Figure 77: Normal force diagram truss 2 with timber bracing

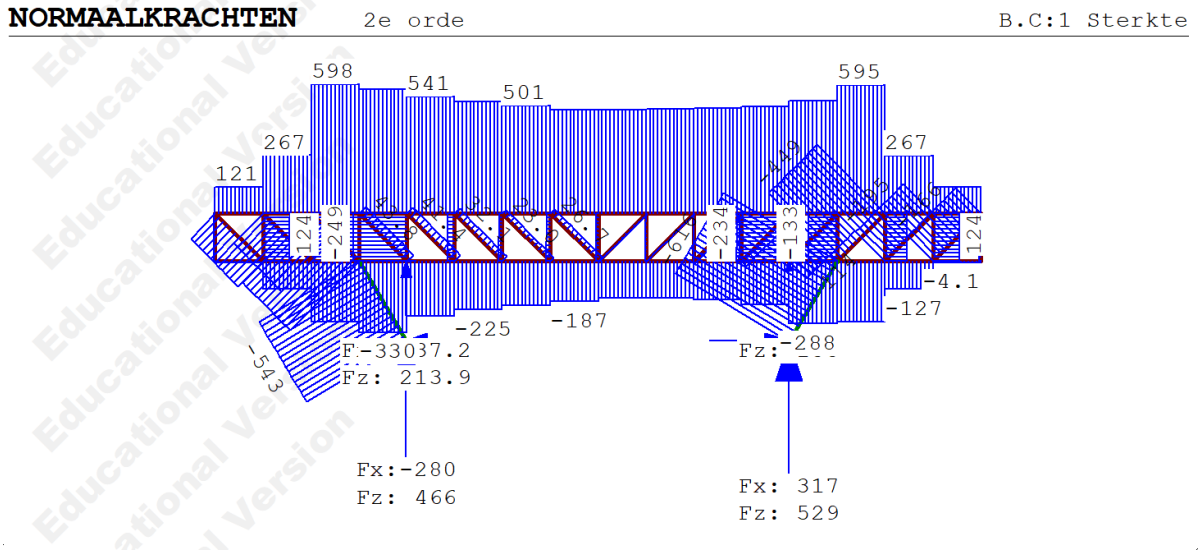


Figure 78: Normal force diagram truss 2 with steel bracing

Truss 2 with bracing

Ultimate Limit State

Quantity	Amount	Unit
$f_{m,k}$	24	N/mm²
$f_{c,0,k}$	21	N/mm²
$f_{t,0,k}$	14,5	N/mm²
h	300	mm
c.t.c.	3,6	m
L	6	m
b	300	mm
$f_{v,k}$	3,4	N/mm²
c.t.c. top a	1,8	m
k_{mod}	0,8	-
γ_m	1,3	-

Service Limit State

Quantity	Amount	Unit
E_{c18}	9000	N/mm²

Technosoft Results		
$M_{technosoft}$	50	kNm
$V_{technosoft}$	42,7	kN
$N_{technosoft}$	598	kN
w_{max}	16,9	mm

Quantity	Amount	Unit
M_d	50	kNm
k_h	0,870551	mm
$f_{m,d}$	12,85736	N/mm²
W_d	3888823	mm³
$b_{W,d}$	259,2548	mm
$\sigma_{m,d}$	11,11111	N/mm²
UC	0,864183	

Quantity	Amount	Unit
w_{rep}	57,6	mm
w_{max}	16,9	mm
UC	0,293403	-

Quantity	Amount	Unit
V_d	42,7	kN
k_h	0,870551	mm
$f_{v,d}$	1,82146	N/mm²
$b_{W,d}$	117,2137	mm
$\sigma_{v,d}$	0,711667	N/mm²
UC	0,390712	-
		-

Quantity	Amount	Unit
N_d	598	kN
k_h	0,870551	mm
$f_{t,d}$	7,76799	N/mm²
$b_{W,d}$	256,6086	
$\sigma_{N,d}$	6,644444	
UC	0,855362	

Figure 79: Hand calculation truss 2 with bracing

2. Increase truss height

The height of the truss is making the roof higher. However, this is not going to change the architectural design drastically. When the height of the truss is reducing the size of the elements, this would reduce the weight on top of the columns, due to a lower self-weight of the truss.

First the truss will be increase height with 11%. This would mean that the truss will have a c.t.c. from the bottom to the top chord of 2.0 meter instead of 1.8 meter. In Technosoft, a new calculation is made with the new height. It turns out that when the truss is higher, buckling arises. In Technosoft, there is a button automatically fix the buckling. With the use of this button, the normal force diagram in Figure 80 is to be seen. Again 500 x 300 mm is used as the section for the elements. Other diagrams are shown in the Appendix.

The maximum moment turns out to be 910 kN, instead of the 1005 kN calculated before. This is 9.5% lower then and gives a Unity Check of 0.86 for the normal forces (Figure 81). This means that the section should not be reduced in dimensions.

Increasing the truss height more, would eventually still change the architectural view to much which is why a higher c.t.c. of the chords is not checked.

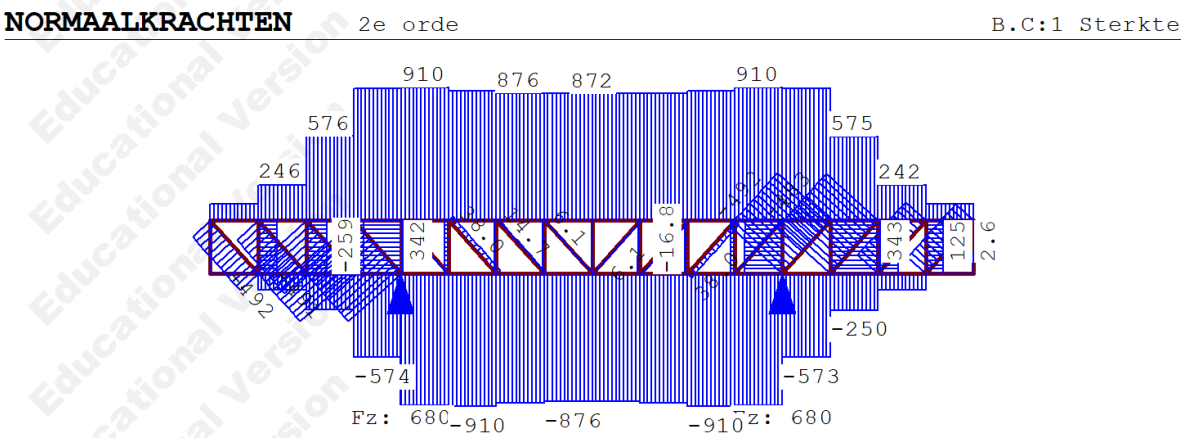


Figure 80: Normal force diagram of truss 2 with c.t.c. of 2 meter

Truss 2 with height of 2.0 meter

Ultimate Limit State

Quantity	Amount	Unit
$f_{m,k}$	24	N/mm²
$f_{c,0,k}$	21	N/mm²
$f_{t,0,k}$	14,5	N/mm²
h	500	mm
c.t.c.	3,6	m
L	6	m
b	300	mm
$f_{v,k}$	3,4	N/mm²
c.t.c. top a	1,8	m
k_{mod}	0,8	-
γ_m	1,3	-

Service Limit State

Quantity	Amount	Unit
E_{c18}	9000	N/mm²

Technosoft Results		
$M_{technosoft}$	55	kNm
$V_{technosoft}$	29,7	kN
$N_{technosoft}$	910	kN
w_{max}	28,3	mm

Quantity	Amount	Unit
M_d	55	kNm
k_h	0,786003	mm
$f_{m,d}$	11,60866	N/mm²
W_d	4737842	mm³
$b_{W,d}$	113,7082	mm
$\sigma_{m,d}$	4,4	N/mm²
UC	0,379027	

Quantity	Amount	Unit
w_{rep}	57,6	mm
w_{max}	28,3	mm
UC	0,491319	-

Quantity	Amount	Unit
V_d	29,7	kN
k_h	0,786003	mm
$f_{v,d}$	1,64456	N/mm²
$b_{W,d}$	54,17862	mm
$\sigma_{v,d}$	0,297	N/mm²
UC	0,180595	-

Quantity	Amount	Unit
N_d	910	kN
k_h	0,786003	mm
$f_{t,d}$	7,013566	N/mm²
$b_{W,d}$	259,4971	
$\sigma_{N,d}$	6,066667	
UC	0,86499	

Figure 81: Hand calculations of truss 2 with c.t.c. of 2 meter

Conclusion of the research

It turns out that the use of braces is the most influencing adjustment of the design. Adding the braces to give an extra support on the first quarter of the cantilever, results in a maximum normal force which is almost half of the maximum normal force of the truss without braces. The braces could be incorporated in the design without changing the design drastically. Using a longer brace which would function as an extra support at the middle of the cantilever would probably reduce the maximum normal force even more. However, this adjustment would change the space in the side box too much.

Increasing the height of the truss has a significant small difference in forces. When increasing the height of the truss with 11% the normal force decreases with 9.5%. However, for a significant larger change in normal forces the truss probably needs to be two times higher. This does give a lot of change in the shape of the building. Next to that, it turned out that buckling of the elements would already be too high for a truss with a c.t.c. of the chords of 2 meter, with the same dimensions used for all the elements (500 x 300).

10. Conclusion

In conclusion, the focus on lightweight construction methods and materials presents a persuasive case for sustainable and efficient building practices. The text emphasizes the positive impact of using lighter materials, such as timber, lightweight aluminum framing, and structural insulated panels, on energy consumption and resource utilization. This approach not only reduces the overall energy footprint but also minimizes the need for extensive resources like concrete, steel, and traditional brick.

Moreover, the discussion on new and advanced building materials, coupled with lightweight technologies, underscores the potential for creating innovative and flexible structures with unique designs. The ability to achieve larger spans due to lower loads of lightweight materials demonstrates the versatility and adaptability of modern construction methods. The specific goal was the application of creating a lightweight construction building on a bridge. The challenges posed by constructing on a bridge, including limited access for cranes and vehicles, are addressed through the adoption of prefabricated elements and lightweight building techniques. The real-world example of designing a building on a bridge over the Dommel at the TU/e campus further illustrates the practical implementation of these principles.

The sections delve into a detailed analysis of the project, covering aspects like environmental considerations, analysis of the current bridge, and the formulation of project requirements. The multifaceted analysis sets the stage for the design process, where the text explores various design concepts based on references from existing structures. The chosen concept, inspired by the KeenSystems headquarters, stands out for its open, large cantilever structure, transparency, and integration with the surrounding environment.

The design and construction of the building presented in this document involve a comprehensive consideration of materialization, load transfer, dimensioning, and stability. The choice of materials, including steel, concrete, and timber, was thoroughly examined, with timber emerging as the preferred material for most of the structure. Timber's lightweight nature, renewability, low carbon emissions, and fire safety qualities align well with the design constraints imposed by the bridge's load capacity. The load transfer analysis was needed for the vertical distribution of forces, leading to the dimensioning of various structural elements. Timber beams, steel cables, roof beams, trusses, and columns were carefully sized to ensure the structure's stability and safety under both permanent and variable loads. Hand calculations were performed, and the results were validated using Technosoft, to result in a proper structural design.

Stability considerations were addressed through two primary principles: the use of portals created by columns and beams and the incorporation of stiff timber frame walls. These principles, along with the structural elements, work collectively to distribute horizontal forces, such as wind, and ensure the stability of the building. The construction process involves prefabrication of elements, mindful of transportation constraints due to the location of the bridge. A reinforced concrete foundation, prefabricated box elements, trusses, cables, and roof beams are assembled on-site to complete the structure.

In summary, the design, dimensioning, and construction processes explained in the report, approach to creating a structurally thought-out and aesthetically pleasing building. The careful selection of materials, load analysis, and stability considerations contribute to a well-rounded design that meets both functional and architectural requirements. The only part left for further investigation is the dimensions of both trusses which are not applicable by using c24 timber.

11. Discussion

In the discussion the obstacles will be described that came up during the process of the design. Different things could be done in more efficient or other way. These things could be taken into account for further investigation.

First of all, the trusses would be better if they have the same height. This way one of the trusses has more height which means smaller dimensions could be used. In terms of space it does not matter because one of the trusses is already high. Also the connections of the trusses would be easier to make with the same height. The reason for the difference in height now is because the connections of the elements should be placed at the nods of the truss. The problem therefore started in the dimensions of the floor beams hanging below the truss. One of them is 3 meters, the other is 3.6 meters. Therefore one of the trusses is divided into squares of 1.5x1.5 and the other truss is divided into squares of 1.8x1.8.

While this would have helped if the trusses were at the same height, there is probably an even better solution. The trusses perpendicular to each other could better be placed on top of each other to make the connections easier, and to cut truss 1 on points with a lower moment. This does give a higher box at the sides, which could in fact lead to an even better architectural view. The boxes at the sides would now stand out even more. In terms of height it does not really matter, because this does not mean that more material is used.

As an architectural choice it would be more spectacular if the building was turned 90°. This way the two floating boxes would hang over the water instead of over the road. It should be kept in mind that there should be a cycle and walk path through the building. Also, the building would stand all over the span of the bridge and therefore not directly on the foundation. The point loads in the column are way too large to stand on the spanning part of the bridge

For the detail of the truss, it is chosen to make a connection with three bolts per element. The reason for this is a reference in which the same principle was applied. However, the manufacturing of this could be difficult because of the large bolts of M24. While most of the truss is made in the factory, it is manageable to use these bolts. However, some connections will be made on side which means it would be easier to have more smaller bolts instead of a few large ones.

For the beams in the middle box, which are spanning 6 meter, the dimensions have to be quite large. This counts for all the floor beams in the box except the ones placed on the foundation. Because of the middle strip the beams are divided in two parts and thus 3 meter each. This would be better for the floors on top as well in terms of extra costs and extra used material used for the 6 meter beams. While the extra middle concrete strip is already placed, a load bearing wall could be placed on top of that. However, the idea was to have open spaces in the middle box as well. Therefore a column beam structure could be placed in the middle which supports the floor beams in the middle.

The last thing is calculation mistake. The factor 2.8 which is taken for the variable loads on the floors in the building is coming from the factor 4.0 for the function ‘bijeenkomstgebouwen’. The factor 0.7 which is coming from Ψ_1, which is also applied in Technosoft. Therefore the c.t.c. distance should be multiplied by 4.0 and not 2.8. This means that for all calculations now the factor 0.7 is applied twice, which means that the loads and therefore the results of the dimensions should be higher. Next time the loads of the façade should be taken into account as well.

12. References

- AllesVoorBouwen. Fermacell Vloerelement 30 mm. 150x50cm Steenwol 10 mm. AllesVoorBouwen. https://allesvoorbouwen.nl/nl_NL/p/Fermacell-Vloerelement-30-mm.-150x50cm-Steenwol-10-mm-2E32-/268#:~:text=Gewicht%3A%2026%20kg%2Fm2
- A. Ghavami, "Steel and Composite Structures: Behaviour and Design for Fire Safety," CRC Press, 2002.
- American Institute of Steel Construction (AISC), "Steel Construction Manual," 15th Edition, 2017.
- Harris, C., & McCrone, E. (2006). "Sustainable Construction." Taylor & Francis.
- H. R. Trebilcock, "Steel Structures: Practical Design Studies," Wiley, 1998.
- Houthandel Van Deuveren. (Year, Month Day). OSB 3 Plaat 18mm 122x244cm. Houthandel Van Deuveren. https://www.houthandelvandeuveren.nl/a-49415264/osb-3-plaat/osb-3-plaat-18mm-122x244cm/#:~:text=Het%20gewicht%20van%20een%20OSB,18mm%20OSB%20%2B%2F%2D%2010kg%20weegt.
- Neville, A. M. (2011). "Properties of Concrete." Pearson Education.
- NEN-EN 1991: Eurocode 1, Belastingen op constructies
- NEN-EN 1995: Eurocode 5, Ontwerp en berekening van houtconstructies
- Mehta, P. K., & Monteiro, P. J. M. (2006). "Concrete: Microstructure, Properties, and Materials." McGraw-Hill Education
- Bhat, I. K., & Hall, M. R. (2013). "Timber Construction Manual." Wiley.
- Saint-Gobain Tape Solutions. (2022, August 15). How Lightweight Solutions Can Change the Construction Industry. Sponsored by Saint-Gobain Tape Solutions. Retrieved January 10, 2024, from: https://www.azom.com/article.aspx?ArticleID=21605
- Schierle, C. (2002). "Aluminum Window Frames: A Comprehensive Review of Their Characteristics, Performance, and Applications." Journal of Architectural Engineering, ASCE.
- Wikipedia contributors. (2022, July). Bridge. Wikipedia.

Appendix

Floor beam 1

Floor beam 1

Variable loads/live loads	**kN/m²**		Ultimate Limit State									
Roof	1,00		**Quantity**	**Amount**	**Unit**	**Quantity**	**Amount**	**Unit**	**Quantity**	**Amount**	**Unit**
Snow	0,70		$f_{m,k}$	18	N/mm²	M_d	4,08701919	kNm	V_d	4,54113	kN
Wind	0,46		k_{mod}	0,8	-	k_h	0,944087511	mm	k_h	0,94409	mm
Water	-		γ_m	1,3	-	$f_{m,d}$	10,45758474	N/mm²	$f_{v,d}$	1,97532	N/mm²
Floor (exhibition function)	2,80		h	200	mm	W_d	390818,6537	mm³	$b_{W,d}$	17,242	mm
			c.t.c.	0,488	m	$b_{W,d}$	58,62279806	mm	$\sigma_{m,d}$	0,54061	N/mm²
Material masses	**kg/m³**		L	3,6	m	$\sigma_{m,d}$	9,730998071	N/mm²	UC	0,27368	-
Timber C24	460		b	63	mm	UC	0,930520604	-			
Steel	7850		$f_{v,k}$	3,4	N/mm²						
Permanent loads/dead loads floor	**kg/m²**	**kN/m²**	Service Limit State								
Floor finishing	15,00	0,14715	**Quantity**	**Amount**	**Unit**						
Estrich Fermacell floor	26,00	0,25506	E_{c18}	9000	N/mm²						
OSB	10,00	0,09810									
Rockwool	-	-	**Quantity**	**Amount**	**Unit**						
Battens	-	-	M_r	3,49134665	kNm						
Aluminum finishing	16,38	0,16069	I_{rep}	36368194,27	mm⁴						
Timber beams	15,00	0,14715	$b_{I,rep}$	54,5522914	mm						
Total	82,38	0,80815	w_{rep}	12,46909518	mm						
			UC	0,865909387	-						

Figure 82: Hand calculation of floor beam 1

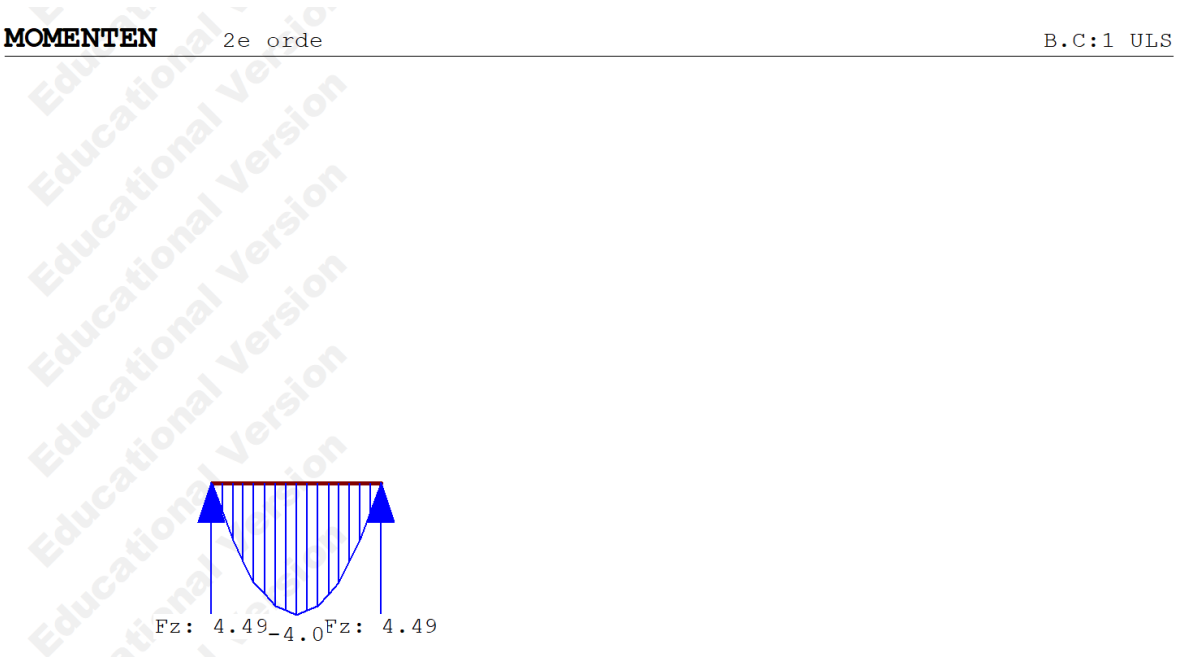


Figure 83: Moment diagram of floor beam 1

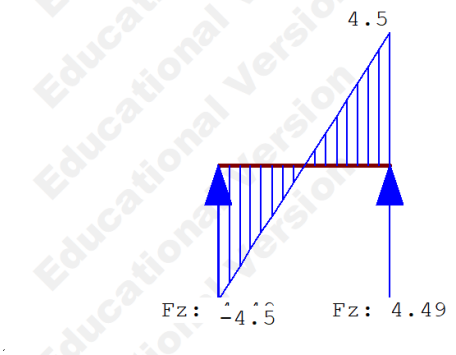


Figure 84: Shear force diagram of floor beam 1

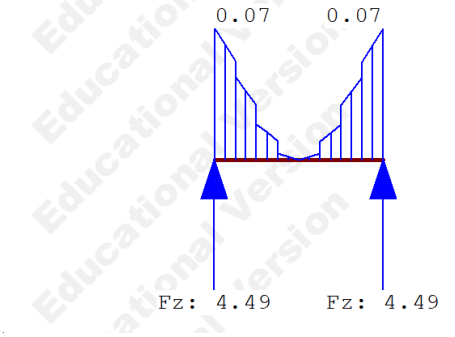


Figure 85: Normal force diagram of floor beam 1

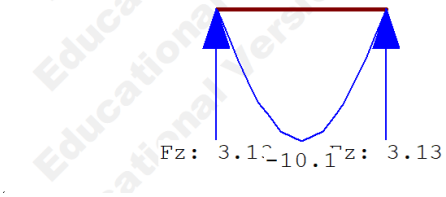


Figure 86: Deflection diagram of floor beam 1

Floor beam 2

Floor beam 2

Variable loads/live loads	**kN/m²**		$V_{ownweight}$	0,12647		Ultimate Limit State								
Roof	1,00		$V_{d,permanent}$	0,85185		**Quantity**	**Amount**	**Unit**	**Quantity**	**Amount**	**Unit**	**Quantity**	**Amount**	**Unit**
Snow	0,70		$V_{d,variable}$	3,68928		$f_{m,k}$	24	N/mm²	M_d	14,7513216	kNm	V_d	19,6684	kN
Wind	0,46			4,54113		k_{mod}	0,8	-	k_h	0,87348178	mm	k_h	0,87348	mm
Water	-					γ_m	1,3	-	$f_{m,d}$	12,9006539	N/mm²	$f_{v,d}$	1,82759	N/mm²
Floor (exhibition function)	2,80		$q_{permanent}$	1,7037	kN/m	h	295	mm	W_d	1143455,34	mm³	$b_{W,d}$	54,7217	mm
			$q_{variable}$	7,37856	kN/m	c.t.c.	0,488	m	$b_{W,d}$	78,8363346	mm	$\sigma_{m,d}$	1,05273	N/mm²
Material masses	**kg/m³**					L	3	m	$\sigma_{m,d}$	10,705687	N/mm²	UC	0,57602	-
Timber C24	460		L	3,6		b	95	mm	UC	0,82985615	-			
Steel	7850					$f_{v,k}$	3,4	N/mm²						
Permanent loads/dead loads floor	**kg/m²**	**kN/m²**				Service Limit State								
Floor finishing	15,00	0,14715				**Quantity**	**Amount**	**Unit**	**Quantity**	**Amount**	**Unit**			
Estrich Fermacell floor	26,00	0,25506				E_{c18}	11000	N/mm²	M_r	12,1342159	kNm			
OSB	10,00	0,09810							I_{rep}	86180511	mm⁴			
Rockwool	-	-							$b_{I,rep}$	40,2832279	mm			
Battens	-	-							w_{rep}	5,08840773	mm			
Aluminum finishing	16,38	0,16069							UC	0,42403398	-			
Timber beams	15,00	0,14715												
Total	82,38	0,80815												

Figure 87: Hand calculations of floor beam 2

MOMENTEN 2e orde B.C:1 ULS

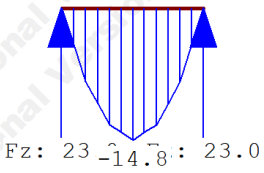


Figure 88: Moment diagram of floor beam 2

DWARSKRACHTEN 2e orde B.C:1 ULS

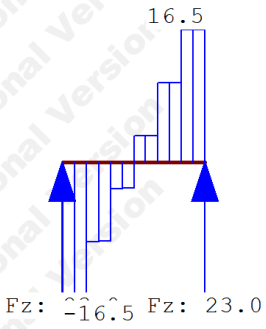


Figure 89: Shear force diagram of floor beam 2



Figure 90: Normal force diagram of floor beam 2



Figure 91: Deflection diagram of floor beam 2

Steel cable

Steel cable

Variable loads/live loads	**kN/m²**		**Floorbeams**			**Perpindicular beams**		
Roof	1,00							
Snow	0,70		h	0,2	m	h	0,3	m
Wind	0,46		b	0,063	m	b	0,1	m
Water	-		L	3,6	m	L	3	m
Floor (exhibition function)	2,80		n	7		n	2	
Material masses	**kg/m³**		146,0592	kg		82,8	kg	
Timber C24	460							
Steel s235	7850		**Total weight of beams**			**Total weight for one steel cable**		
Permanent loads/dead loads floor	**kg/m²**	**kN/m²**	2,245108752	kN		39,623885	kN	
Floor finishing	15,00	0,14715						
Estrich Fermacell floor	26,00	0,25506	**Total weight of loads**			**Steel s235**		
OSB	10,00	0,09810						
Rockwool	-	-	37,37878	kN		235	N/mm²	
Battens	-	-						
Aluminum finishing	16,38	0,160688	168,6122766	mm²		r	7,326046	
Timber beams	0,00	0				d	14,65209	
Total	67,38	0,660998						

Figure 92: Hand calculations steel cable

Roof beam

Roof beam

Variable loads/live loads	**kN/m²**		Ultimate Limit State									
Roof	1,00		**Quantity**	**Amount**	**Unit**		**Quantity**	**Amount**	**Unit**		**Quantity**	**Amount**	**Unit**
Snow	0,70		$f_{m,k}$	18	N/mm²		M_d	1,90597	kNm		V_d	2,11775	kN
Wind	0,46		k_{mod}	0,8	-		k_h	0,96964	mm		k_h	0,96964	mm
Water	-		γ_m	1,3	-		$f_{m,d}$	10,7406	N/mm²		$f_{v,d}$	2,02879	N/mm²
Floor (exhibition function)	2,80		h	175	mm		W_d	177455	mm³		$b_{W,d}$	8,94729	mm
			c.t.c.	0,488	m		$b_{W,d}$	34,7666	mm		$\sigma_{m,d}$	0,36304	N/mm²
Material masses	**kg/m³**		L	3,6	m		$\sigma_{m,d}$	7,46831	N/mm²		UC	0,17895	-
Timber C24	460		b	50	mm		UC	0,69533	-				
Steel	7850		$f_{v,k}$	3,4	N/mm²								
Permanent loads/dead loads floor	**kg/m²**	**kN/m²**	Service Limit State									
Floor finishing	15,00	0,14715	**Quantity**	**Amount**	**Unit**							
Estrich Fermacell floor	26,00	0,25506	E_{c18}	9000	N/mm²							
OSB	10,00	0,09810										
Rockwool	-	-	**Quantity**	**Amount**	**Unit**							
Battens	-	-	M_r	1,9907847	kNm							
Aluminum finishing	16,38	0,16069	I_{rep}	20737341	mm⁴							
Timber beams	10,00	0,0981	$b_{I,rep}$	46,432297	mm							
Total	77,38	0,7591	w_{rep}	13,372501	mm							
			UC	0,9286459	-							

Figure 93: Hand calculations roof beam

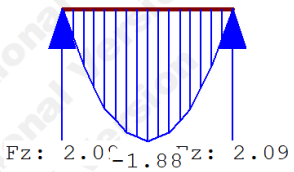


Figure 94: Moment diagram roof beam

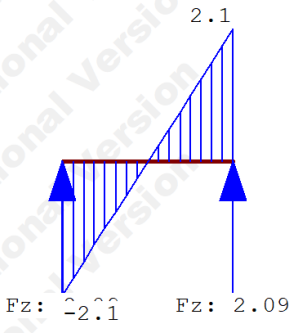


Figure 95: Shear force diagram roof beam

NORMAALKRACHTEN 2e orde B.C:1 ULS

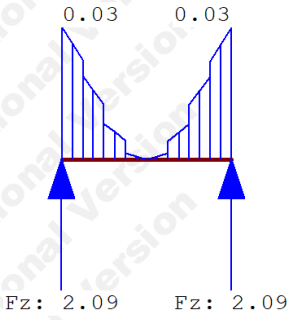


Figure 96: Normal force diagram roof beam

VERPLAATSINGEN 1e orde [mm] B.C:2 SLS

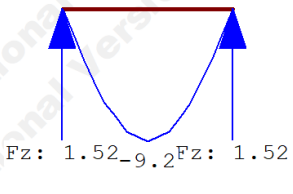


Figure 97: Deflection diagram roof beam

Truss 1

Truss 1

Variable loads/live loads	kN/m²	
Roof	1,00	
Snow	0,70	
Wind	0,46	
Water	-	
Floor (exhibition function)	2,80	
Material masses	**kg/m³**	
Timber C24	460	
Steel	7850	
Permanent loads/dead loads floor	**kg/m²**	**kN/m²**
Floor finishing	15,00	0,14715
Estrich Fermacell floor	26,00	0,25506
OSB	10,00	0,09810
Rockwool	-	-
Battens	-	-
Aluminum finishing	16,38	0,16069
Timber beams	15,00	0,14715
Timber perpencular beams	7,500	0,07358
Total	82,38	0,88172
Weight steel cable	**kg**	**kN**
	12	0,11772

Permanent loads/dead loads floor	kg/m²	kN/m²
Floor finishing	15,00	0,14715
Estrich Fermacell floor	26,00	0,25506
OSB	10,00	0,09810
Rockwool	-	-
Battens	-	-
Aluminum finishing	16,38	0,16069
Timber beams	15,00	0,14715
Timber perpencular beams	7,500	0,07358
Total	82,38	0,88172
Weight steel cable	**kg**	**kN**
	12	0,11772

Permanent loads/dead loads floor	kg/m²	kN/m²
Floor finishing	15,00	0,14715
Estrich Fermacell floor	26,00	0,25506
OSB	10,00	0,09810
Rockwool	-	-
Battens	-	-
Aluminum finishing	16,38	0,16069
Timber beams	10,00	0,0981
Total	77,38	0,7591
Loads on truss		
Variable point load	**kN**	
Middle point loads	30,24	
Side point loads	15,12	
Average loads	25,92	
Permanent point load	**kN**	
Middle point loads	9,64033	
Side point loads	4,87902	
Average loads	8,27995	
Distributed load	**kN/m**	
Variable	3,60	
Permanent	2,73275	
Total	**kN/m**	
Total variable load	12,24	
Total permanent load	5,99274	

Ultimate Limit State		
Quantity	**Amount**	**Unit**
$f_{m,k}$	24	N/mm²
$f_{c,0,k}$	21	-
$f_{t,0,k}$	14,5	-
h	300	mm
c.t.c.	3,6	m
L	6	m
b	150	mm
$f_{v,k}$	3,4	N/mm²
c.t.c. top and bottom chord	1,5	
k_{mod}	0,8	
γ_m	1,3	
Service Limit State		
Quantity	**Amount**	**Unit**
$E_{0,05}$	9000	N/mm²
Hand calculation		
M_d	459,9231	kNm
Axial forces	306,6154	
$A_{bottom\ chord}$	14600,73	mm²
$A_{top\ chord}$	21145,89	mm²
V_d	153,3077	kN
N	216,8098	kN
$A_{diagonal}$	10324,28	mm²

Technosoft Results		
$M_{technosoft}$	9,3	kNm
$V_{technosoft}$	12	kN
$N_{technosoft}$	333	kN
w_{max}	19,2	mm
Quantity	**Amount**	**Unit**
M_d	9,3	kNm
k_h	0,87055	mm
$f_{m,d}$	12,8574	N/mm²
W_d	723321	mm³
$b_{W,d}$	48,2214	mm
$\sigma_{m,d}$	4,13333	N/mm²
Quantity	**Amount**	**Unit**
w_{top}	48	mm
w_{max}	19,2	mm
UC	0,4	-

Quantity	Amount	Unit
V_d	12	kN
k_h	0,87055	mm
$f_{v,d}$	1,82146	N/mm²
$b_{W,d}$	32,9406	mm
$\sigma_{v,d}$	0,4	N/mm²
UC	0,2196	-
UC	0,32148	-

Quantity	Amount	Unit
N_d	333	kN
k_h	0,870550563	mm
$f_{c,d}$	7,767989642	N/mm²
$b_{W,d}$	142,8941143	
$\sigma_{N,d}$	7,4	
UC	0,952627429	

Figure 98: Hand calculations truss 1

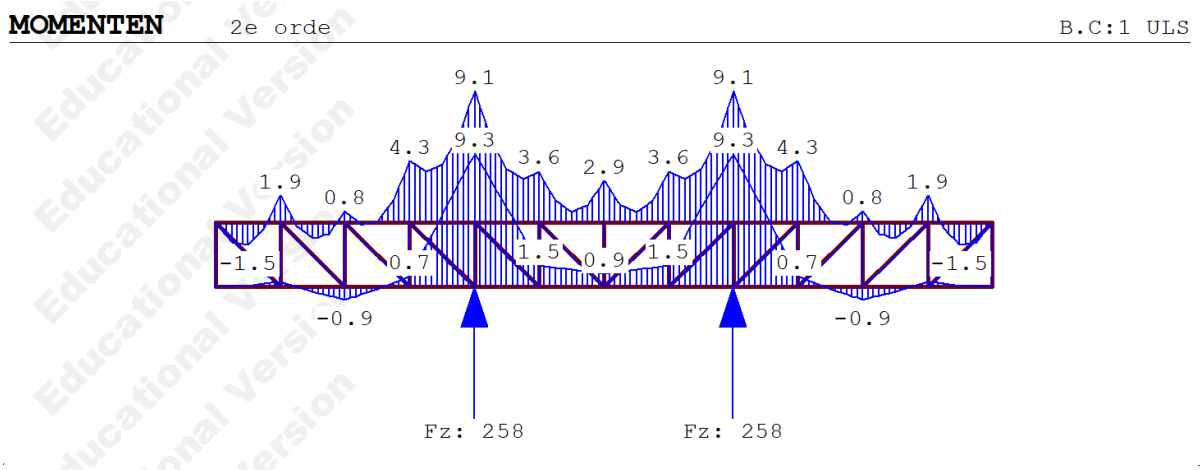


Figure 99: Moment diagram truss 1

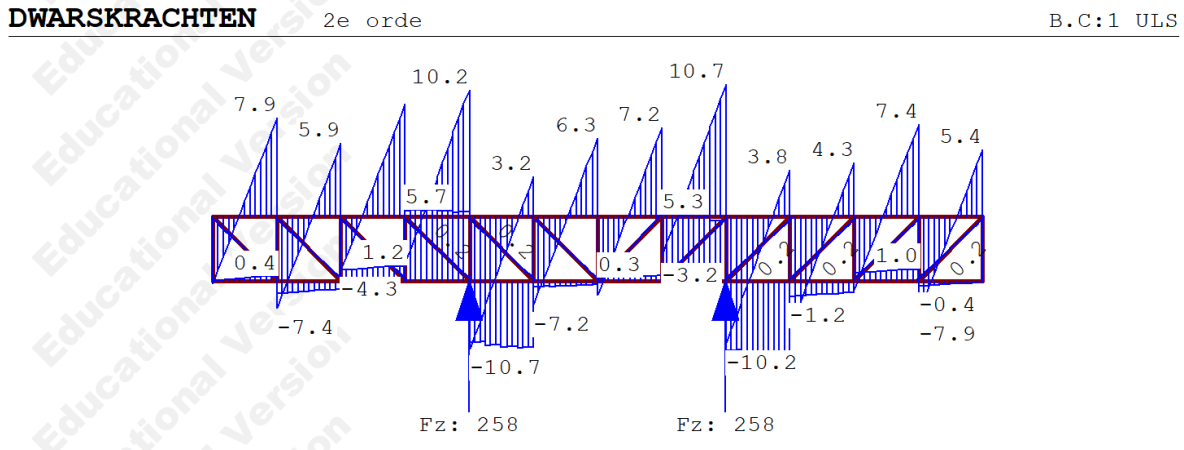


Figure 100: Shear force diagram truss 1

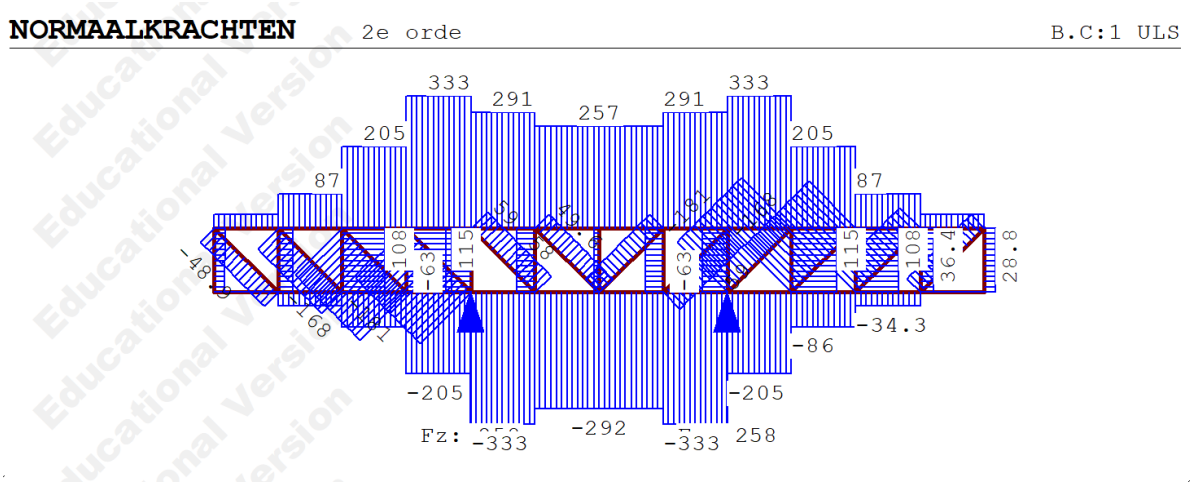


Figure 101: Normal force diagram truss 1

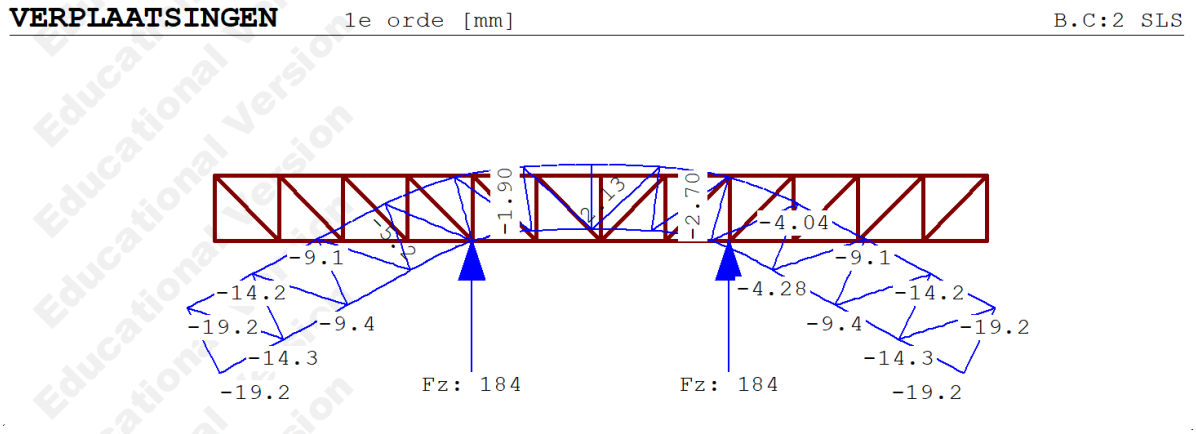


Figure 102: Deflection diagram truss 1

Truss 2

Truss 2

Variable loads/live loads	kN/m²	
Roof	1,00	
Snow	0,70	
Wind	0,46	
Water	-	
Floor (exhibition function)	2,80	

Material masses	kg/m³
Timber C24	460
Steel	7850

Permanent loads/dead loads floor	kg/m²	kN/m²
Floor finishing	15,00	0,14715
Estrich Fermacell floor	26,00	0,25506
OSB	10,00	0,09810
Rockwool	-	-
Battens	-	-
Aluminum finishing	16,38	0,16069
Timber beams	15,00	0,14715
Timber perpencular beams	7,500	0,07358
Total	82,38	0,88172

Weight steel cable	kg	kN
	12	0,11772

Permanent loads/dead loads floor	kg/m²	kN/m²
Floor finishing	15,00	0,14715
Estrich Fermacell floor	26,00	0,25506
OSB	10,00	0,09810
Rockwool	-	-
Battens	-	-
Aluminum finishing	16,38	0,16069
Timber beams	10,00	0,0981
Total	77,38	0,7591

Loads on truss	
Variable point load	**kN**
Middle point loads	30,24
Side point loads	15,12
Average loads	25,92
Permanent point load	**kN**
Middle point loads	9,64033
Side point loads	4,87902
Average loads	8,27995
Distributed load	**kN/m**
Variable	3,60
Permanent	2,73275
Total	**kN/m**
Total variable load	12,24
Total permanent load	5,99274

Ultimate Limit State		
Quantity	**Amount**	**Unit**
$f_{m,k}$	24	N/mm²
$f_{c,0,k}$	21	-
$f_{t,0,k}$	14,5	-
h	500	mm
c.t.c.	3,6	m
L	6	m
b	300	mm
$f_{v,k}$	3,4	N/mm²
c.t.c. top and bottom chord	1,5	
k_{mod}	0,8	
γ_m	1,3	

Service Limit State		
Quantity	**Amount**	**Unit**
$E_{c,0}$	9000	N/mm²

Point load 3 meter from support	kN
Variable point load	123,12
Permanent point load	53,57461
Point load 6 meter from support	**kN**
Variable point load	61,56
Permanent point load	26,7873

Technosoft Results		
$M_{technosoft}$	21,7	kNm
$V_{technosoft}$	11,7	kN
$N_{technosoft}$	1012	kN
w_{max}	19,2	mm

Quantity	Amount	Unit
M_d	21,7	kNm
k_h	0,786	mm
$f_{m,d}$	11,6087	N/mm²
W_d	1869294	mm³
$b_{W,d}$	44,8631	mm
$\sigma_{m,d}$	1,736	N/mm²
UC	0,14954	

Quantity	Amount	Unit
V_d	11,7	kN
k_h	0,786	mm
$f_{v,d}$	1,64456	N/mm²
$b_{W,d}$	21,3431	mm
$\sigma_{v,d}$	0,117	N/mm²
UC	0,07114	-

Quantity	Amount	Unit
N_d	1012	kN
k_h	0,786	mm
$f_{t,d}$	7,01357	N/mm²
$b_{W,d}$	288,584	
$\sigma_{N,d}$	6,74667	
UC	0,96195	

Quantity	Amount	Unit
w_{inst}	57,6	mm
w_{max}	57	mm
UC	0,98958	-

Figure 103: Hand calculations truss 2

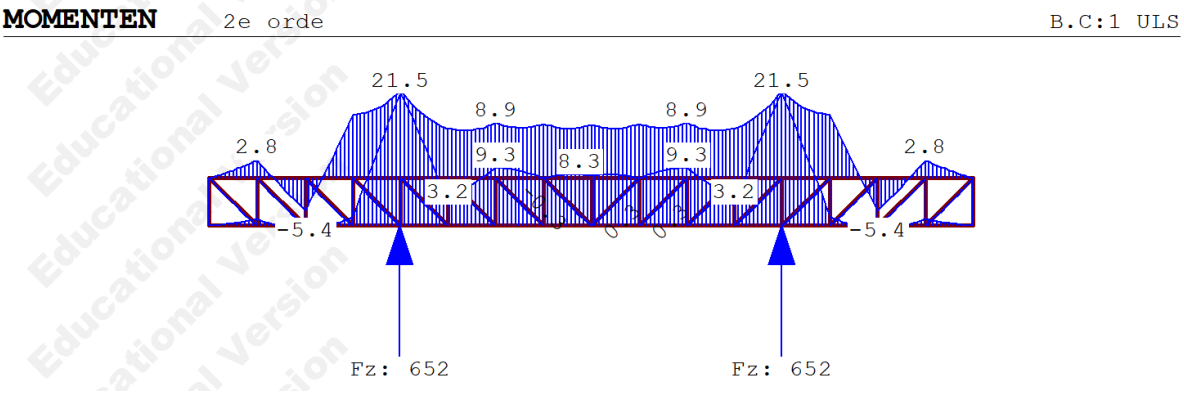


Figure 104: Moment diagram truss 2

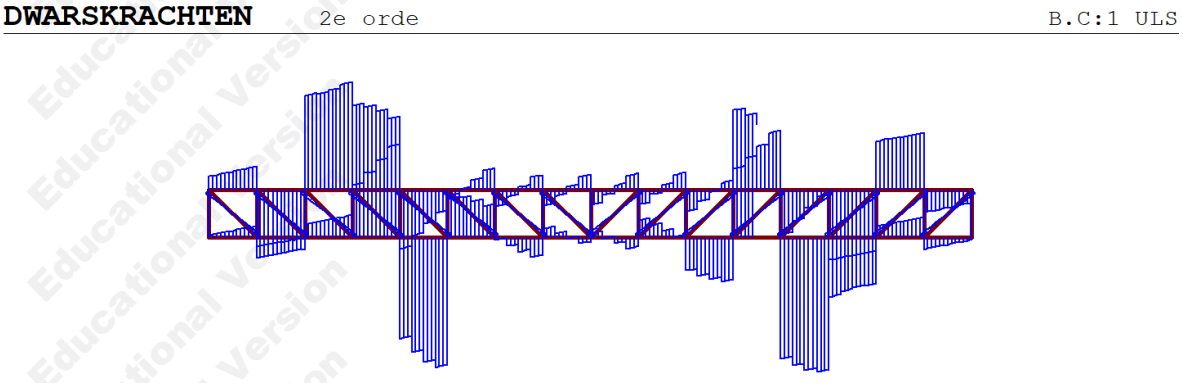


Figure 105: Shear force diagram truss 2

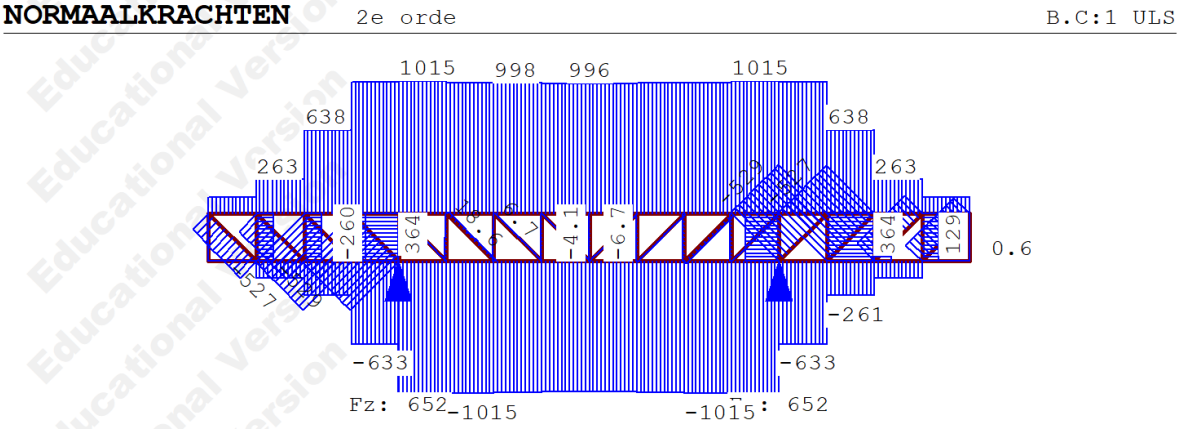


Figure 106: Normal force diagram truss 2

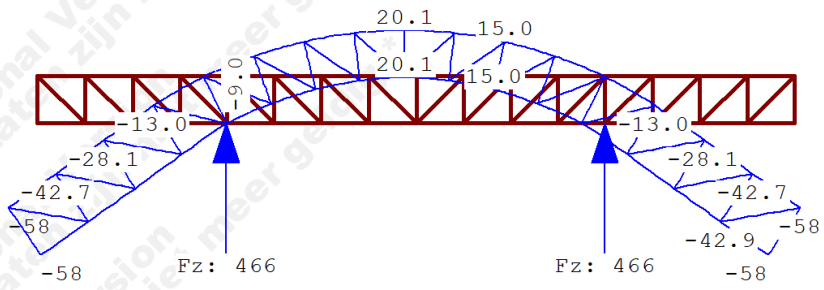


Figure 107: Deflection diagram truss 2

Column

Column						
Variable loads/live loads	**kN/m²**			**Reaction force Technosoft**		
Roof	1,00			R	659000	N
Snow	0,70					
Wind	0,46			**Compressive force timber**		
Water	-			$f_{c,0,k}$	21	N/mm²
Floor (exhibition function)	2,80					
				Area of cross section		
Material masses	**kg/m³**			A	31380,95	
Timber C24	460					
Steel s235	7850					
Permanent loads/dead loads floor	**kg/m²**	**kN/m²**				
Floor finishing	15,00	0,14715				
Estrich Fermacell floor	26,00	0,25506				
OSB	10,00	0,09810				
Rockwool	-	-				
Battens	-	-				
Aluminum finishing	16,38	0,160688				
Timber beams	0,00	0				
Total	67,38	0,660998				

Figure 108:Hand calculations column

Truss with bracing

Truss 2 with bracing

Ultimate Limit State

Quantity	Amount	Unit
$f_{m,k}$	24	N/mm²
$f_{c,0,k}$	21	N/mm²
$f_{t,0,k}$	14,5	N/mm²
h	300	mm
c.t.c.	3,6	m
L	6	m
b	300	mm
$f_{v,k}$	3,4	N/mm²
c.t.c. top a	1,8	m
k_{mod}	0,8	-
γ_m	1,3	-

Service Limit State

Quantity	Amount	Unit
E_{c18}	9000	N/mm²

Technosoft Results		
$M_{technosoft}$	50	kNm
$V_{technosoft}$	42,7	kN
$N_{technosoft}$	600	kN
w_{max}	16,9	mm

Quantity	Amount	Unit
M_d	50	kNm
k_h	0,870551	mm
$f_{m,d}$	12,85736	N/mm²
W_d	3888823	mm³
$b_{W,d}$	259,2548	mm
$\sigma_{m,d}$	11,11111	N/mm²
UC	0,864183	

Quantity	Amount	Unit
w_{rep}	57,6	mm
w_{max}	16,9	mm
UC	0,293403	-

Quantity	Amount	Unit
V_d	42,7	kN
k_h	0,870551	mm
$f_{v,d}$	1,82146	N/mm²
$b_{W,d}$	117,2137	mm
$\sigma_{v,d}$	0,711667	N/mm²
UC	0,390712	-

Quantity	Amount	Unit
N_d	600	kN
k_h	0,870551	mm
$f_{t,d}$	7,76799	N/mm²
$b_{W,d}$	257,4669	
$\sigma_{N,d}$	6,666667	
UC	0,858223	

Figure 109: Hand calculations truss with bracing

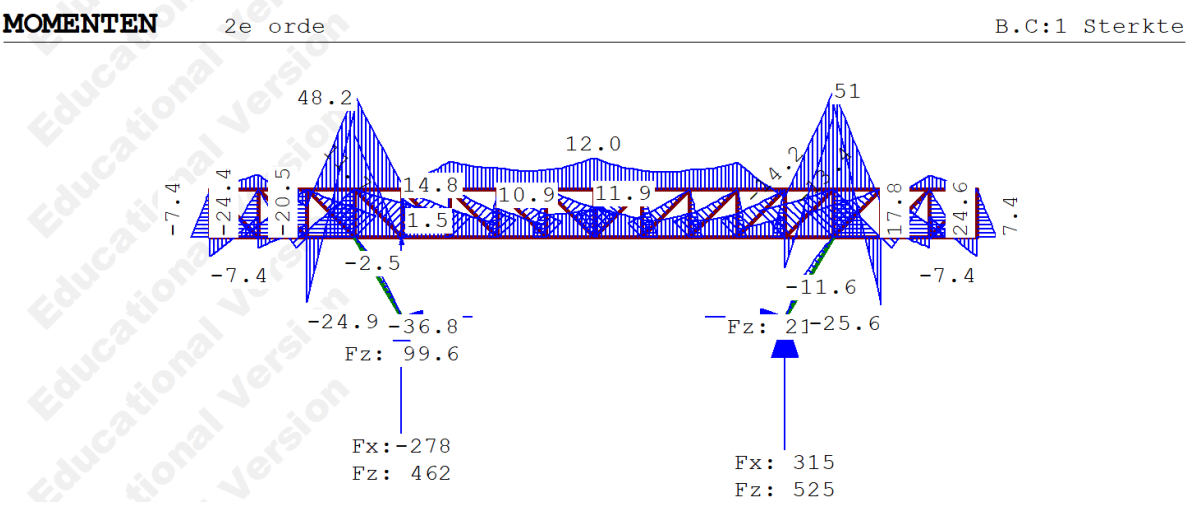


Figure 110: Moment diagram truss with bracing

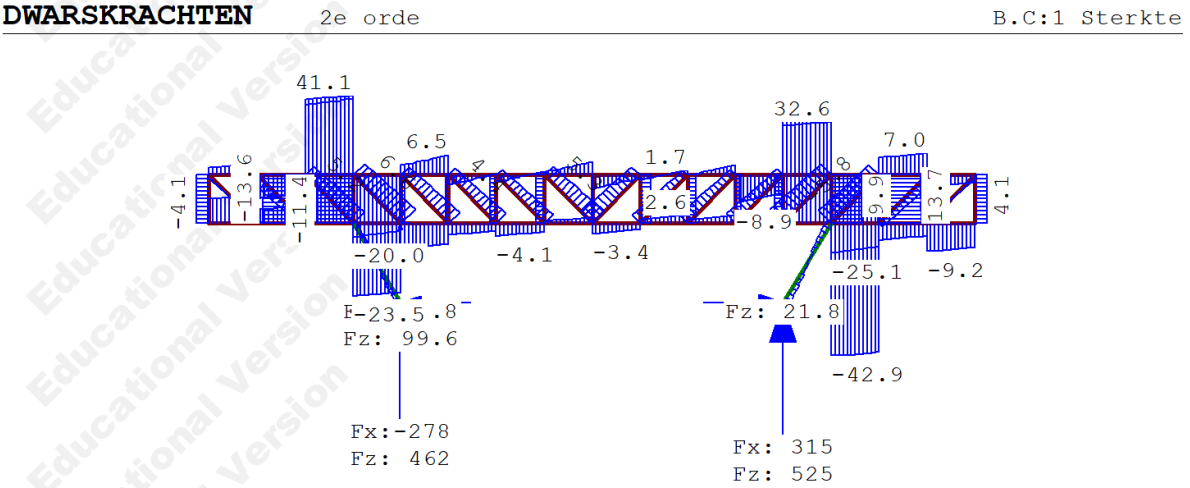


Figure 111: Shear force diagram truss with bracing

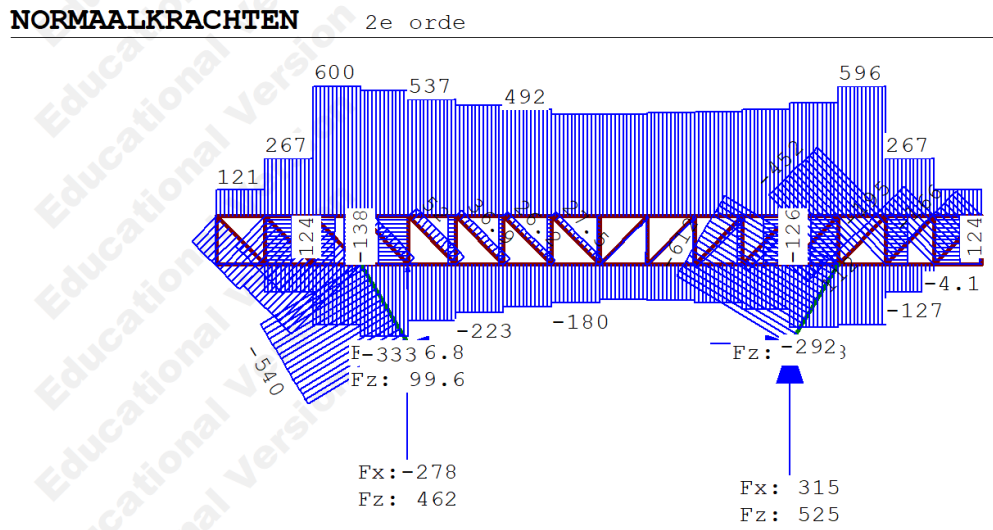


Figure 112: Normal force diagram truss with bracing

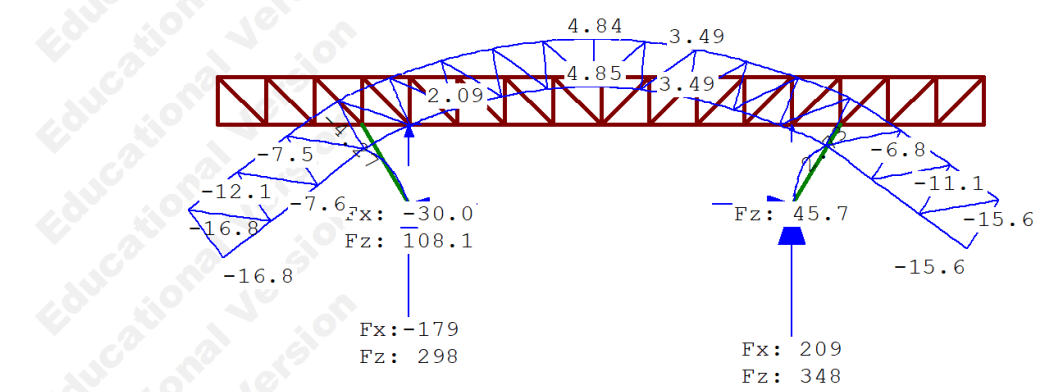


Figure 113: Deflection diagram truss with bracing

Truss with c.t.c. distance of top and bottom chord equal to 2.0 meter

Truss 2 with height of 2.0 meter

Ultimate Limit State

Quantity	Amount	Unit
$f_{m,k}$	24	N/mm^2
$f_{c,0,k}$	21	N/mm^2
$f_{t,0,k}$	14,5	N/mm^2
h	500	mm
c.t.c.	3,6	m
L	6	m
b	300	mm
$f_{v,k}$	3,4	N/mm^2
c.t.c. top a	1,8	m
k_{mod}	0,8	-
γ_m	1,3	-

Service Limit State

Quantity	Amount	Unit
E_{c18}	9000	N/mm^2

Technosoft Results		
$M_{technosoft}$	55	kNm
$V_{technosoft}$	29,7	kN
$N_{technosoft}$	910	kN
w_{max}	28,3	mm

Quantity	Amount	Unit
M_d	55	kNm
k_h	0,786003	mm
$f_{m,d}$	11,60866	N/mm^2
W_d	4737842	mm^3
$b_{W,d}$	113,7082	mm
$\sigma_{m,d}$	4,4	N/mm^2
UC	0,379027	

Quantity	Amount	Unit
w_{rep}	57,6	mm
w_{max}	28,3	mm
UC	0,491319	-

Quantity	Amount	Unit
V_d	29,7	kN
k_h	0,786003	mm
$f_{v,d}$	1,64456	N/mm^2
$b_{W,d}$	54,17862	mm
$\sigma_{v,d}$	0,297	N/mm^2
UC	0,180595	-

Quantity	Amount	Unit
N_d	910	kN
k_h	0,786003	mm
$f_{t,d}$	7,013566	N/mm^2
$b_{W,d}$	259,4971	
$\sigma_{N,d}$	6,066667	
UC	0,86499	

Figure 114: Hand calculations truss with height of 2.0 meter

MOMENTEN 2e orde B.C:1 Sterkte

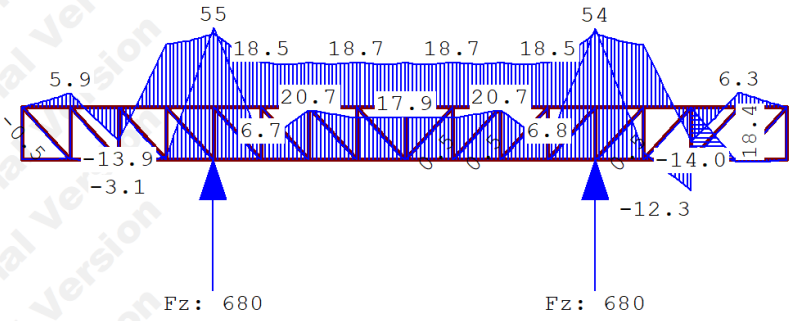


Figure 115: Moment diagram truss with height of 2.0 meter

DWARSKRACHTEN 2e orde B.C:1 Sterkte

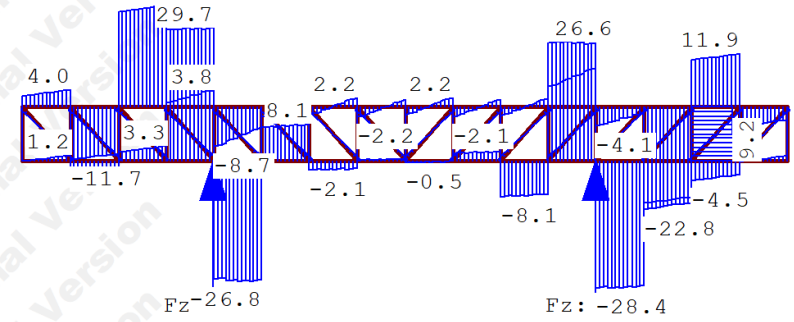


Figure 116: Shear force diagram truss with height of 2.0 meter

NORMAALKRACHTEN 2e orde B.C:1 Sterkte

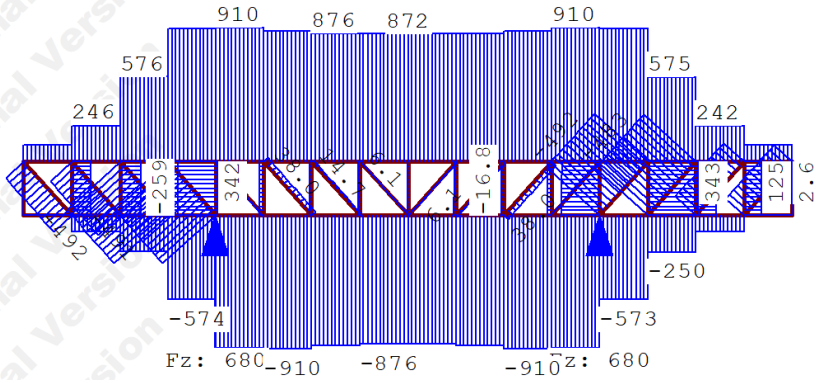


Figure 117: Normal force diagram with height of 2.0 meter

VERPLAATSINGEN 1e orde [mm] B.C:2 Verplaatsing

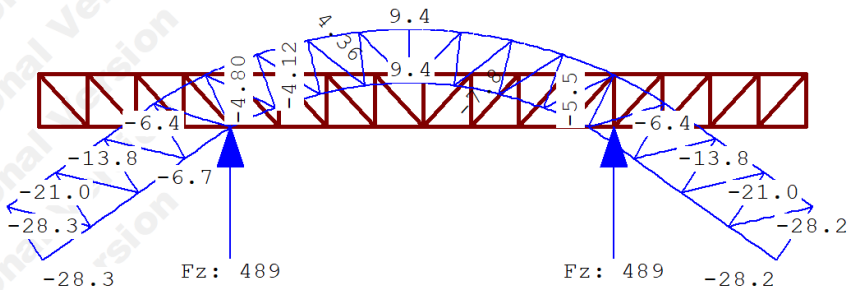


Figure 118: Deflection truss with height of 2.0 meter



Figure 119: Render 1

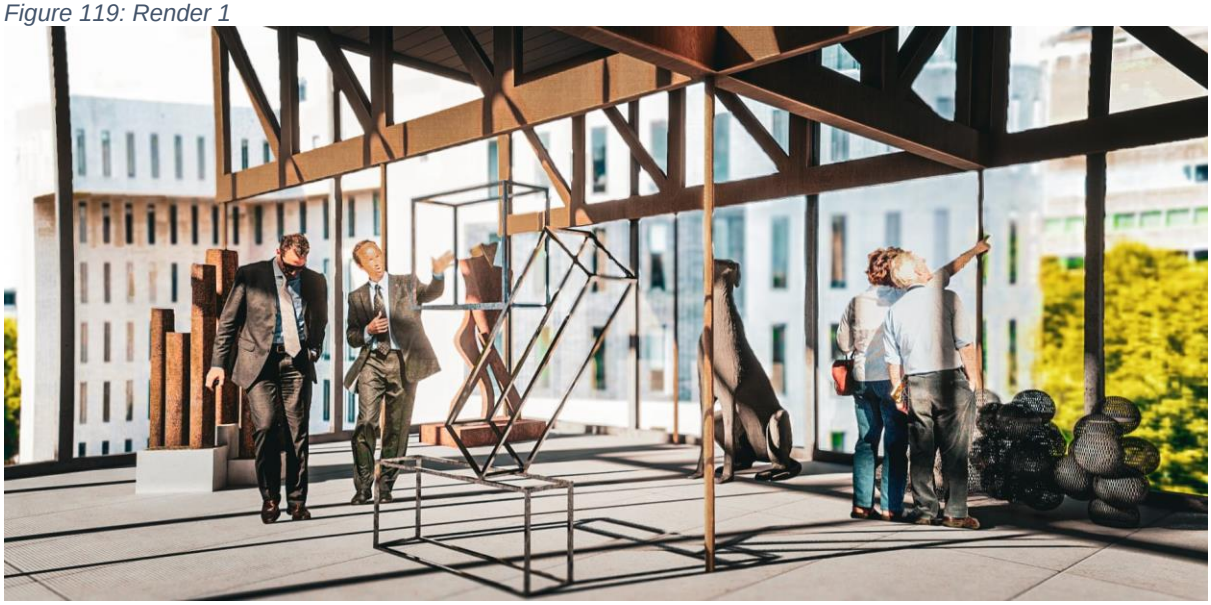


Figure 120: Render 2



Figure 121: Render 3



Figure 122: Render 4



Figure 123: Render 5



Figure 124: Render 6



Figure 125: Render 7



Figure 126: Render 8



Figure 127: Render 9